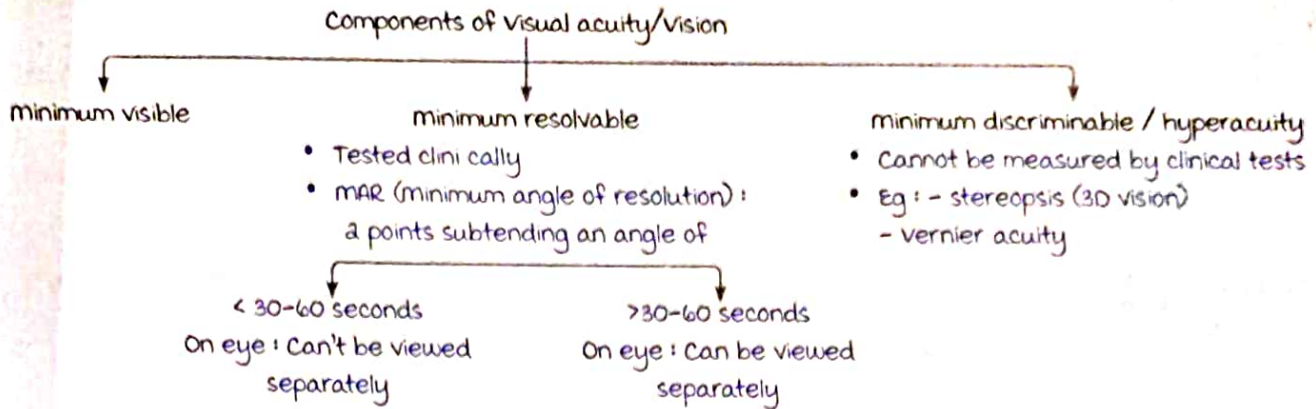


TESTS FOR VISION AND NORMAL OPTICS OF EYES

----- Active space -----



Tests for Vision

00:07:40

TESTS FOR FAR VISION

I. Snellen's chart :

- minimum 7 rows of letters.
- Number of letters per row : variable.
- Each eye tested separately.

Fraction at the rows :

eg :- $\frac{6}{24}$ → Distance of examination (in m).
 $\frac{6}{24}$ → Respective distance: minimum distance at which normal eye can visualize the row.

Procedure :

Patient positioned at 6m/20ft. from chart

Serial attempts to read all rows until smallest visible row $\xrightarrow{\text{If visible}}$ Record acuity
 - eg : 6/9

If even 6/60 is not visible at 6m

move patient closer in increments of 1m until larger row is visible $\xrightarrow{\text{If visible}}$ Record acuity
 - eg : 3/60 if largest row seen at 3m

Finger counting close to face $\xrightarrow{\text{If visible}}$ FCCF +

If negative

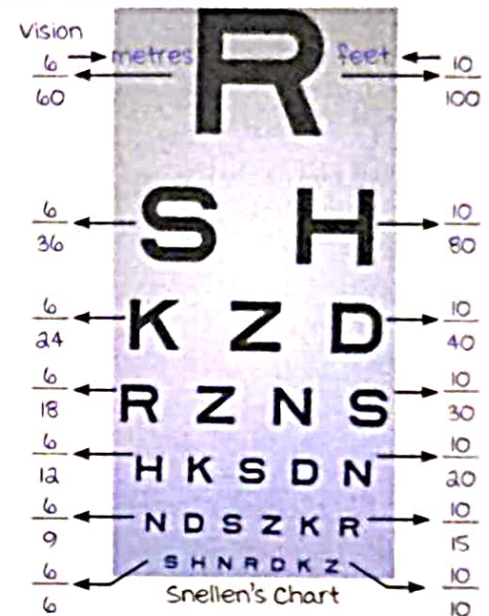
Perception of hand movements tested $\xrightarrow{\text{If visible}}$ Hm (+)

If not perceived

Perception of light tested $\xrightarrow{\text{If visible}}$ PL +

If negative

Patient blind.



----- Active space -----

4. Tests for verbal fx - school children (age 2-5 yrs):

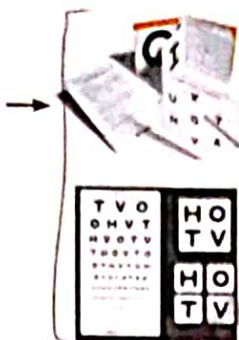
Do not require communication by patient.

Types

Based on matching of letters:

- a. **Sheridan Gardiner**: HOTV test
- Child matches letters on the cards/tiles

- b. **STYCAR**: Screening in young children & the intellectually disabled



Based on matching of pictures:

- a. Key Picture test:



- b. Broken wheel test



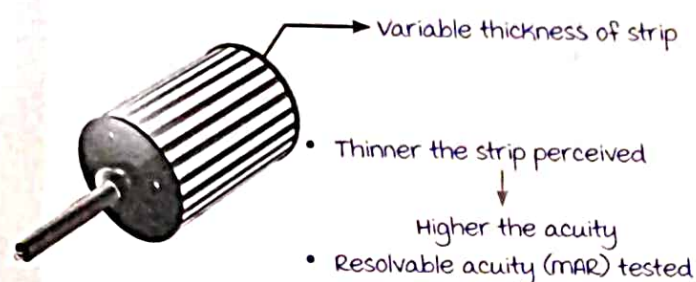
- c. Allen Picture card pre school test

5. Tests for infants

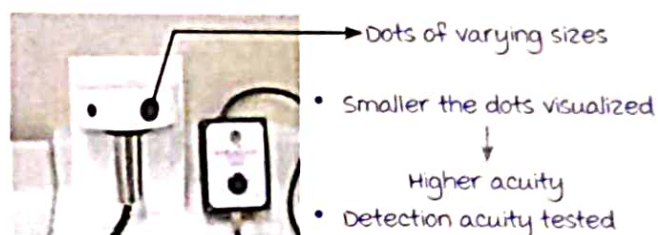
- a. Optokinetic Nystagmus (OKN) tests:

- Done with rotating drum → Infant fixates gaze on dot/strip as it appears & disappears.

OKN Drum:



Carford drum:



- b. Preferential Looking Test (FLT):

- An infant prefers to look at striped over plain areas.
- Progressive narrowing of stripes.
- Thinner the stripes visualized → ↑ acuity.

Types of FLT

- c. Cardiff Acuity Test (CAT):
used in both infants & toddlers.
Based on vanishing optotype.



CAT



Teller acuity cards



Keeler gratings

- d. Visual evoked potential (VEP):

Principle: Light stimulus → visual pathway → ↑ visual cortex activity → Normal VEP.

----- Active space ----- e. Indirect assessment :

Reflex	Components	Time of development
Blink reflex	- Stimulus : Illumination of eyes - Response : Blinking of eye	Present since birth
menace reflex	Stimulus : Approach of an object near eye Response : Closure of eyes	Develops around 5 months of age

TESTS FOR NEAR VISION

use : for diagnosis of presbyopia

Distance of examination: 30-35 cm (comfortable reading distance).

a. Snellen's near chart (m/c) : N_6 (best), N_9 , N_{12} , N_{18} , N_{24} ,
 N_{36} (worst).

b. Jaeger's chart : Results between $J_1 \rightarrow J_7$.

c. Roman test type

Note : least distance of distinct vision (Near point) = 25cm.



Snellen's near chart

TESTS FOR CONTRAST SENSITIVITY

use : Diseases like cataract

Contrast sensitivity affected before visual acuity

Tests : mnemonic $\rightarrow$ CRAP BF

Cambridge low contrast gratings

Regan chart

Arden gratings

Pelli-Robson chart (m/c) $\rightarrow$ • Letters as triplets

Bailey-Lovie contrast chart • Distance of examination : 1m

FACT chart.



Normal Optics of Eye

00:58:42

----- Active space -----

Function of eye : converging of light rays onto retina.

Areas with refractive power :

a. Gullstrand's Concept :

	Cornea		Lens	
	Anterior surface	Radius of curvature	Anterior surface	Posterior surface
Refractive powers	+48.8 D	-5.88 D	+19 D	
Radius of curvature	7.7 mm	6.8 mm	10 mm	-6 mm

Note : Power at posterior surfaces of cornea & lens are in the negative as light is moving from denser to rarer medium.

b. Listing's reduced eye :

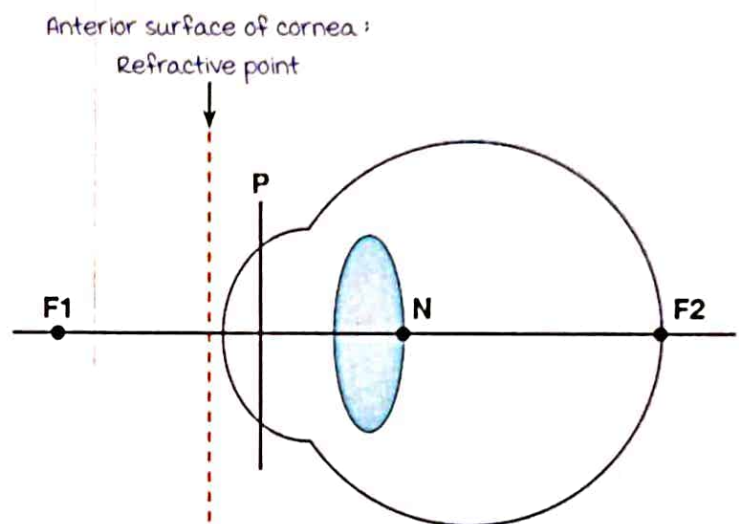
- Total power of :
 - Eye : +58.6 D
 - Cornea : +43 D
 - Lens : +19 D

• Application :

Alter axial length

N, P, F alteration

Aberrant image formation
(not on retina)



P : Principle point (1.35mm from reference)

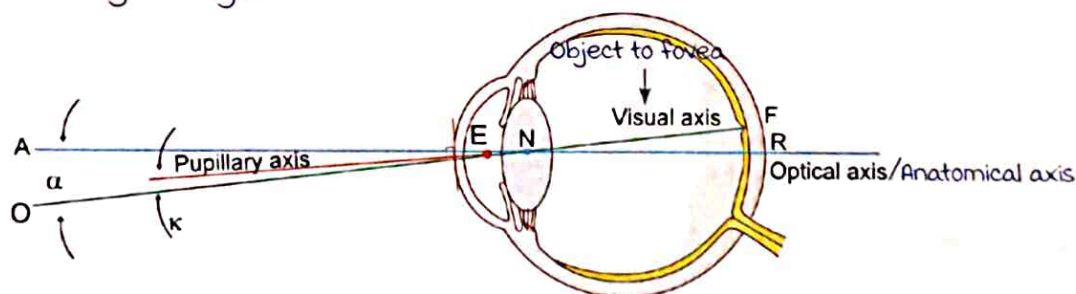
N : nodal point (7.0mm from reference)

F_a : Focal point for real image (24.13mm from reference)

F_v : Focal point for virtual image (-15.7mm)

Factors affecting refracting Power :

- Radius of curvature.
- Difference in refractive index b/w 2 media.
- Axial length of eye.



a. Angle kappa (κ) : B/w visual & pupillary axis.

- Normally $\leq 5^\circ$ (if $\uparrow\uparrow$ or $\downarrow\downarrow$ $\rightarrow$ appearance of deviated eyeball).

b. Angle alpha (α) : B/w visual axis & anatomical axis.

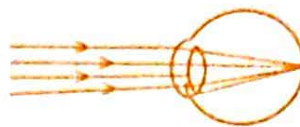
----- Active space -----

MYOPIA AND HYPERMETROPIA

Emmetropia :

Normal refractory power of eye (+58.6 D).

Parallel rays from infinity



Converge on retina

Ametropia :

Abnormal refractory power.

mono-ocular

Present in monocular and binocular vision.

Spherical

Corrected by spherical lens.

- myopia.
- Hypermetropia.

Cylindrical

Corrected by cylindrical lens.

Astigmatism.

Binocular

- Pathology contributed to by both eyes.
- Present only in binocular vision.
- Disappears in monocular vision.
- Eg:
 - Anisokonia.
 - Anisometropia.

Pathogenesis and Mechanism of Spherical Refractory Errors

00:06:30

myopia :

Pathogenesis :

↑ Refractory power of eye

↓ Early convergence of rays.

↓ Image formed anterior to retina.

Corrective lenses : (-) lenses.

Hypermetropia :

Pathogenesis :

↓ Refractory power of eye

↓ converging of rays.

↓ Image formed behind retina.

Corrective lenses : (+) lenses.

mechanisms of change in refractive power of eye :

----- Active space -----

	myopia	Hypermetropia
a) Axial length (AL) (Normally 24mm)	Axial myopia (m/c) : • ↑ Axial length. • 1mm ↑ in AL = 3D added to power. • Eg : Buphthalmos.	Axial hypermetropia : • ↓ Axial length. • 1mm ↓ in AL = 3d subtracted from power.
b) Curvature of cornea/lens (Radius of curvature : ROC)	Curvatural myopia : • ↑ Curvature of cornea/ lens : ↓ ROC. • Eg : Keratoconus/Lenticonus.	Curvatural hypermetropia : ↓ curvature of cornea/lens : ↑ ROC.
c) Refractive index (RI) of lens > cornea.	Index myopia : ↑ RI of lens. Eg: Nuclear cataract (sclerosis).	Index hypermetropia : ↓ RI
d) Position of lens (Power = 1/ focal length)	Positional myopia : • Lens positioned anteriorly. • Focal length : ↓ • Power of lens : ↑ • Eg: Weill marchesani syndrome → Lens moves forwards → ↓ focal length & ↑ power	Positional hypermetropia : • Lens positioned posteriorly. • Focal length : ↑ • Power of lens : ↓

Presentations of Spherical Refractory Errors

00:20:09

SYMPTOMS

In myopia :

- Loss of far vision : Near/short-sightedness.
- Floaters present in high myopia : D/t vitreous syneresis/liquefaction.
- Half closure/Half squinting of eyes
 - Cutting off of peripheral rays → Central rays fall straight onto retina → Improves vision

In hypermetropia :

Loss of near vision (far sightedness) : Clinically no difficulty in near vision d/t over-accomodation → Presents as tired eyes/headache : Asthenopia.

----- Active space -----

EXAMINATION SIGNS

	myopia	Hypermetropia
Size of eyeball	Large ($\uparrow$ AL) $\rightarrow$ Pseudoproptosis	Small ($\downarrow$ AL)
Anterior chamber	Deeper than normal	Shallower than normal : Prone to angle closure glaucoma.
Squint	Far point comes near to eye $\downarrow$ Accommodation not required for near vision (Convergence does not occur after) $\downarrow$ Convergence insufficiency $\downarrow$ Propensity to deviate outwards : <u>Exophoria.</u>	Overaccommodation $\downarrow$ Overconvergence $\downarrow$ <u>Esotropia</u>
Angle Kappa	Lesser than normal $\rightarrow$ <u>Pseudoesotropia</u> : Appearance of inward deviation of eye	more than normal ($>5^\circ$) $\rightarrow$ <u>Pseudoexotropia</u> : Appearance of outward deviation of eye.

FUNDOSCOPY SIGNS

Fundus : View of all structures visible together in posterior segment of eye.

- vitreous.
- Retina. } Transparent
- Choroid : Red.

Hypermetropia :

- Shot silk fundus : Shiny appearance.
- Pseudo papillitis.

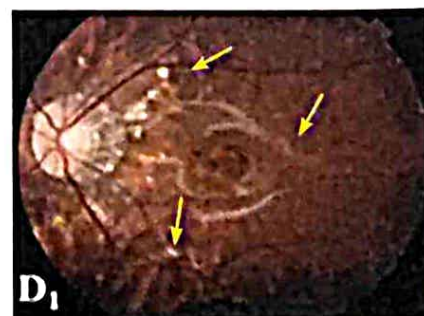
myopia :

- Signs seen if pathological/degenerative myopia present :

- AL > 26 mm OR
- Refractive power $> +6$ D.

a. Tessellated/Tigroid appearance :

- Diffuse attenuation of retinal pigment epithelium.
- Choroidal red color more visible.

**D₁**

Tessellated/Tigroid appearance

b. Temporal myopic crescent :

Disparity in stretching of eyeball

Chorioretinal atrophy

underlying sclera visible (white)

c. Lacquer cracks :

Fine breaks in Bruch's membrane
(innermost layer of choroid)

underlying sclera visible (yellowish)

d. Foster Fuch spots : Areas of subretinal hemorrhage at macula.

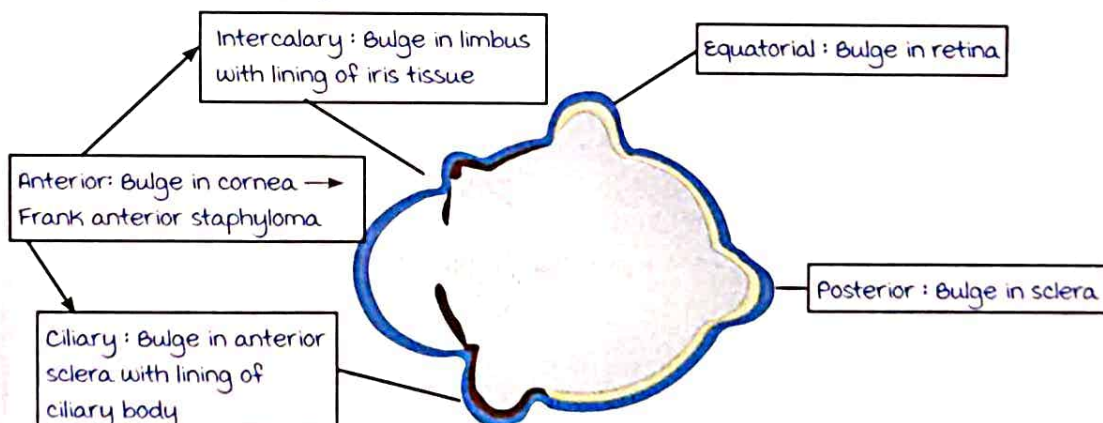
e. Posterior staphyloma : In sclera.

↑AL → Scleral stretching → Scleral thinning
 → Bulging d/t normal IOP → Staphyloma
 (Ectatic bulge in sclera).

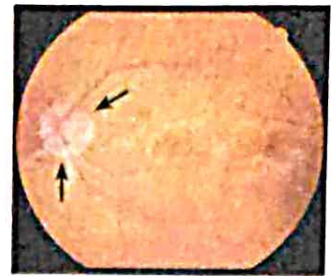
f. Rhegmatogenous retinal detachment :

- m/c complication of pathological myopia.
- ↑AL → retinal stretching.
- Break in retina (detachment)
- Rupture of vessels : vitreous hemorrhage.
- Supertraction : vitreous pulls retina over optic disc (OD) → OD covered by retina → Tilted OD (seen in very high myopia).

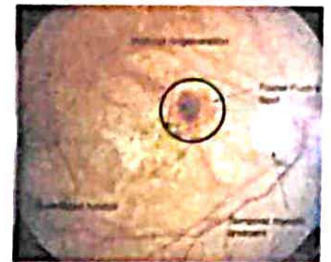
Note : Types of staphyloma



----- Active space -----



Temporal myopic crescent



Foster Fuch's spots



Lacquer cracks

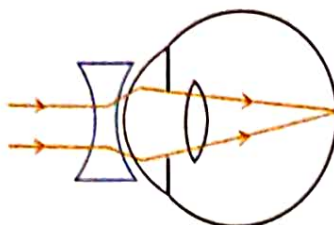
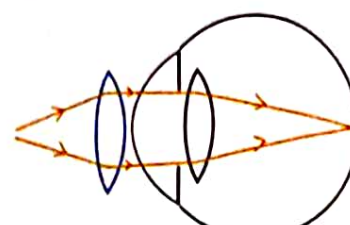
Posterior staphyloma



myopic fundus

----- Active space -----

Spectacles & Lenses

Treatment	myopia	Hypermetropia
Spectacles : Spherical	 Concave/diverging glasses (with negative power)	 Convex/converging lens (with positive power)
Contact lenses	Soft lenses • Preferred in high myopia because of high powered concave glasses. - Are thicker. - minify image.	

Surgery for Hypermetropia

01:00:08

Rarely performed d/t :

- Incidence of hypermetropia < myopia.
- very high hypermetropia is rare.

1. Thermal laser keratoplasty : THC : YAG laser.
(THC : Thallium Holmium Chromium)

2. Hyperopic LASIK.

3. Conductive keratoplasty : using radiofrequency

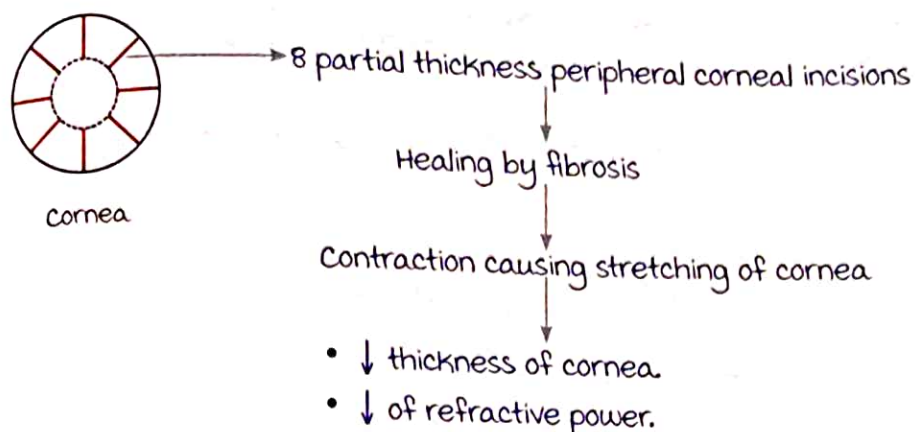
Surgery for Myopia

01:03:11

RADIAL KERATOTOMY

Principle : ↓ corneal curvature → ↓ refractive power.

procedure :



- Not done currently d/t unpredictable results.

LASER SURGERY

1. PRK : Photo Refractive Keratectomy.

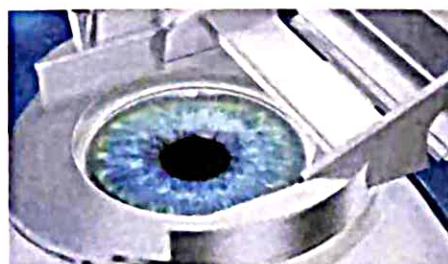
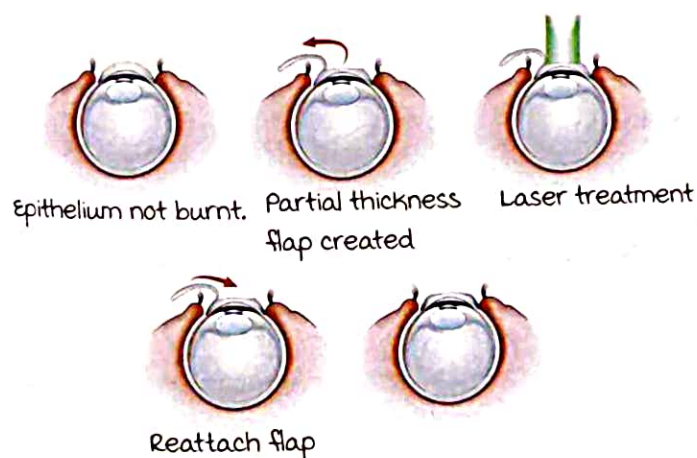
2. LASIK : Laser Assisted In-Situ Keratomileusis

Excimer laser used.

Excimer laser :

- Argon fluoride laser.
- Wavelength : 193nm.
- MOA : Photablation → Central thinning of cornea → Flattening of cornea → ↓ Refractive power.

LASIK procedure :



----- Active space -----

- 10-12 microns ablated for each dioptre.
- 250 micron stromal bed residual thickness.

C/I to LASIK :

- Corneal thickness $< 450\mu\text{m}$ pre-operation : $\uparrow$ chance of breaching residual thickness barrier.
- Age < 18 yrs.
- Refraction is not stable for at least 12 months pre-op.

3. SMILE : Small Incision Lenticule Extraction :

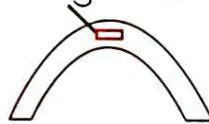
- Laser used : Nd : Glass
- Mechanism : Photodisruption.
- Flapless surgery $\rightarrow$ No side effects from flap creation.

Procedure of SMILE :

1. Patient positioned near machine.
2. Machine creates suction around cornea :
Ensures eye does not move.
3. Laser fixed:

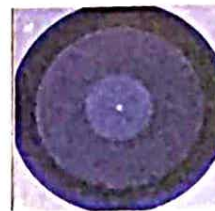
2 incisions given near stroma :

- Deep.
- Superficial.



Free scleral stromal button created : Lenticule.

4. Corneal Incision: Lenticule pulled out $\rightarrow$ Thinned cornea.



Eye visualised



Stabilisation by suction



Pulling out of the lenticule

ICL SURGERY

Implantable collamer lens (ICL)

- made of hydrogel.
- Implanted in the sulcus anterior to lens.

Indication : myopia $> 8D$ exclusively (high myopia).

- LASIK in high myopia $\rightarrow$ $\uparrow$ chance of breaching stromal residual bed thickness criteria.

INTACS

Intra Corneal Stromal Ring Segments

- Can correct 2-5D pf myopia (low power myopia)

2 semi-circular rings
implanted $\rightarrow$ Stretches &
flattens cornea.



Note : Convex v/s concave lens.

----- Active space -----

	Convex lens	Concave lens
Refraction	• Converging lens. • (+) lens.	• Diverging lens. • (-) lens.
Identification	• High power lenses: - Thick in centre. - Object appears magnified. • On moving convex lens : Object moves in opposite direction.	• High power lens : - Thick at periphery. - Object appears minified. • On moving concave lens : Object moves in same direction.
uses	• Correction of hypermetropia, aphakia, presbyopia. • Ophthalmoscopy (+20D in indirect).	Correction of myopia.
Image	• Real and inverted (m/c). • may produce virtual and erect image.	Always virtual, erect, diminished image.

----- Active space -----

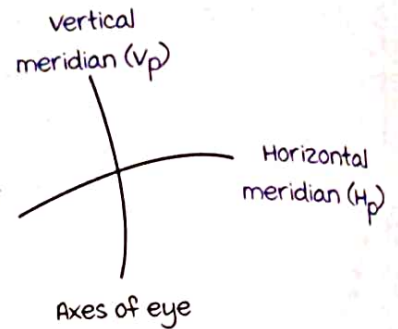
ASTIGMATISM, READING OF SPECTACLE PRESCRIPTION AND BINOCULAR ERRORS

00:00:13

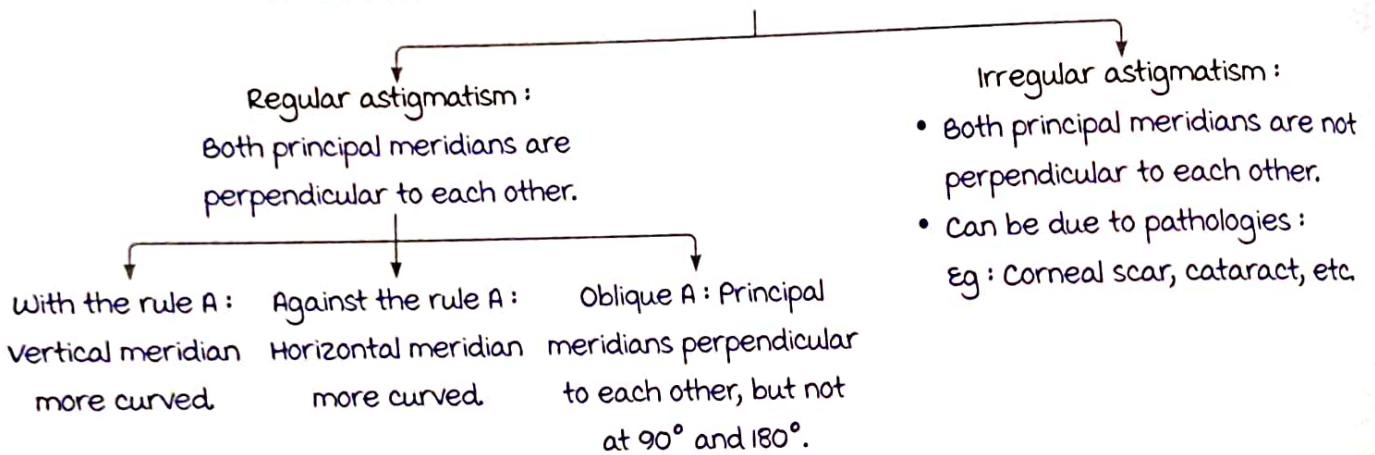
Astigmatism (A)

Overview :

- Cylindrical refractory error.
- Different refractory power in both principal meridians/axes of eye.
(In spherical refractory errors $\rightarrow$ Same power in both axes).



CLASSIFICATION



REFRACTIVE TYPES

	Axis 1	Axis 2
Simple myopic	Emmetropic	Myopic
Compound myopic	Myopic	
Simple hypermetropic	Emmetropic	Hypermetropic
Compound hypermetropic	Hypermetropic	
mixed astigmatism	Myopic	Hypermetropic

CLINICAL FEATURES

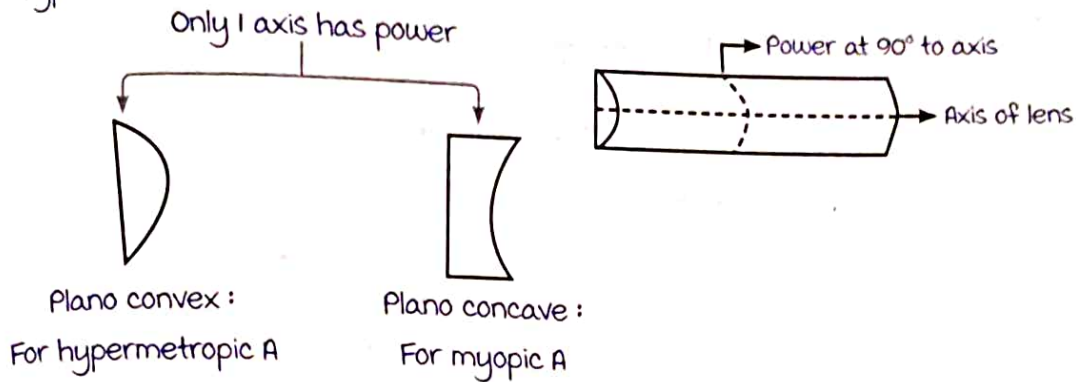
1. Loss of vision : Loss of sharpness (All letters seem to stretch/merge).
2. Asthenopia.

TREATMENT

1. Spectacles :

cylindrical lenses.

- Have power 90° to their axis.
- Types :

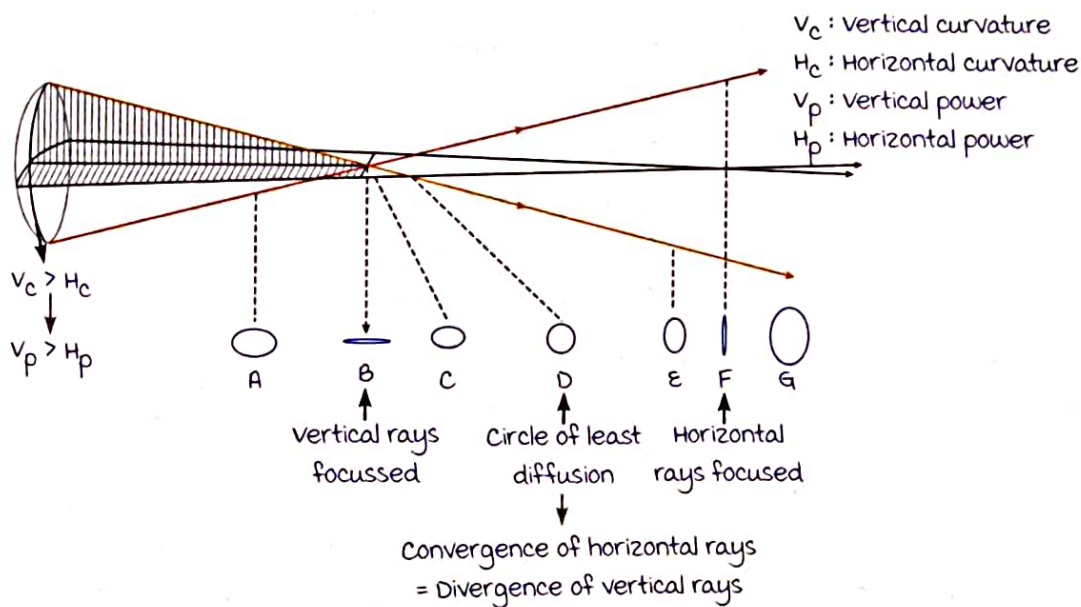


a. Toric Contact Lenses :

A lens with 2 surfaces with different powers.

Sturm's Conoid :

Arrangement of light rays refracted through a toric surface.



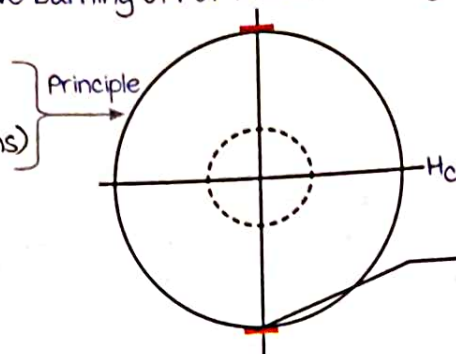
Principle : Converges rays so circle of last diffusion lies at retina $\rightarrow$ Clear image visualized.

----- Active space ----- 3. Surgery :

a. Astigmatism Lasik : Selective burning off of the axis with higher curvature in the cornea.

b. Accurate keratotomy

c. LRI (Limbal relaxing incisions)



Eg :
With the rule astigmatism ($V_c > H_c$)

Partial thickness incision at 90° & 180° on cornea/limbus

Heals by fibrosis

vertical curvature stretched

Reduced V_c

Deciphering Spectacle Prescriptions

00:21:17

Format of prescription :

	Sph	Cyl	Axis
RE			
LE			

- RE : Right eye.
- LE : Left eye.
- Sph : Spherical power $\xrightarrow{\text{For}}$ myopia/hypermetropia.
- Cyl : Cylindrical power $\xrightarrow{\text{For}}$ Astigmatism.
- (+) power prescribed for hypermetropia.
- (-) power prescribed for myopia.

Spherical and cylindrical lenses :

	Lens type	Example of prescription
myopia	- sph	-1D sph
Simple myopic astigmatism	- cyl	-1D cyl
Compound myopic astigmatism	- sph/- cyl	-1D sph/-1.5D cyl
Hypermetropia	+ sph	+0.75D sph
Simple hypermetropic astigmatism	+ cyl	+1.5D cyl
Compound hypermetropic astigmatism	+ sph/+ cyl	+0.5D sph/+1 D cyl
mixed astigmatism	- sph/+ cyl	-1D sph/+2D cyl
	+ sph/- cyl	+1D sph/-2D cyl

simple transformation :

done to ensure that cyl power > sph power

eg :

- In a prescription $-x$ D sph / $+y$ D cyl @ 180° where $x > y$.

↓
Apply simple transformation.

- New sph : Old sph + old cyl.
- New cyl : $-(\text{Old cyl})$.
- New axis : Perpendicular to old axis.
- New prescription : $(y-x)$ D sph / $-y$ D cyl @ 90° .

With and Against the rule astigmatism :

Inferred using the axis of cylindrical lens being prescribed.

	180°	90°
With the rule	- lens (c)	+ lens (v)
Against the rule	+ lens (v)	- lens (c)

C : Convex lens.

V : Concave lens.

- Eg : -1 D cyl @ 180° (cylindrical lens with axis of 180°)

↑ power at 90° (As power of cylindrical lens is perpendicular to its axis)

↓
vertical power > horizontal power : **With the rule astigmatism.**

Binocular Refractory Errors

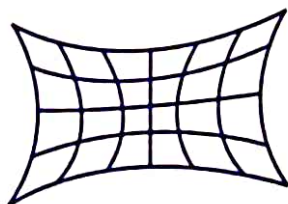
00:46:45

ANISEIKONIA

- **Different shape** (Distortion) **and/or size of an image** between both eyes.
- Occurs d/t varying thickness of glasses :
 - Convex lens : Thicker at center.
 - Concave lens : Thick at periphery.

Pin cushion defect : Seen with very high plus (+) correction.

Barrel defect : Seen with high minus (-) correction.



Pin cushion defect



Barrel defect

----- Active space -----

ANISOMETROPIA

- Different refractory power b/w both eyes.
- Significant if difference > 2.5 D.

magnification rule for convex lenses for 1D $\rightarrow$ 2% image magnification $\xrightarrow{\text{if significant anisometropia}}$ varying sizes of images from each eye $\rightarrow$ Failure to fuse 2 images : Diplopia.

Rx : Surgical correction.

REFRACTION : HOW TO PRESCRIBE GLASSES AND APHAKIA

----- Active space -----

Refraction : Procedure of determining and correction of refractive errors.

Types :

Objective refraction :

- performed **first**.
- Independent of patient responses.

Subjective refraction :

- Performed **second**.
- Patient's response taken into account.

Objective Refraction

00:02:21

performed in **Retinoscopy/Skiascopy/Shadow test**.

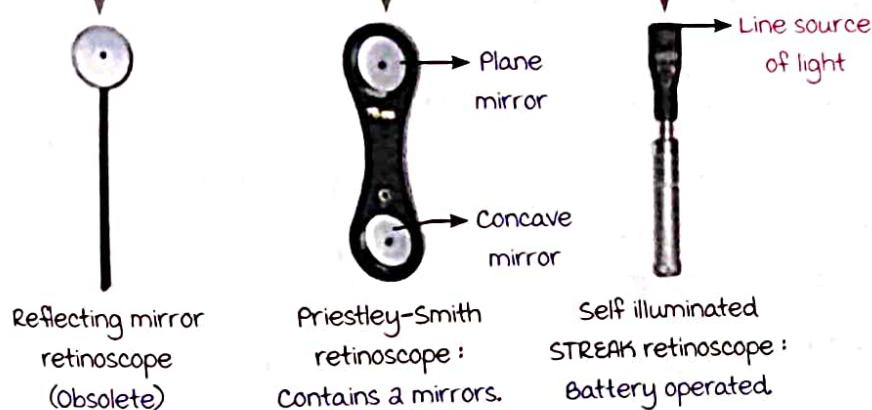
methods :

Retinoscopy methods

manual retinoscopy

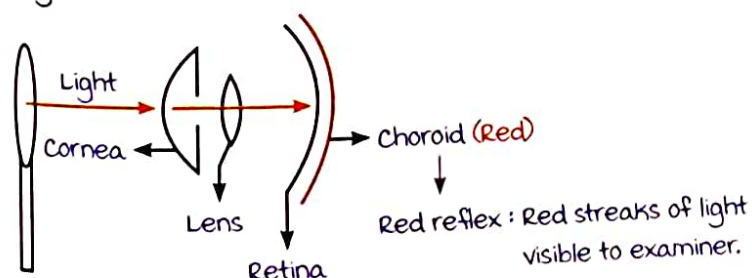
Automated refractometers

Instruments employed



PROCEDURE OF RETINOSCOPY

using self illuminated STREAK retinoscope.

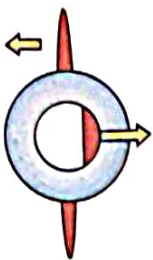
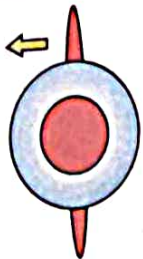
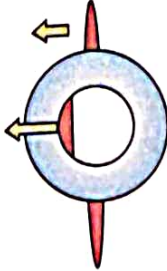
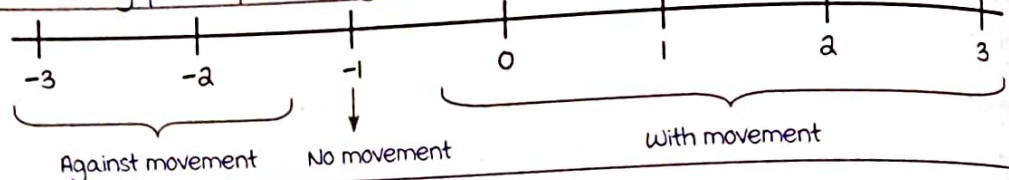


Self Illuminating Streak retinoscope

----- Active space ----- Distance of examination : 1 metre.

Method : Examiner moves retinoscope → Direction of movement of red streak compared to movement of retinoscope.

Findings :

	Against movement	No movement	With movement
Appearance			
Dx	> 1D myopia	1D myopia	< 1D myopia / Emmetropia / Hypermetropia
Refractive power scale			

RETINOSCOPY VALUE

Determined by neutralization.

Neutralization :

Procedure : movement of red reflex seen on retinoscopy

↓
Place lens in front of eye during retinoscopy

Power of lens at which movement ceases (neutralization) is noted.

Influence : Power of lens at neutralisation = RV.

- R.V. is 0 at +1D (if no lens used).

Corrected Retinoscopy value (CRV) :

Takes into account :

1. Variation in distance of examination (d) : $C_i = 1/d$
2. Addition of cycloplegics/mydriatics :
 - Correction for tonal allowance.
 - $C_a =$
 - a. 1 : Atropine
 - b. 0.75 : Cyclopentolate
 - c. 0.50 : Homatropine

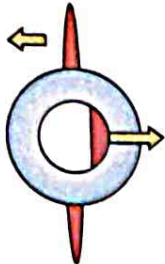
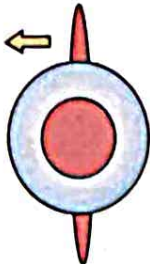
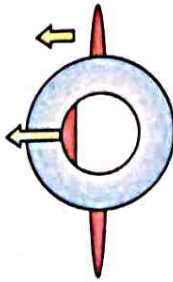
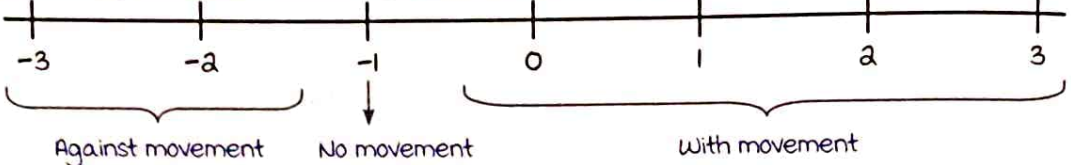
Formula : $CRV = RV - C_i - C_a$

- CRV can be calculated in different axes.

----- Active space ----- Distance of examination : 1 metre.

Method : Examiner moves retinoscope → Direction of movement of red streak compared to movement of retinoscope.

Findings :

	Against movement	No movement	With movement
Appearance			
Dx	> 1D myopia	1D myopia	< 1D myopia / Emmetropia / Hypermetropia
Refractive power scale			

RETINOSCOPY VALUE

Determined by neutralization.

Neutralization :

Procedure : movement of red reflex seen on retinoscopy

↓
Place lens in front of eye during retinoscopy

↓
Power of lens at which movement ceases (neutralization) is noted.

Influence : Power of lens at neutralisation = RV.

- R.V. is 0 at +1D (if no lens used).

Corrected Retinoscopy value (CRV) :

Takes into account :

1. Variation in distance of examination (d) : $C_d = 1/d$
2. Addition of cycloplegics/mydriatics :
 - Correction for tonal allowance.
 - $C_a =$
 - a. 1 : Atropine
 - b. 0.75 : Cyclopentolate
 - c. 0.50 : Homatropine

Formula : $CRV = RV - C_d - C_a$

- CRV can be calculated in different axes.

Note : cycloplegia always done during retinoscopy (D/t interference from accommodation).

- In children < 5 yrs : DOC → 1% Atropine eye ointment (TDS for 3 days).
- In adults :
 - DOC : 0.5% - 1% Cyclopentolate eye drops.
 - m/c used : Tropicamide (D/t quick and short acting).

----- Active space -----

Prescribing spectacles

00:27:00

method :

Step 1 : Choose any axis as 'S'.

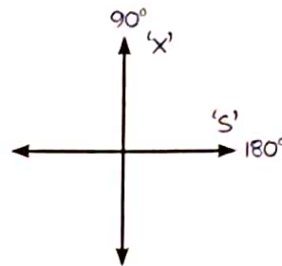
Step 2 : make other axis as 'X'.

Step 3 : Calculate for prescription.

Format for prescription of spectacles :

- Spherical CRV/Cylindrical CRV @ axis of spherical.

S/X-S at axis of S or X/S-X at axis of X



Where X and S are CRVs in their respective axes

Subjective refraction

00:37:07

Assesses if patient is comfortable with power calculated on objective refraction.

Adjustment of refraction :

- Trial and error method.
- Fogging technique.

Refinement of refraction

Cylinder :

- Jackson cross cylinder test.
- Astigmatic fan test.

Sphere :

- Duochrome test.
- Pinhole testing.

Binocular balancing :

- Duochrome test with fogging.
- Prism dissociation method.

ADJUSTMENT OF REFRACTION

1. Trial and error method :

- Primary method using a trial frame.
- Lens power altered till Snellen's chart is clearly visible.
- Unocular testing.



Trial frame :

Contains multiple slots for spherical & cylindrical lenses.

2. Fogging method :

Use :

- To overcome abnormally high accommodation.
- Before cycloplegia.

----- Active space ----- Procedure : Placing very high convex lens (+ power).
 ↓
 Relaxes ciliary muscles (Relaxes accommodation).
 ↓
 True refractive error can be assessed.

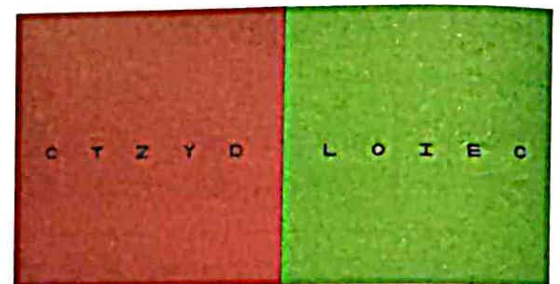
REFINEMENT OF REFRACTION

Detection of minute errors in prescription.

Refinement of spheres :

1. Duochrome test :

Appearance	Inference
Letters equally clear against both backgrounds	Correct spherical power prescribed
Letters on red background appear clearer	Residual myopia : Indicates undercorrection
Letters on green background appear clearer	Residual HMT : Indicates overcorrection



Duochrome test

2. Pinhole test :



Pinhole : 1 mm diameter

Uses	Procedure	Results & Inference
Refinement/ Confirmation of spherical power	After prescription of spectacles ↓ Pinhole kept in front of eye with spectacles.	If vision improves further ↓ Indicates residual refractory error upto 4D.
To determine presence/absence of refractory error	When patient complains of LOV ↓ Pinhole kept in front of eye.	• Vision improves : Indicates refractory error as cause of LOV. • Vision does not improve: Indicates other ocular pathology as cause of LOV.
Perform 2-point discrimination test / macular function test	Disc with 2 pinholes 2mm apart ↓ Single light source shown to patient.	Normal macula : Patient sees 2 separated light points.
Low vision aid	Multiple holed pin hole used	-

Refinement for cylinders :**1. Jackson's Cross Cylinder Test :**

- Power :
-2.5 D Sph/+0.5 D Cyl or -0.5 D Sph/+1 D Cyl.
- Create effect if 2 cross cylinders :
 - minus cylinder on 90° axis.
 - Plus cylinder on 180° axis.

2. Astigmatic Fan Test :

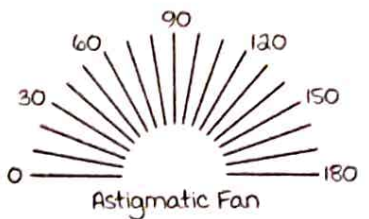
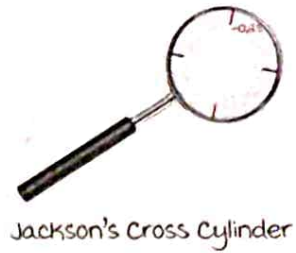
method : view fan with spectacles.

All lines appear equally dark/clear

Any line looks darker/clearer

Appropriate prescription provided

Refinement needed in cylindrical prescription.

**Stenopaic Slit (SS)**

00:55:15

USES**1. Diagnosis of astigmatism :**

Procedure : SS kept in front of eye → Rotated slowly for 360°.

Observation & Inference : If patient sees clear image at a particular point

Indicates presence of astigmatism.

**2. Confirmation of power of astigmatism :**

Procedure : Cut hole in axis of cylinder while patient wears spectacles prescribed.

Observation & Inference : If vision improves → Cylindrical power needs correction.

3. Fincham's test :

uses	Procedure	Observation & Inference
Differentiate b/w cataract and glaucoma (Both present with halos around light)	Stenopic slit rotated in front of eye	• Halos remain intact : Indicative of ACG. • Halos appear broken → Dx : Cataract
Determine median for optical iridectomy	vision compromised d/t corneal opacity. ↓ SS rotated to determine meridian least affected by corneal opacity.	Iridectomy performed along meridian

----- Active space ----- Note :

Empirical meridian for iridectomy — { Inferonasal → For patients involved in near work.
Superotemporal → For patients involved in far work.

Amaurosis

01:06:49

Definition : Complete loss of vision in one/both eyes without any organic cause.
(No signs on ocular examination)

Note : Amblyopia → Partial LOV.

TYPES

Bilateral amaurosis :

Causes :

- Acute nephritis.
- Uremic amaurosis :
 - B/L complete LOV.
 - Pupillary reflexes intact as only higher visual centres affected.

Amaurosis Fugax :

Transient Complete LOV.

Causes :

- Thrombo embolic episodes (strokes) → m/c cause.
- CRAO (Central retinal artery occlusion).
- Papilloedema.

Aphakia

01:12:10

Absence of lens.

Causes :

- Surgical.
- Traumatic.
- Congenital.

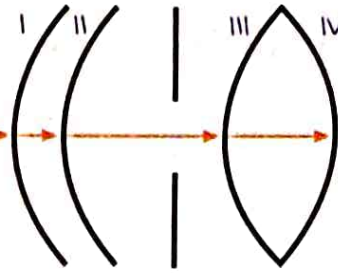
SYMPTOMS

- LOV for far and near vision (No accommodation).
- Refractory error : Hypermetropia (Low power of eye).
- AC deeper : Iris moves posteriorly d/t absence of lens.
- Irido-donesis (Trembling iris) : D/t lack of support posteriorly.
- Purkinje images : 2 visible.

Note : Purkinje images

• **Beam of light**

when cast on eye :
Reflects, forming purkinje images.



----- Active space -----

• Normal : 4 purkinje images.

I : Anterior surface of cornea

II : Posterior surface of cornea

III : Anterior surface of lens

IV : Posterior surface of lens

Convex surface → Erect image.

Concave surface → Inverted image.

TREATMENT

a. IOL implantation : Treatment of choice.

b. Soft contact lens

c. Spectacles :

- Spherical convex (+) power.
- maximum possible +10 D to +13 D : Heavy, large lens.

Side effects of high power spectacles :

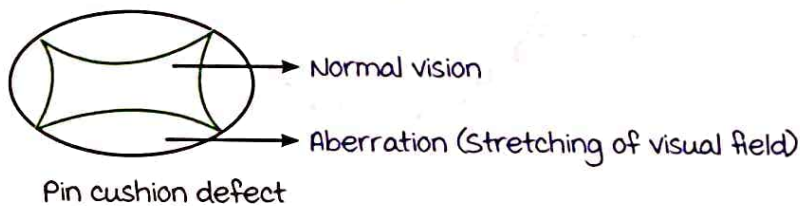
3. magnification : upto 20% magnification (For +10 D spectacles)

↓
Abberant perception of distance of objects

↓
Objects appear nearer.

4. Cosmetic side effect : Eyes look large (Frog eyes).

5. Spherical aberration : Pin cushion defect.



6. Prismatic aberration : Jack in box phenomenon/Roving ring scotoma.

- mechanism : ↑ Thickness of lens

↓
Edges behave as a prism

↓
multiple refractions occurring near edges

- Only central clear vision.
- Peripheral vision compromised.

----- Active space -----

ANATOMY AND INVESTIGATIONS OF RETINA

00:01:33

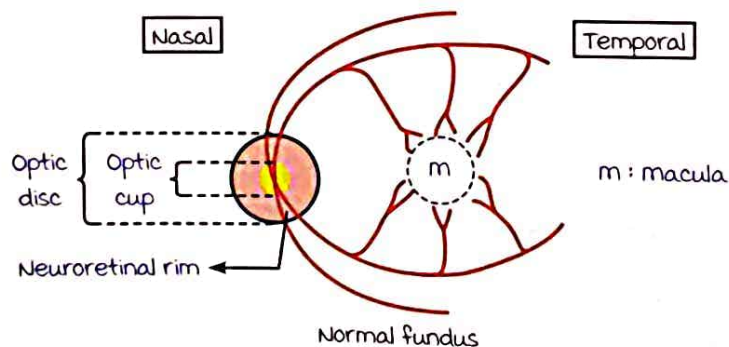
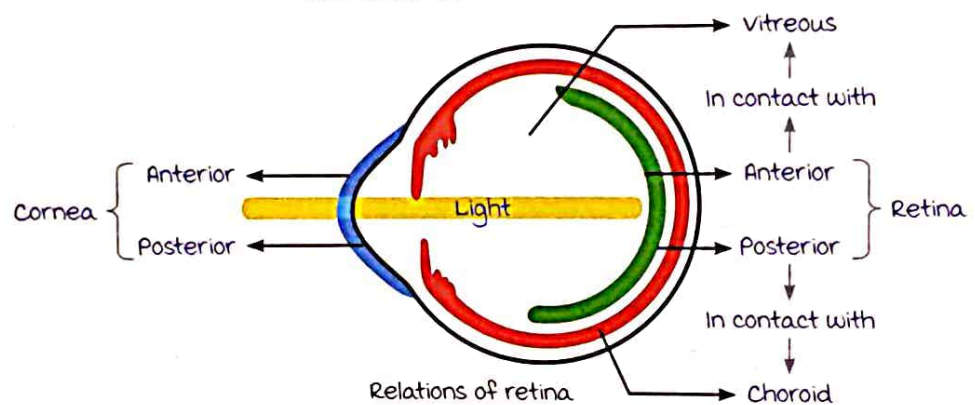
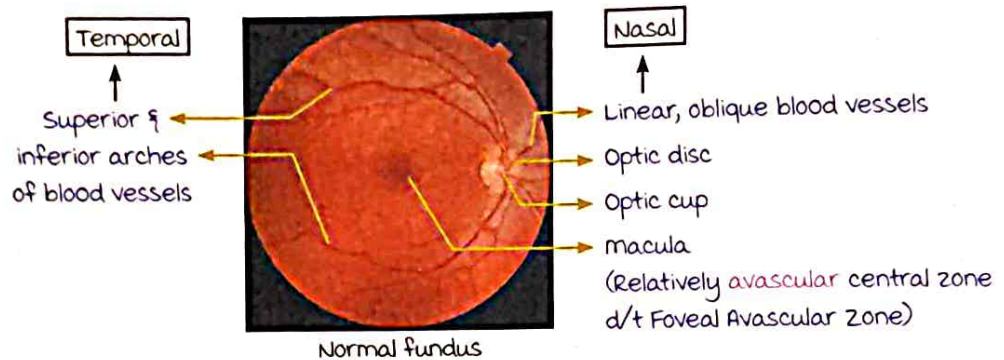
Anatomy

Colour of retina : Transparent.

Number of layers : 10.

Gross anatomy :

- Optic disc.
- macula For central vision; Central part : Fovea.
Location : Temporal to optic disc.
- Ora serrata : Anterior termination of retina.
- Part of 'fundus' (view of all structures in posterior segment).



Layers of Retina :

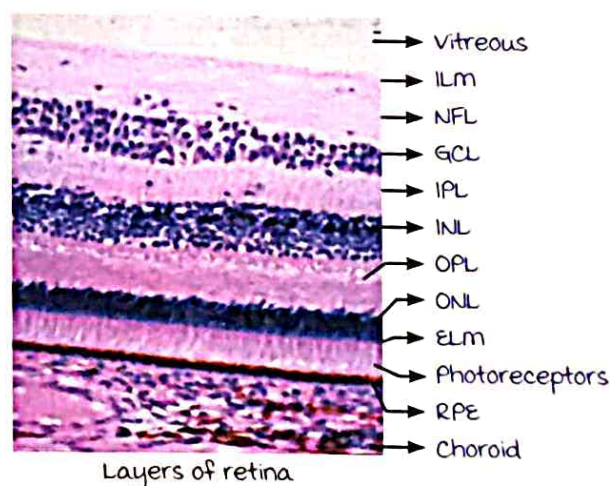
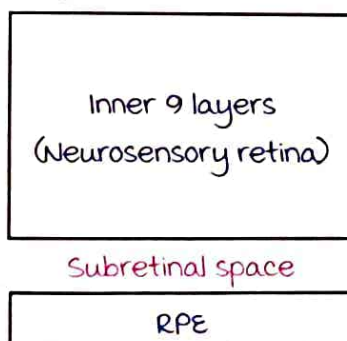
----- Active space -----

Ten layers : mnemonic → **IN6 [IOP]** (Innermost to outermost).

Layer	Feature
Internal limiting membrane (ILM)	• True basement membrane. • Formed by footplates of muller cells (Structural supporting cells).
Nerve fiber layer (NFL)	• Forms optic nerve (Collection of axons of ganglion cells). • 3rd order neurons.
Ganglion cell layer (GCL)	• Contains ganglion cells : 3rd order neurons. • Initial site of generation of action potential.
Inner plexiform layer (IPL)	Synapse between ganglion cell & bipolar cells.
Inner nuclear layer (INL)	Contains bipolar cells : 2 nd order neurons.
Outer plexiform layer (OPL)	• Synapse between bipolar cells & photoreceptors. • Endothelial cells of capillaries → Form inner blood retinal barrier.
Outer nuclear layer (ONL)	
External limiting membrane (ELM)	Not a true basement membrane.
Photoreceptors	Rods & cones (1st order neurons). ↓ Receives light (After it passes through 8 layers). ↓ Generates local graded potential → Conducted back to nerve fiber layer.
Retinal Pigment epithelium (RPE)	• Phagocytic function. • Forms outer blood retinal barrier → Prevents collection of fluid in subretinal space.

Neuroglial cells (Structural supporting cells) in retina :

- Astrocytes
- muller cells
- microglia



----- Active space -----

Blood supply of retina :

- Inner 6 layers : Central retinal artery
 - Outer 4 layers : Short posterior ciliary artery
- } From **ophthalmic artery**
(Branch of internal carotid artery)
- (Also supplies choroid & posterior sclera).
- macula : Supplied by central retinal artery ± Cilioretinal artery
(Branch of short posterior ciliary artery)

Investigations

00:35:09

Ophthalmoscopy :

	Direct	Indirect
Instrument used		
magnification	15 times	3-5 times
External lens	Not used (Convex lens (+) within instrument)	Convex lens : Power $\rightarrow +20D$
Image formed	Virtual & erect	Real and Inverted
Stereopsis/ Depth perception	Absent (monocular examination)	Present (Binocular examination)
Distance of examination	Close to patient's eye	An arm's length
Patient's position	Standing	Supine
Area visualized	• Smaller • Central fundus : Optic disc + macula • 2 disc diameters • 10° field of view	• Larger • Central + Peripheral fundus : Optic disc + macula + ora serrata • 8 disc diameters • 45° field of view
Evaluation with hazy media	Not possible	Possible
Indication	Pathology involving a specific part of retina. E.g. Glaucoma	Pathology involving any part of retina. E.g. Retinal detachment.



----- Active space -----

Fundus Fluorescein Angiography (FFA) :

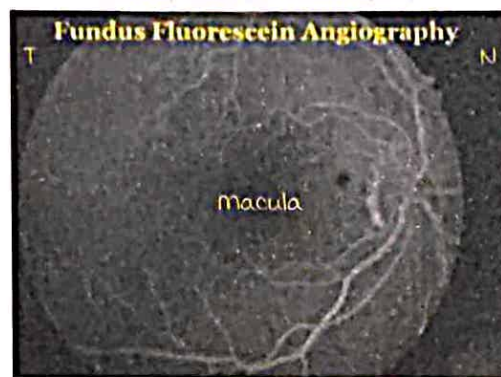
Procedure :

5 ml of 10% solution **Fluorescein dye** (water-soluble orange coloured dye) injected into antecubital vein.

↓
Blue light illumination.

↓
Excitation of fluorescein molecules at 520-530 nm.

↓
Blood vessels visualized.

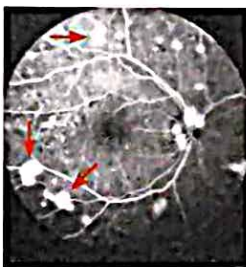


Normal FFA

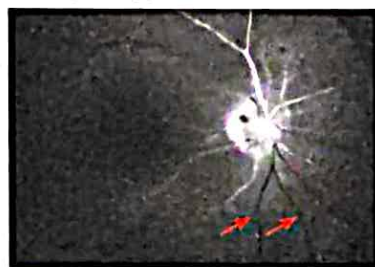
use :

To diagnose

↓
vascular disorders



Leakage of dye



Blocked vessel

↓
macular disorders



macula : Bright

Indo-Cyanine Green Angiography (ICGA) :

Dye used : Indocyanine green.

- 98% protein bound → Better retention.
- Fluoresces at 805 nm → Better penetration.

use : For visualizing **choroidal circulation**.

C/I : Iodine hypersensitivity.

----- Active space -----

B-scan ultrasonography :

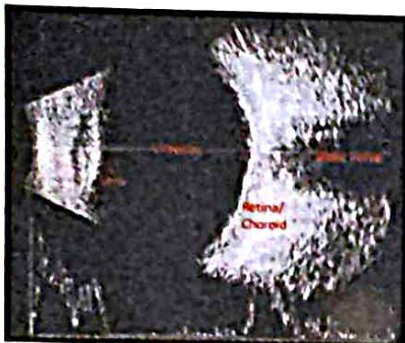
Brightness scan.

Mechanism : uses 7.5-10 MHz ultrasonic waves.

Generates echoes →

- Hyperechoic area (Bright/white). E.g. **Retina (Anterior)** and **Choroid (Posterior)**
- Hypoechoic/anechoic areas (Dark/black). E.g. **Vitreous**.

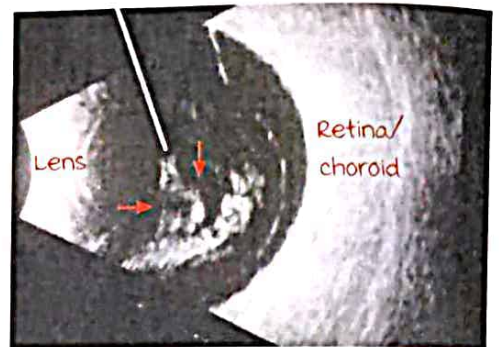
Indication : Patients with **ocular media opacity** → Difficult ophthalmoscopy.
(Corneal opacity/cataract)



Normal B-scan



Retinal detachment



Vitreous hemorrhage

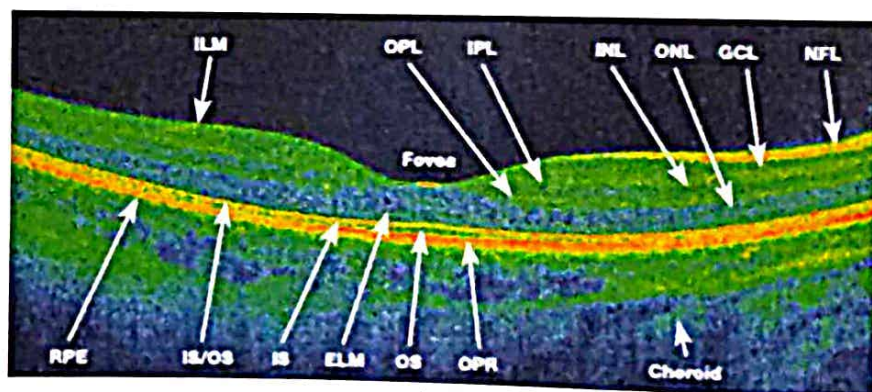
Optical Coherence Tomography (OCT) :

Principle : Low coherence interferometry

Optical biopsy → Visualizes microstructures of retina (To confirm diagnosis).

Landmarks :

- Fovea : Depressed d/t absence of ganglion cell layer.
- RPE : Red.



OCT



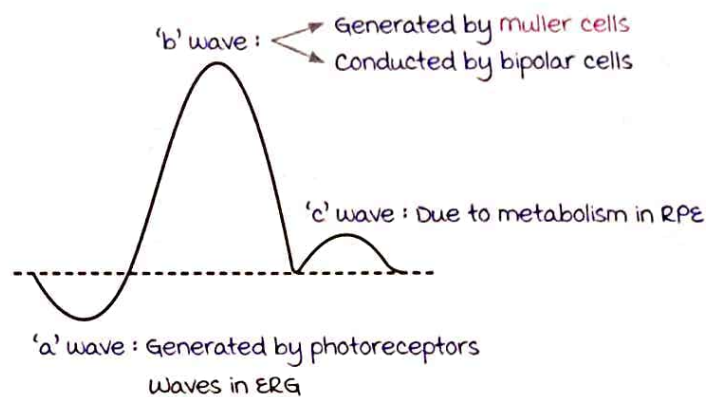
----- Active space -----

Electroretinography (ERG):

- Graphical record of action potentials.
- Source: Panretina.

Special ERG:

1. Early Receptor Potential (ERP) } Generated by **cones** only
 2. Multifocal ERG } (Function of macula)
 3. Pattern ERG
- Source: Ganglion cells → Represents optic nerve function
 - Use: To diagnose glaucoma.



Electro-oculography (EOG):

- Record of standing potential.
 - Produced d/t voltage difference across apical and basal segments of RPE cells.
 - Source:
 1. Photoreceptors
 2. RPE
 } **Outer retina**

- Use: To diagnose diseases of outer retina.

- measured by **Arden's ratio**:

$$\frac{\text{Light peak (Highest value of EOG in well-lit conditions)}}{\text{Dark trough (Lowest value of EOG in dark)}} > 1.85 \text{ or } 185\% \text{ (Normal)}$$

RETINOBLASTOMA AND MACULAR DISORDERS

----- Active space -----

Retinal pathologies :

1. Retinoblastoma (Intraocular tumor).
2. Macular diseases → Central loss of vision.
3. Fundus dystrophies → Disorders of cones
→ Disorders of rods
4. Vascular disorders.
5. Retinal detachment.

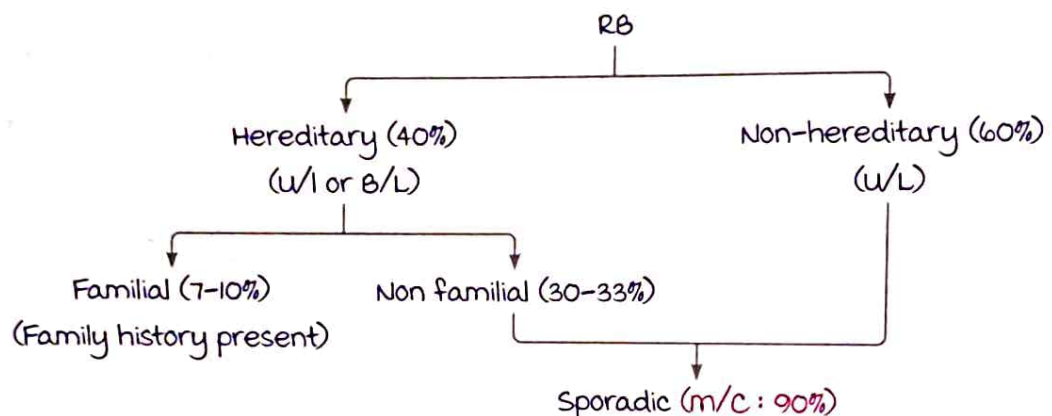
Retinoblastoma (RB)

00:02:10

m/c primary intraocular malignancy.

Incidence : 1 : 17,000

Inheritance :



Hereditary RB :

- mode of inheritance : AD
- Gene involved : **RB-1** → Tumor suppressor gene
→ Location : Chr 13q14

• Knudson's 2-Hit hypothesis :

RB → Genetically : AD
→ Phenotypically : AR
First mutation : Before birth (Rr)

↓
Second mutation : After birth (rr) → Loss of heterozygosity leads to malignant transformation & presentation.

- **Trilateral RB** : B/L RB + **Pinealoblastoma/PNET** (Primitive neuroendocrine tumor)

----- Active space -----

m/c malignancy associated with RB.

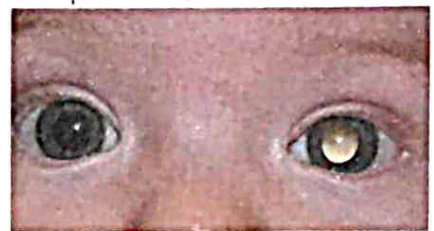
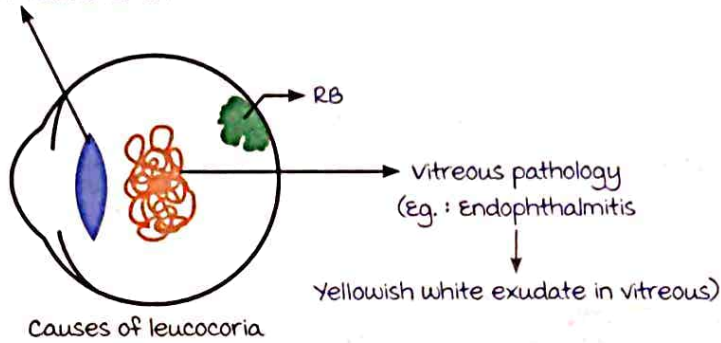
- m/c associated 2° malignancy : **Osteosarcoma**.

Clinical features :

1. Leucocoria : (m/c)

- white pupillary reflex.
- D/t loss of transparency of structures behind pupil.
- Sign with various diagnoses.

mature cortical cataract



Leucocoria

m/c cause

- u/L : RB → m/c presenting age : <3 years.
- B/L : Cataract.

2. Squint (2nd m/c) → Esotropia.

3. 2° Glaucoma with buphthalmos (3rd m/c).

D/t seeding from tumor. Enlarged eyeball.

Other features : D/t tumor invasion. Eg. : Iris invasion → Iritis.

Routes of Spread :

- **Direct (m/c)** : Through
 - Optic nerve (m/c)
 - Sclera
- Hematogenous.
- Lymphatic.

Growth pattern :

Endophytic : Grows into vitreous.

Exophytic : Grows towards choroid.

Retrolaminar :

- Worst prognosis.
- Spread via optic nerve to orbit & cortex → Intracranial metastases.



Endophytic growth

----- Active space -----

Investigations :

IOC : **CT** > X-ray (Since RB is m/c cause of **intraocular calcifications**)



Can also detect intracranial metastases.

MRI : To assess for extracranial metastases.

LDH : ↑ed levels in aqueous humor.

Biopsy (Excisional type) :

- **Gold standard investigation.**
- Sample : From optic nerve.



HPE done.

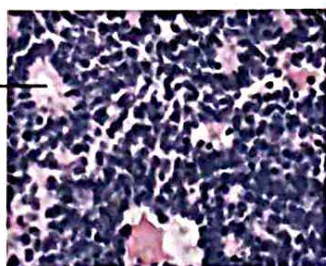
HPE features of biopsy :

- **Flexner Wintersteiner rosettes** (Pathognomonic).
- Homer-Wright rosettes (Pseudorosette) : Also seen in
 - Neuroblastoma.
 - Ewing's sarcoma.
- Fleurettes.

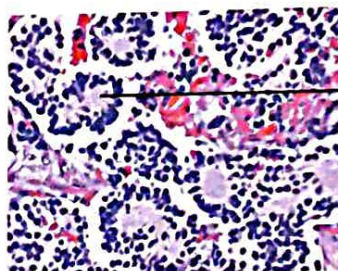


Endophytic growth on MRI

Central empty space surrounded by cells with peripherally arranged nuclei.



Rosettes in RB



Pseudorosettes

Central entanglement of neural processes irregularly arranged cells & nuclei.

Treatment :

- **Enucleation** : TOC (Removal of whole eyeball with optic nerve).
- Chemotherapy : **CEV**
 - Carboplatin
 - Etoposide
 - Vincristine

Based on staging :

Stage	Treatment
G1	Cryotherapy/laser photocoagulation
G2	Carboplatin + Vincristine
G3	Carboplatin + Etoposide + Vincristine
G4	
G5	Enucleation

Grade 5 :

If any (+)

- Tumour occupying >50% of total eye volume.
- vitreous hemorrhage (+).
- Rubeosis iris (Neovascularisation of iris) (+).

Macular Disease

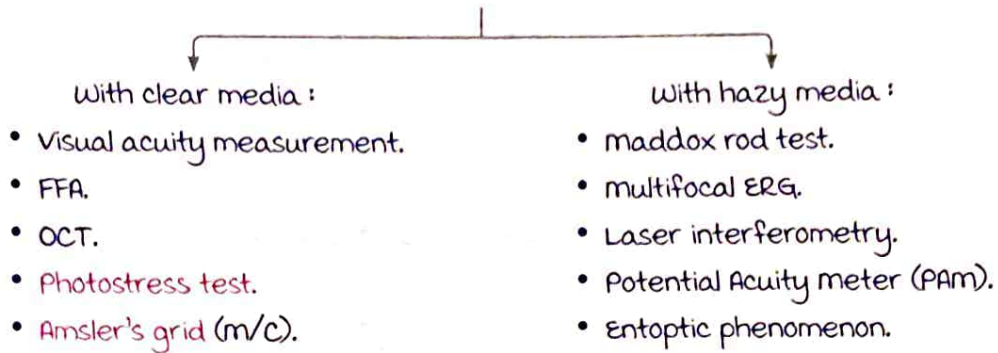
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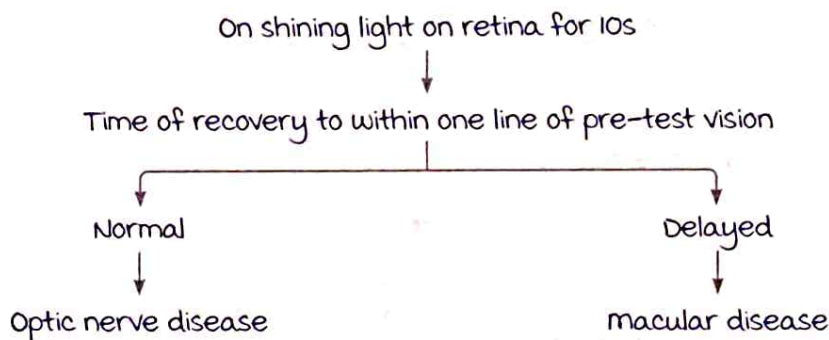
Early presentation : D/t central loss of vision.

If untreated → Chronic damage → Permanent loss of central vision.

Macula Function Tests :

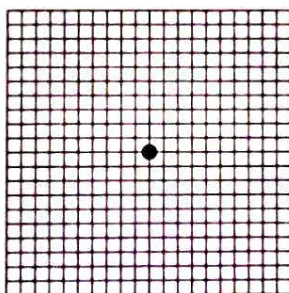


Photostress test :

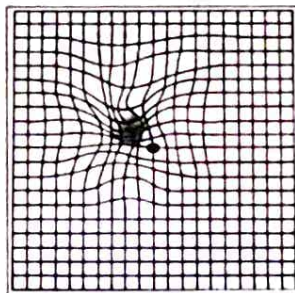


Amsler's grid :

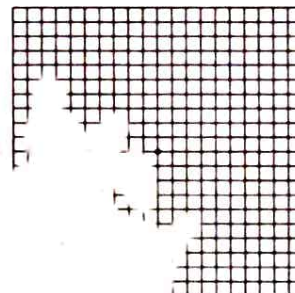
- Square chart with 400 5x5 mm squares.
- Differentiates macular disease from optic nerve disease.



Normal



In macular disease :
wavy lines d/t change
in contour of retina.



In optic nerve disease :
Parts of lines missing.

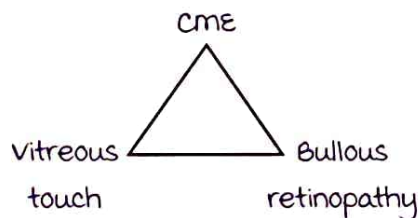
----- Active space -----

CENTRAL SEROUS RETINOPATHY VS CYSTOID MACULAR EDEMA

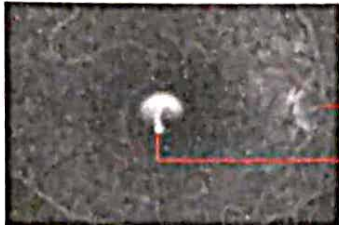
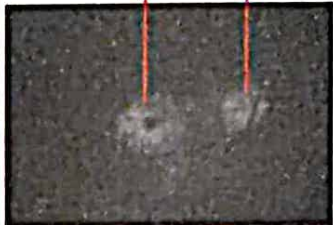
	Central Serous Retinopathy (CSR)	Cystoid macular edema (CME)
Pathology	Collection of fluid at macula in subretinal space d/t defect in RPE.	Collection of fluid at macula in outer plexiform layer (6 th layer).
Risk factors	• males (25-55 years). • Type A personality (↑stress). • Steroids. • Cushing's syndrome. • H.pylori infection	mnemonic : RUN for PRIDE. • Retinitis pigmentosa. • uveitis. • Niacin/nicotine. • Prostaglandin analogues. • Central Retinal vein occlusion. • Irvin Gass syndrome. • Diabetic retinopathy. • Epinephrine in aphakia.
Clinical features	1. Loss of central vision. 2. metamorphopsia (Distorted image).	

Irvin Gass syndrome :

Complication of cataract surgery.



	CSR	CME
OCT	Loss of foveal depression.	
	Shallow neurosensory retinal detachment in chronic cases. 	Cystic spaces in OPL in chronic cases.

	CSR	CME
FFA (Pattern of pooling/leakage of dye)	- Ink blot appearance. - Smoke stack or umbrella appearance. 	Flower petal/petaloid appearance 
Treatment	Self-resolving. Steroid: C/I. - CSR persistent > 4 months or - Recurrent or - Visual field defect in other eye ↓ a. Laser photocoagulation b. Photodynamic therapy : using verteporfin ↓ On shining light of 689 nm, drug acts.	Steroids (mainstay)

----- Active space -----

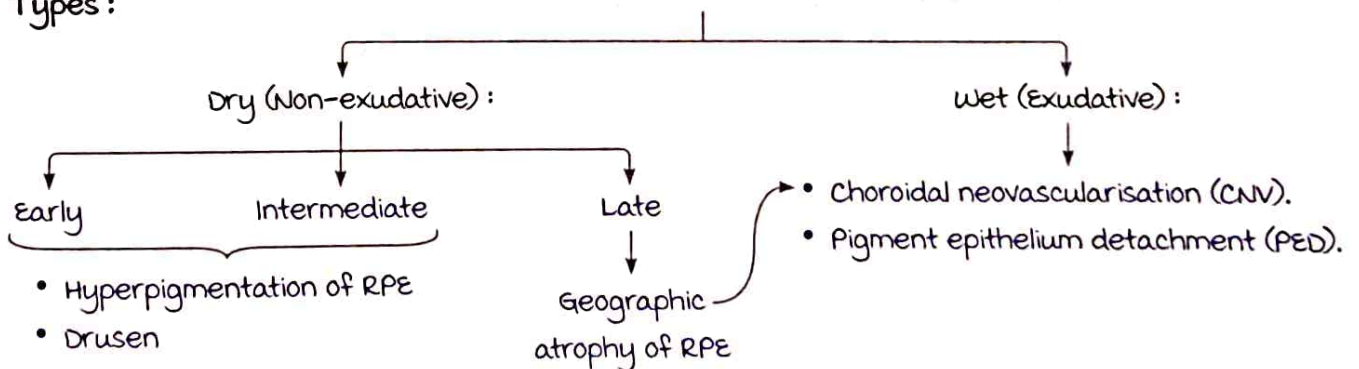
Age Related Macular Degeneration (ARMD)

01:01:12

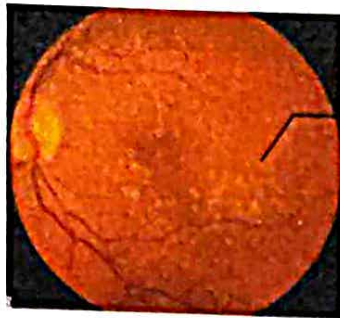
Risk factors :

- Age >50 years.
- Smoking.
- Blue iris.
- High fat diet.

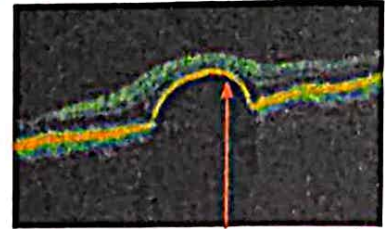
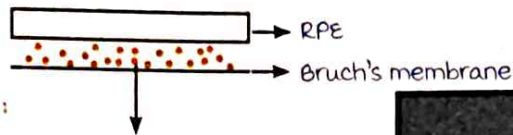
Types :



----- Active space -----

**Drusen :**

Extracellular eosinophilic deposits between RPE (Outermost layer of retina) and Bruch's membrane (innermost layer of choroid)



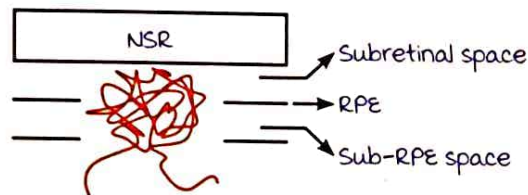
Pigment epithelium detachment (PED)

Choroidal neovascularisation :**Pathogenesis :**

Damage to RPE & Bruch's membrane

vessels grow into subretinal space leading to hemorrhage

Forms fibrovascular complex k/a CNVM
(Choroidal Neovascular membrane)



Classical CNVM → IOC : FFA
(In subretinal space)

Occult CNVM → IOC : ICGA
(In sub-RPE space)

Treatment :

Dry ARMD	Wet ARMD
Antioxidants (AREDS-2 study) Fixed dose combination of : Vitamin C & E + Zinc & copper + Lutein & Zea-xanthin (Carotenoids)	a. In subfoveal CNVM : • Anti-VEGF drugs : TOC 1. Bevacizumab 2. Ranibizumab } Inhibits VEGFA isoform 3. Aflibercept : Soluble decoy receptor for VEGF 4. Brolucizumab • Photodynamic therapy
	b. In extrafoveal CNVM : Laser photocoagulation



Other Macular Disorders

01:15:31

----- Active space -----

Epiretinal membrane : AKA cellophane maculopathy.

Angioid streaks :

Find crack in Bruch's membrane.

Causes : PEPSI

- Pseudoxanthoma elasticum
- Ehler Danlos syndrome
- Paget's disease
- Sick cell disease
- Idiopathic

Hypotony maculopathy :

D/t IOP < 5 mmHg

Causes :

- Excessive drainage after glaucoma surgery.
- Retinal detachment.
- Penetrating trauma.
- Chronic uveitis.

----- Active space -----

DYSTROPHIES OF FUNDUS : RETINITIS PIGMENTOSA AND OTHERS

Affects photoreceptor retinal pigment epithelium (RPE) :
Choriocapillaris complex → Degeneration of rods & cones.

Physiology of rods & cones :

	Rods	Cones
Vision	Night (scotopic) vision.	Day color (photopic) vision.
	Peripheral vision : D/t max concentration in perifoveal area.	Central vision : D/t max concentration at fovea.
Defect	Night blindness/Nyctalopia.	Color/day blindness/Hemeralopia/
	Peripheral loss of vision.	Central scotoma.
Causes	Of nyctalopia: mnemonic : OMG! (CRAP)_a • Oguchi disease. • myopia. • Gyrate atrophy. • Congenital stationary night blindness. • Cataract : Cortical. • Retinal detachment. • Retinitis pigmentosa. • Albinopunctatus : Fundus. • vitamin A deficiency. • Phenothiazine toxicity. • Post-refractive surgery.	Of hemeralopia : mnemonic : C₃A₃T • Cone dystrophy. • Central cataract (nuclear & posterior subcapsular). • Cohen syndrome. • Achromatopsia. • Aniridia. • Albinism. • Trimethadione.

Retinitis Pigmentosa

00:07:35

m/c hereditary fundus dystrophy.

Inheritance :

- Autosomal recessive : m/c mode of inheritance.
- Autosomal dominant : Best prognosis.
- X-linked recessive : Worst prognosis.

Pathophysiology : Rod-cone dystrophy (rods affected first.)

----- Active space -----

CLINICAL FEATURES**Symptoms :**

- Nyctalopia.
- Peripheral visual field loss.
 - mid-peripheral superior scotoma → Ring/doughnut scotoma → Tubular/tunnel vision.

Signs :

1. Arteriolar attenuation.

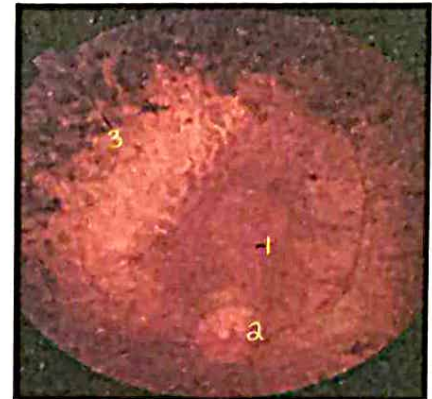


2. Bony spicule (perivascular) pigmentation.

3. Waxy, pale optic disc d/t consecutive optic atrophy.

4. Tapetal reflex : Shining reflex at macula.

- Seen in female carriers of XLR inheritance.



Fundoscopy

MANAGEMENT**Investigation :**

Electroretinography (ERG) : IOC.

Treatment :

Not available in India.

Argus II retinal prosthesis :

- Restores central 20° vision.
- Done in complete loss of vision.

Voretigene neparvovec :

- used in biallelic RPE-65 gene mutation.
- Adeno-associated virus vector based gene therapy.

SYSTEMIC ASSOCIATIONS

- Lawrence moon Bell syndrome (m/c) : RP + intellectual disability + obesity + polydactyly.
- usher's syndrome : RP + Deafness.
- Refsum disease : RP + Cerebellar ataxia + Polyneuropathy.
- Bassen Kornzweig syndrome : RP + abetalipoproteinemia + acanthocytosis (spur cells).

----- Active space -----

Best's Macular Dystrophy

00:23:17

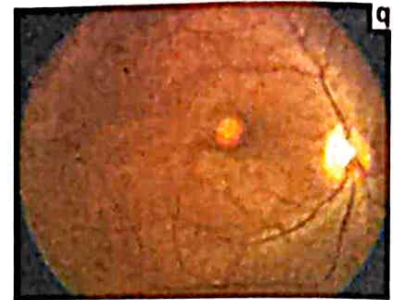
Autosomal dominant.

CLINICAL FEATURES**Symptoms :**

- Hemeralopia.
- Central scotoma.

Signs :

- Lipofuscin accumulation.
- Fundus : vitelliform lesion (egg yolk appearance)
- Pseudohypopyon : Collection of fluid in anterior chamber.
 - Vitelliruptive stage.



vitelliform lesion

MANAGEMENT**Investigation :**

- EOG : IOC
 - Arden's ratio : <1.5 or 150% (Normal : >1.85).
- ERG : Normal.

Stargardt's Disease

00:28:10

- Autosomal recessive.
- m/c juvenile macular dystrophy.

CLINICAL FEATURES**Symptoms :**

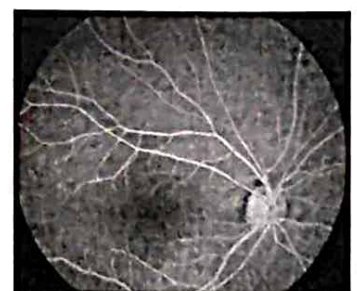
- Hemeralopia.
- Central scotoma.

Signs :

Bull's eye maculopathy



Dark choroid/ Silent choroid sign



Normal FFA : Background fluorescence d/t dye in choroidal vessels.



Note : Chloroquine toxicity

----- Active space -----

1. Pre-maculopathy (reversible) : Normal vision + Central scotoma.

2. maculopathy : Bull's eye lesion.

• Seen if:

- Daily dose : $>3.5\text{mg/kg}$ or 250mg .

- Cumulative dose : $>460\text{ g}$.

- Duration of R_x : $>5\text{yrs}$.

- Renal/liver dysfunction.

• Clinical features :

- Diminished visual acuity.

- Irreversible.

3. Vortex keratopathy.

Congenital Night Blindness Disorder

00:36:10

Aka Infantile onset nyctalopia.

Causes :

Causes	ERG	Fundus
Congenital stationary night blindness (CSNB)	Negative	Normal
Oguchi disease	Normal	Abnormal
Enhanced S cone syndrome	Negative	
Fundus albipunctatus	Absent rod responses	

Oguchi disease :

• Damage in conduction (No damage to rod cells).

• Mizuo-Nakamura phenomenon :

- Golden yellow fundus in bright conditions.

- Dark adapted retina in dark conditions.

Miscellaneous

00:41:09

X-LINKED RETINOSCHISIS

• Split in retina between nerve fibre layer & rest of retina.

• O/E : Bicycle wheel maculopathy.

STICKLER'S SYNDROME

m/c inherited cause of RD in children.



----- Active space -----

RETINAL VASCULAR DISORDERS : PART 1

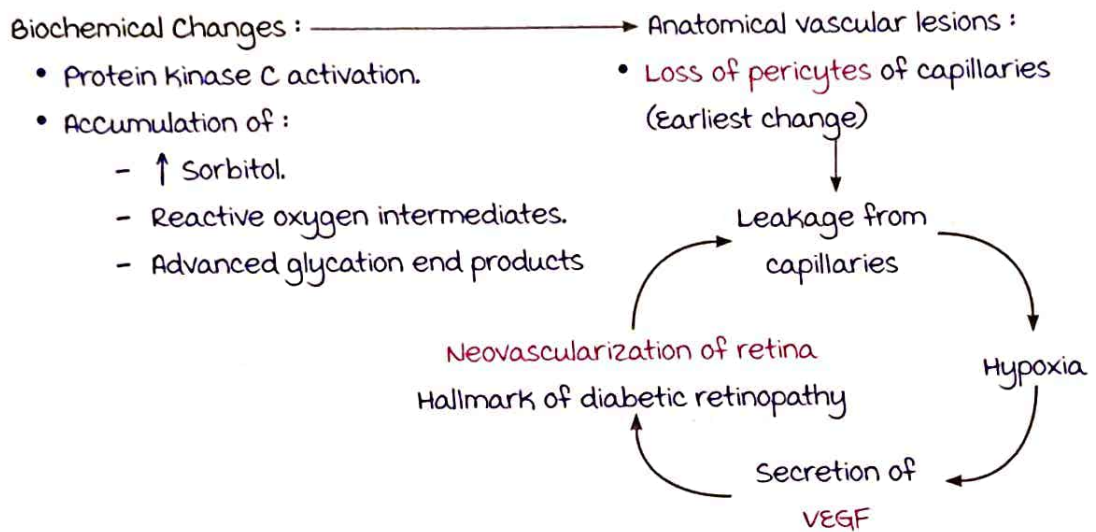
Diabetic Retinopathy

00:01:42

RISK FACTORS

- Duration of diabetes $\propto$ risk.
 - Time of screening
 - Type 1 : After 5yrs
 - Type 2 : As soon as possible
- $\uparrow$ HbA1c.
- Pregnancy.
- Hypertension & hyperlipidemia.
- Diabetic nephropathy.

PATHOGENESIS



Classification & Clinical Features of Diabetic Retinopathy

00:13:17

----- Active space -----

ETDRS (EARLY TREATMENT DIABETIC RETINOPATHY STUDY)

Non proliferative diabetic retinopathy (NPDR)

1. **microaneurysms** : Earliest sign seen in inner nuclear layers

O/E : Tiny red dots

FFA : Tiny hyperfluorescent dots

2. Hemorrhage :

- **Flame shaped** in superficial nerve fibre layer.
- **Dot and blot** hemorrhage in outer Plexiform layer.

FFA : Hyperfluorescent.

3. Exudates : Lipid deposits in outer plexiform layer

O/E : **Yellowish** irregular deposits

FFA : **Hypo**fluorescent

4. Cotton wool spots (soft exudates) :

Collections of neuronal debris in nerve fibre layer.

O/E : Large **Yellowish** deposits, regular & smooth edges

FFA : **Hypo**fluorescent.

5. Venous tortuosity :

- Due to hypoxia
 - Beading.
 - Looping.



6. IRMA : Intra-retinal microvascular abnormalities

- Shunt vessels bypassing capillary bed.
- Precursor to PDR.

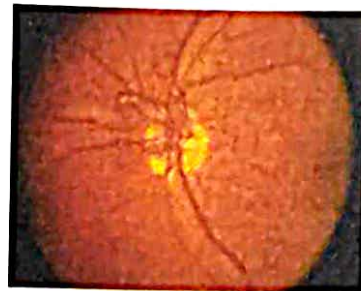
Proliferative diabetic retinopathy (PDR)

Neo-vascularization (NV) :

- O/E : Tortuous tuft of vessel.
- FFA : **Hyper**fluorescent.

NVD :

NV at disc or within 1 disc diameter



NVD

NVE :

NV elsewhere

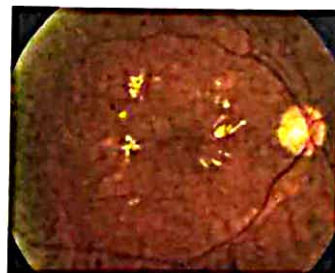


FFA imaging

1. NVD.

2. microaneurysms.

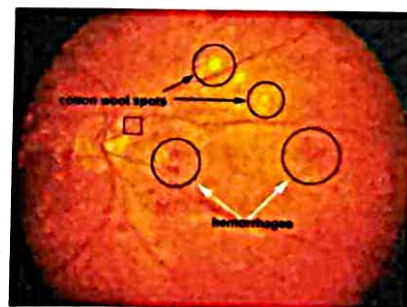
3. Hyperfluorescent exudates.



Exudates



Hemorrhages (1) & exudates (2)



Cotton wool spots & hemorrhages

----- Active space -----

Treatment of Diabetic Retinopathy

00:32:48

NPDR

- Glycemic control.
- Follow-up

NPDR stage	Follow up in
mild	9-12 months
moderate	6 months
Severe	4 months
Very severe	2 months

4-2-1 rule of NPDR :

- 4 quadrants hemorrhage.
- 2 quadrants venous beading.
- 1 quadrant IRMA :
 - Any 1 : severe NPDR.
 - ≥ 2 : very severe NPDR.

PDR

- **Pan retinal photocoagulation : TOC**
 - Argon green laser
 - multiple sittings.
 - S/E : visual field defects.
- Anti VEGF drugs
Ranibizumab, Bevacizumab, Aflibercept, Brolucizumab

Indication for treatment :

- NVD $> 1/3$ disc area.
- NVD with vitreous hemorrhage.
- NVE $> 1/2$ disc area with vitreous hemorrhage.

Complications of Diabetic Retinopathy

00:42:07

1. Diabetic macular edema :

- D/t leakage from capillaries.
- **m/c cause of loss of vision in NPDR (DR).**

Rx :

- Focal laser photocoagulation : TOC.**
- Intravitreal triamcinolone.

Indications :

- Ischemic maculopathy.
- Pseudophakic patient.
- Coexisting cardiovascular disease.

3. Anti VEGF drugs :

- Indication : Centre involving edema within $300\mu\text{m}$ of + foveal thickness
centre of macula $> 250\mu\text{m}$

Indication for Rx :**Clinically significant macular edema (CSME) :**

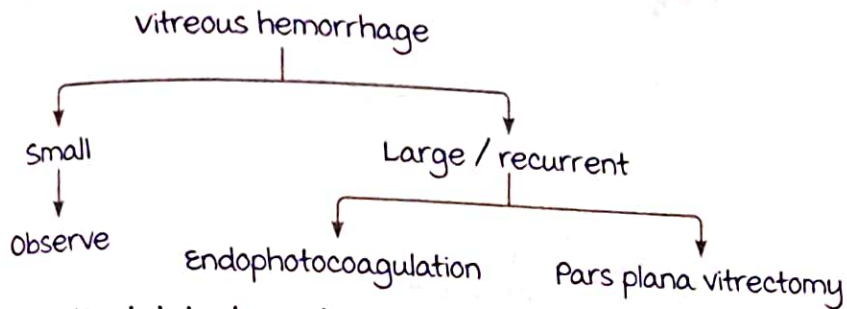
- Retinal thickening within $500\mu\text{m}$ of centre of macula.
- Exudates within $500\mu\text{m}$ of centre of macula (retinal thickening anywhere).
- Retinal thickening or exudates (> 1 DD size) within 1DD ($1500\mu\text{m}$) of centre of macula.

2. vitreous hemorrhage :

- source of hemorrhage: Neovascularization of retina.
- m/c cause of loss of vision in PDR.
- m/c cause of vitreous hemorrhage
 - in adult : Diabetic retinopathy
 - in young adults (20-40 yrs) : Blunt trauma
- m/c cause of recurrent vitreous hemorrhage : Eale's disease

----- Active space -----

Rx :



3. Retinal detachment :

m/c cause of blindness.

4. Rubeosis iridis :

- Neovascularisation of iris.
- D/t Oxygen demand by posterior segment on anterior segment.

----- Active space -----

RETINAL VASCULAR DISORDERS : PART 2

Hypertensive retinopathy

00:00:09

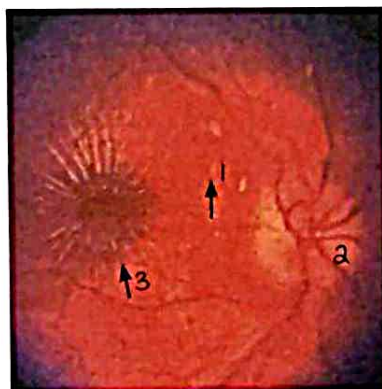
vasoconstriction phase → Sclerotic phase (Atherosclerosis) → Exudative phase.

Clinical features :

Keith-Wagner classification :

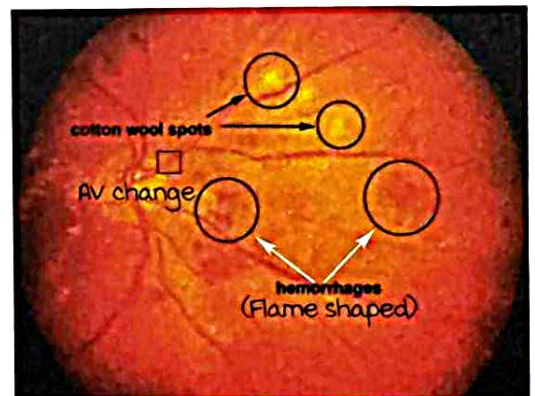
Grade	Characteristics
1	Arteriolar attenuation. • A : V (Diameter) ratio = 1:3 (Normal : 2:3).
2	Salus sign : Deflection of blood vessel.
3	• Bonnet sign : Banking of blood vessel. • Gunn sign : Tapering of blood vessel. • Flame shaped hemorrhages with cotton wool spots.
4	• Copper wiring of arterioles. • Silver wiring of arterioles. • Papilledema.

At AV crossings
in veins



Grade 4 HTN retinopathy

1. Cotton wool spots
2. Swelling of optic disc
3. Exudates : macular star appearance : malignant HTN.



Grade 3 Hypertensive retinopathy

Central Retinal Artery Occlusion (CRAO)

00:09:57

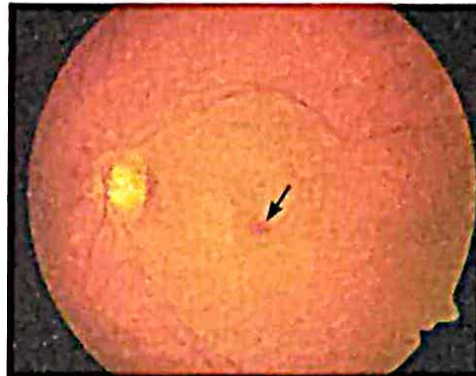
----- Active space -----

CAUSES

- Atherosclerosis : m/c.
- Emboli :
 - Hollenhorst plaque : Cholesterol emboli originating from bifurcation of CCA.
- Giant cell arteritis.

CLINICAL FEATURES

- Sudden, painless loss of vision.
- Optic atrophy (Consecutive).
- RAPD or Marcus Gunn pupil
- Fundus appearance :
 - Pale fundus : D/t retinal edema.
 - Cattle tracking fundus : D/t segmentation of blood column.
 - Cherry red spot at macula : D/t collection of fluid in ganglion cell layers (GCL) of retina except at foveola.



Cherry red spot

CRAO with macular
ciliorretinal artery sparingDye filling ciliorretinal
artery in FFA

Note : Cherry red spots at macula :

Mnemonic : Cherry trees never grow tall in sand, mud & grime.

- | | | |
|--------------------------|---------------------------------|--|
| • CRAO. | } Gangliosidosis
Type 2 | |
| • Trauma-blunt. | | |
| • Niemann picks disease. | | |
| • Gangliosidosis Type 1. | | |
| | • Tay Sachs disease. | |
| | • Sandhoffs disease. | |
| | • metachromatic leukodystrophy. | |
| | • Gaucher's disease : Type 2. | |

----- Active space -----

TREATMENT

Ocular emergency : Rx within 4 hrs.

24-48 hrs : Complete occlusion.

- Ocular massage : Dislodge emboli.
- Vasodilatation.
 - Sublingual isosorbide nitrate.

- Carbogen : 95% O_a + 5% CO_a → Hypercarbia → $\uparrow$ PCO_a → $\downarrow$ pH → vasodilatation.
Respiratory acidosis
- $\downarrow$ IOP.
 - IV mannitol.
 - Paracentesis : Aspiration of aqueous.
- Rx of cause.

Central retinal vein occlusion (CRVO)

00:30:05

m/c cause : Hypertension.

1. Non-ischemic stage :

D/t Stasis of blood → $\begin{cases} \text{Hypoxia} \\ \uparrow \text{Vascular permeability} \end{cases} \rightarrow \text{macular edema} \rightarrow \text{Loss of vision}$

Rx :

- Intravitreal triamcinolone.
- 0.7mg Dexamethasone intra-vitreous implants (Geneva study).
- Anti VEGF drugs.

2. Ischemic Stage :

Severe hypoxia → Capillary endothelium damage.

Clinical features :

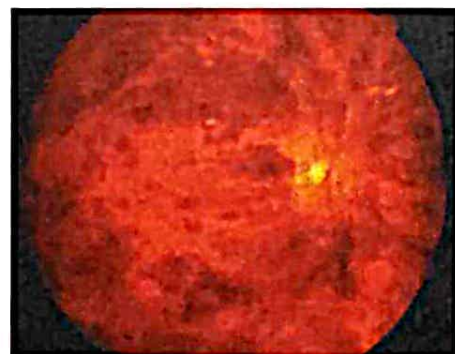
- Severe flame shaped hemorrhages.
- Rubeosis iridis (Neovascularisation of iris) d/t Oxygen demand by posterior segment

Blood leak into anterior chamber → Blocks trabecular meshwork → Neovascular glaucoma :
AKA 100 day glaucoma : Glaucoma progression of 3 months in CRVO.

Rx : Panretinal photocoagulation.

Note :

m/c cause of neovascular glaucoma : Diabetic retinopathy.



Tomato ketchup/splash appearance

BRANCHED RETINAL VEIN OCCLUSION (BRVO)

- Superior part of central retinal vein occluded.
- m/c site : AV crossings.
- m/c quadrant : Superotemporal.



BRVO

----- Active space -----

Retinopathy of prematurity (ROP)

00:41:53

Occurs in :

- Age at birth : ≤ 32 weeks.
- Birth weight : ≤ 1750 grams.

Age of screening :

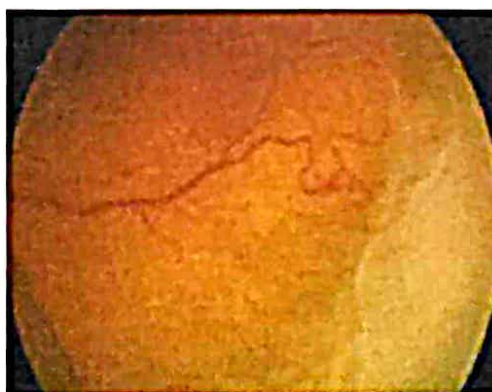
Criteria		Time of screening
Age at birth	Birth weight	
≥ 28 wks	≥ 1200 g	4 weeks after birth
< 28 wks	< 1200 g	2-3 weeks after birth

Stages :

Stages	Features
1	Demarcating line between vascularized retina & peripheral avascular retina
2	Ridge formation at demarcating line
3	Extra retinal neovascularization on the ridge moving into vitreous
4	Partial retinal detachment (RD)
5	Total RD



Stage 2 ROP

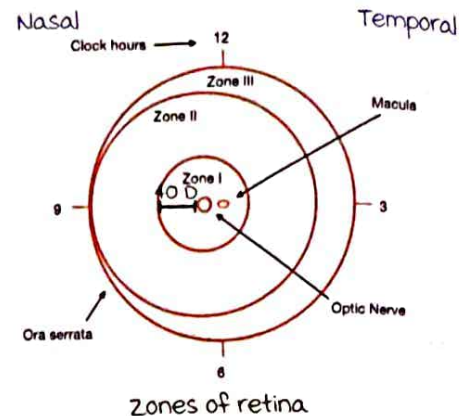


Stage 3 ROP

----- Active space -----

Treatment :**Laser photocoagulation :**

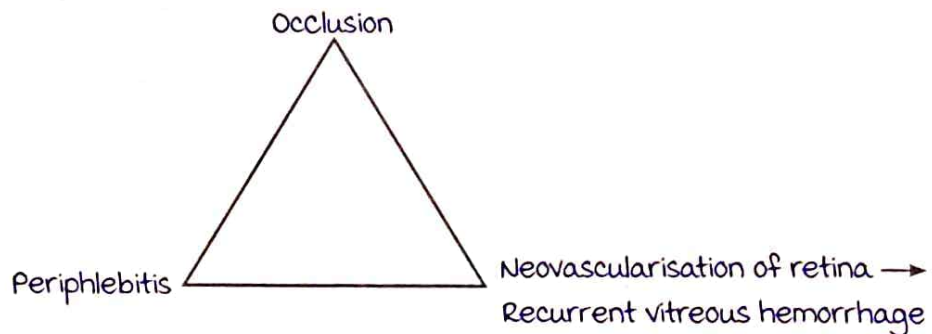
- Indication : Pre-threshold ROP typer I (ETROP study).
 - Zone I + any stage + plus disease (Or)
 - Zone I + stage 3 (Or)
 - Zone 2 + stage 2/3 + plus disease (Or)
- Plus disease : Venous tortuosity & dilatation in ≥ 2 quadrants.



Pars plana vitrectomy : If RD +.

Miscellaneous

00:53:52

EALES' DISEASE

Rx : Panretinal photocoagulation.

COATS' DISEASE

- Idiopathic retinal telangiectasia.
- m > F.
- Unilateral.

Clinical features :

- Painless loss of vision.
- Strabismus.
- Leukocoria.
- Exudative retinal detachment.

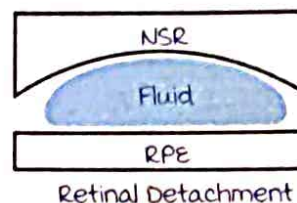
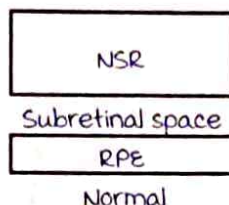
RETINAL DETACHMENT

----- Active space -----

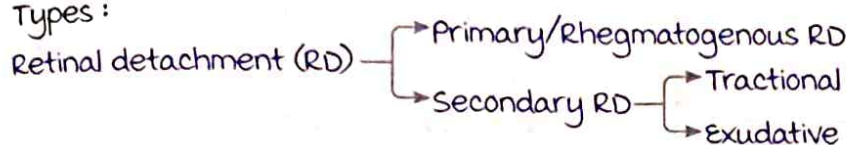
separation of neurosensory retina from retinal pigment epithelium and collection of subretinal fluid.

Investigations :

- Indirect ophthalmoscopy : IOC.
- Optical coherence tomography (OCT).
- B-scan ultrasonography.



Types :

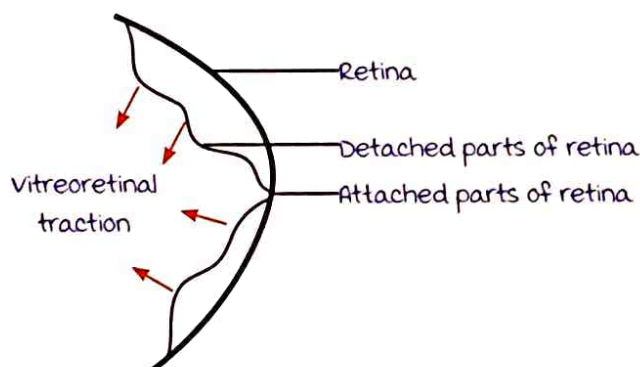


Primary/Rhegmatogenous Retinal Detachment (RRD)

00:03:39

Pathogenesis :

Syneresis (Liquefaction of vitreous)
 ↓
 Posterior vitreous detachment (PVD)
 ↓
 Dynamic vitreoretinal traction from attached parts of vitreous.
 ↓
 Retinal breaks develop.
 ↓
 Fluid enters subretinal space.



Causes :

- Pathological myopia (m/c).
- Blunt trauma.
- Cataract surgery (Aphakia).
- Lattice degeneration of retina.

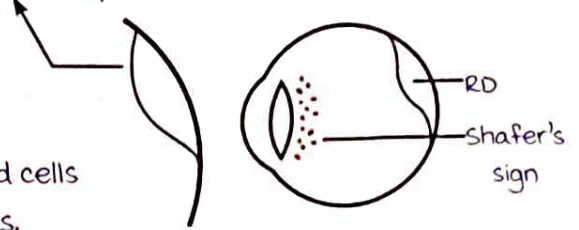
Symptoms :

- Photophobia : Flashes of light (Stretching of nerve fibers → Ectopic neural impulses to brain).
- Floaters : Small black flying dots (d/t vitreal opacities).
- Curtain falling in front of eye.
- Loss of vision : Sudden & painless.



----- Active space ----- Signs :

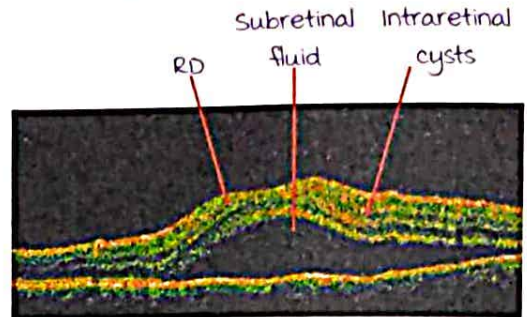
1. Detached retina : Gray/opaque, **convex** shaped.
2. Extent of RD till ora serrata.
3. **Shafer's sign**.
 - Tobacco-dust appearance.
 - D/t deposition of brown pigmented cells at the anterior surface of vitreous.



RD



U shaped retinal tear



Longstanding RRD

m/c location : **Superotemporal quadrant**.

Lincoff rules : Rules to locate breaks in retina.

4. RAPD (Relative afferent pupillary defect).

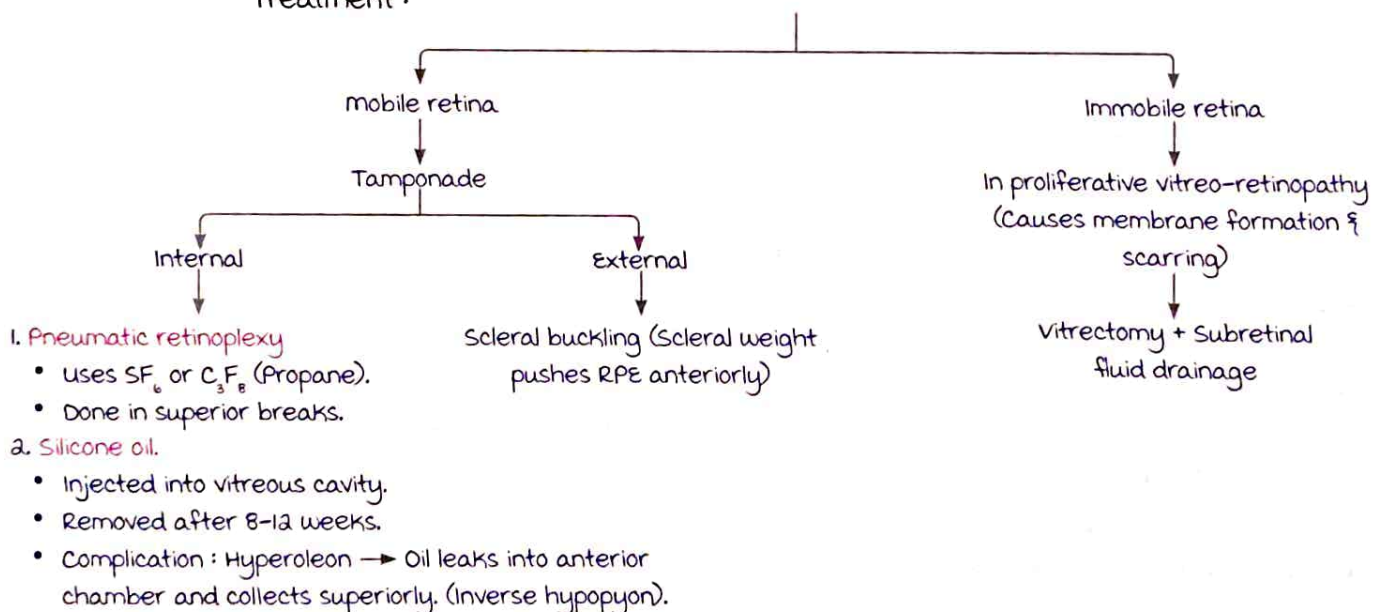
5. In **chronic RD** :

- Retinal atrophy.
- Subretinal demarcation lines k/a **High water marks** (> 3 months).
- Intraretinal cysts. (> 1 year).

management :

Prophylaxis : Laser photocoagulation to seal breaks (usually superotemporal quadrant).

Treatment :



Note : Lincoff rules → used to locate breaks in retina.

Secondary Retinal Detachment

00:25:06

----- Active space -----

	Tractional RD	Exudative RD
Pathology	Static vitreoretinal traction.	Exudation into subretinal space.
Etiology	• Diabetic retinopathy (m/c). • Retinopathy of prematurity. • Penetrating trauma. • Sick cell retinopathy (Roth spots (+) : Hemorrhage with clear centre). O	• Choroidal melanoma (m/c). • Toxemia of pregnancy. • Coat's disease (Idiopathic retinal telangiectasia). • VKH syndrome • Central serous retinopathy.
Clinical Features	• Loss of vision : Gradual/ & painless. • Shape of RD : Concave. • Photopsia. • Floaters. • Extension of RD till ora serrata. } (-)	• Loss of vision : Sudden, painless. • Shape of RD : Convex. • Hallmark features : Shifting fluid (with shifting head position).
Treatment	Treat the cause.	

----- Active space -----

SPECIAL INVESTIGATIONS FOR CORNEA

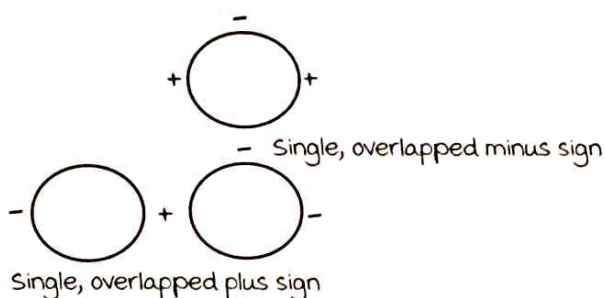
Keratometry

00:02:11

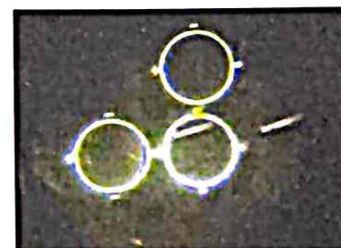
Principle : Anterior surface of cornea acts as a convex mirror.

- Size of image $\propto 1/\text{curvature}$.
- Image seen as keratometric mires.

Correct positioning of mires :



Keratometer



Keratometric mires

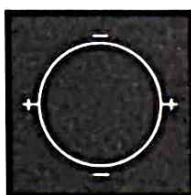
Uses :

1. Measure corneal curvature.

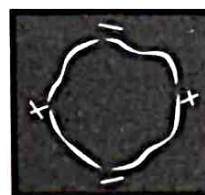
2. Diagnose astigmatism :
 → Horizontally oval mires : With the rule astigmatism.
 → Vertically oval mires : Against the rule astigmatism.

3. To ascertain proper fitting of contact lens.

4. To diagnose keratoconus (pulsating mires seen).



Good fit of lens



Steep fit of lens

Pachymetry

00:10:13

measures corneal thickness.

Normal Corneal Thickness (CT) : 0.5 to 0.6 mm (0.54 mm) or 540 μ .

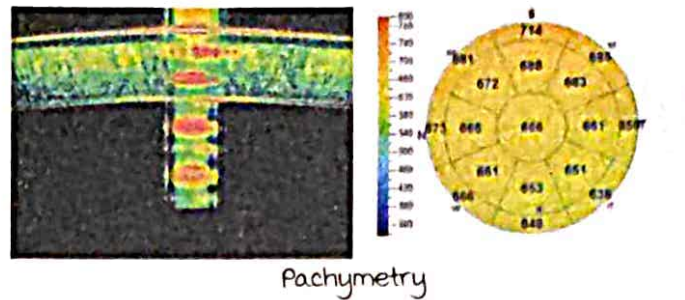
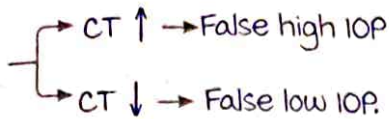
- CT is highest in the centre → Reduces towards periphery.

Uses :

1. Eligibility for LASIK :

- $c/1$ if $CT < 450\mu$.

a. IOP measurement : $1\text{mm Hg IOP} = 10\mu\text{m of CT.}$



Pachymetry

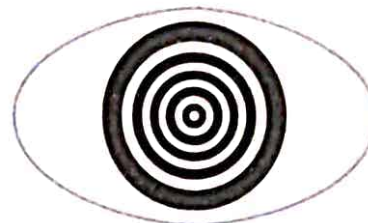
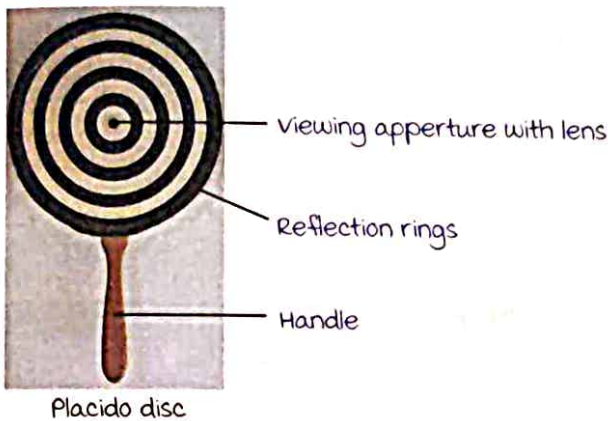
Corneal topography

00:14:00

Examination of corneal surface.

methods used :

1. Placido disc :



Reflection of placido disc : Normal

2. Orbscan : Uses slit scan imaging principle.

3. Pentacam :

measures :

- CT.
- Corneal curvature.
- Anterior segment imaging.
- Topographic elevation maps of anterior surface.
- Topographic elevation maps of posterior surface.

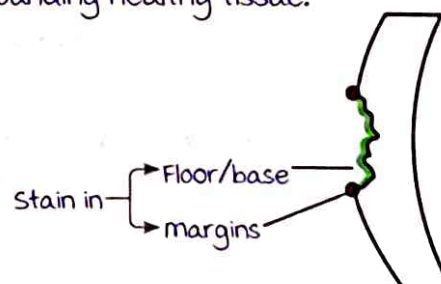
Corneal vital staining

00:18:40

used to visualize and distinguish ulcer from surrounding healthy tissue.

Fluorescein dye :

- Stains area of denuded epithelium (Floor/base of ulcer).
- Visualized under cobalt blue light $\rightarrow$ Green fluorescence.



----- Active space -----

Other uses :

- Goldmann's applanation tonometry.
- Jones' dye disappearance test : Lacrimation assessment.
- Seidel's test (In penetrating trauma).

Rose Bengal dye :

- Stains devitalised/necrotic tissue (margins of ulcer).

Disadvantage : Can be toxic to healthy tissue.

Lissamine green dye :

Stains devitalised/damaged tissue.

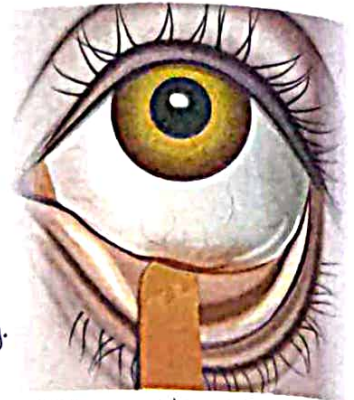
Advantage : Non-toxic to healthy ocular tissue.

Other uses :

m/c used for diagnosis of dry eye by conjunctival staining.

Alcian blue dye :

Stains — mucin deposits
 Filaments.



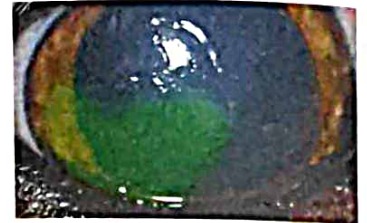
Flourescein dye application



Lissamine green dye



Rose bengal stain



Flourescein dye

Other corneal investigations

00:28:04

Confocal microscopy :

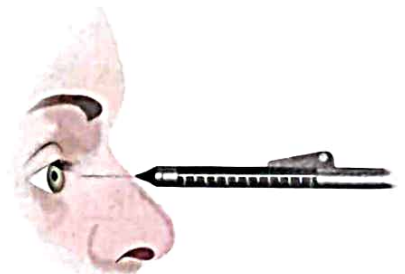
Non-invasive technique for in-vivo imaging of all layers of living cornea.

Corneal aesthesiometer :

Examines corneal sensations.

Instrument : Pen-like instrument with filament (6cm long).

- Length of filament eliciting corneal sensations $\propto$ corneal sensations $\propto$ 1/ anaesthesia.



Corneal aesthesiometer

CORNEAL ULCER AND KERATITIS

----- Active space -----

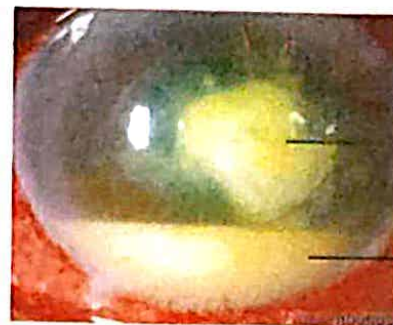
Discontinuity in corneal epithelium + Cellular infiltration & necrosis of underlying layers.

Bacterial Corneal Ulcer

00:02:20

ETIOLOGY

- Staphylococcus aureus : m/c worldwide.
- Pneumococcus : mc in India.
 - Ulcer serpens : Centre → Periphery in zigzag pattern.
 - Hypopyon corneal ulcer.
- Pseudomonas : m/c in Contact lens users.
 - Green discharge.
- Atypical mycobacteria : m/c in h/o LASIK.
- Nocardia : Wreath shaped/Pin head ulcer.



Pneumococcal ulcer

Corneal ulcer

Hypopyon : Collection of pus in anterior chamber

Bacteria invading intact epithelium :

- Neisseria : N. meningitidis, N. gonorrhoea.
- Corynebacterium diphtheria.
- Listeria.
- Shigella.
- Haemophilus aegyptius.



Wreath shaped ulcer



PATHOGENESIS

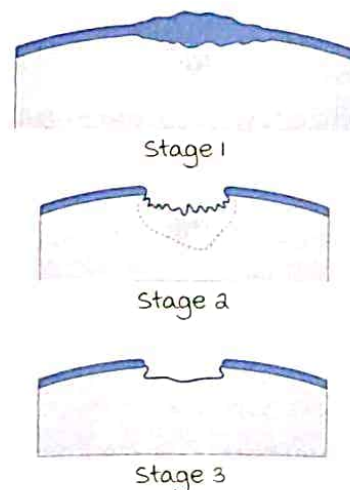
Stage 1-Initial stage :

Saucer shaped ulcer with overhanging margins.

Stage 2-Progressive stage :

Infiltration by PMNs.

- Hypopyon.
- Keratouveitis.
 - mobile : Liquid like pus.
 - Sterile.



----- Active space -----

Stage 3-Regressive stage :

- Smooth floor & edges of ulcer.
- Vascularization +.
- ↓ Inflammatory response.

Stage 4-Scar/Cicatrizization :

Permanent vision loss.



Stage 4

COMPLICATION**1. Ectatic cicatrix :**

- Thinning of Cornea.
- Bulging of cornea d/t ↑ IOP (Temporary/permanent).

2. Anterior staphyloma :

Bulging of Cornea d/t thinning + incarceration of iris tissue.



Lobulated Appearance of Anterior Staphyloma



Descemetocoele

3. Descemetocoele :

Herniation of Descemet's membrane through Corneal ulcer.



Iris Prolapse



Pseudo-cornea

4. Perforation of cornea :

a. Aqueous leak → ↓ IOP.

b. Iris prolapse.

c. Pseudo-cornea : Corneal surface formed d/t plugging of perforation by iris tissue.

5. Endophthalmitis/Panophthalmitis.**MANAGEMENT****Antibiotics :**

- After gram staining & culture of ulcer base, scraping.
- **Fortified antibiotics** : Fortified Cephazolin 5 % + Fortified Tobramycin 1.3 %.

Atropine eye drops : Relieve ciliary spasms.

Impending perforation :

- Oral doxycycline.
- Soft Bandage Contact Lens (BCL).
- Cyanoacrylate glue.

Contraindications :

Steroids : D/t epithelial thinning.

Fungal Corneal Ulcer

00:30:00

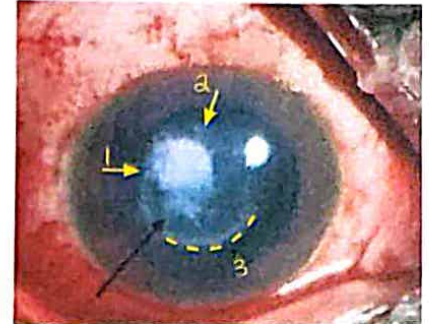
----- Active space -----

- m/c cause : *Aspergillus* (Filamentous fungi with septate hyphae).
 - H/O trauma with vegetative fungi.

Signs :

Out of proportion to symptoms.

- Dry looking ulcer.
 - Hypopyon :
 - Thick, immobile.
 - A sterile.
1. Feathery ulcer margin.
2. Satellite lesions.
3. Weasley Immune Ring :
Antigen-antibody complexes.



Fungal Corneal Ulcer

Treatment :

- Antibiotics :
 - Natamycin 5 % eye drops : DOC for *Aspergillus* *fusarium*.
 - Amphotericin B 0.15 % eye drops.
- Atropine : Supportive Rx.
- Steroids : C/I.

Acanthamoeba Keratitis

00:38:19

Risk factors :

- Unhygienic practices in contact lens users.
 - Contaminated solution, washing in tap water, swimming pools.

Clinical features :

Symptoms out of proportion to signs.

Signs :

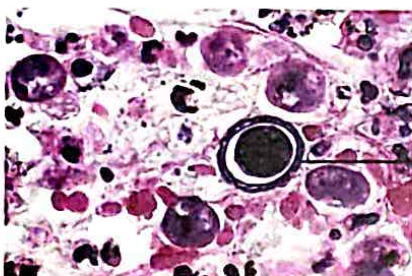
- Pseudodendritic keratitis.
- Radial keratoneuritis : Inflammation of nerves throughout cornea.
- Ring shaped ulcer with abscess : Pathognomonic.



Ring abscess

Investigation :

1. Gram staining :



Double walled
Acanthamoeba cyst

2. Culture media :

Non-nutrient agar enriched with *E. coli*.

----- Active space ----- Treatment :

Drugs inhibiting membrane synthesis :

- PHMB 0.02% (Polyhexamethylene biguanide) : **DOC**.
- Chlorhexidine 0.02%.

Drugs inhibiting DNA synthesis :

Propamidine : m/c.

Viral Keratitis

00:44:58

HERPES SIMPLEX VIRUS (HSV)

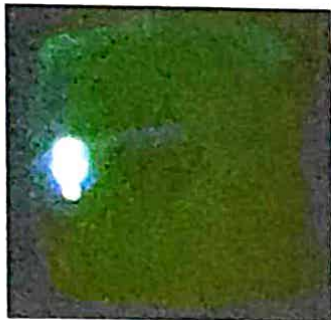
1° infection : Conjunctivitis.

Recurrent infection : Keratitis.

Clinical features :

a. Epithelial lesions : D/t active viral replication.

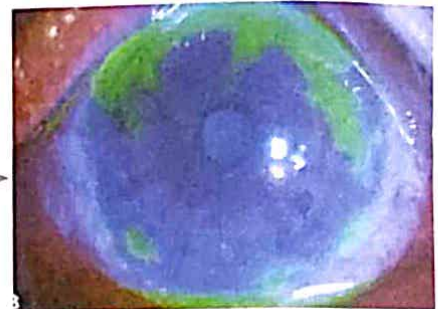
Fluorescence staining :



Superficial punctate
(Dot like) keratitis



Dendritic ulcer :
Branching with terminal buds.



Geographic ulcer

Rx :

- 3% acyclovir eye ointment : 5 times/day x 10-14 days.
- Steroids : C/I.

Note : Other d/d's for Dendritic ulcer :

- Acanthamoeba Keratitis.
- Tyrosinemia.
- HZV Keratitis.
- Contact lens overuse.

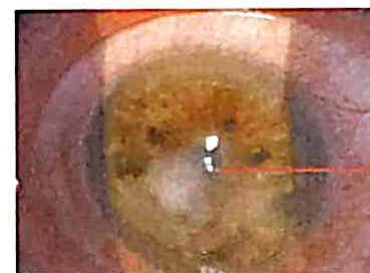
b. Stromal Keratitis :

- Immune Stromal Keratitis :
D/t **type III hypersensitivity**.
Occurs following epithelial keratitis.

Rx : 3% Acyclovir ointment + Topical steroids.

- Necrotizing Stromal Keratitis :
most severe form of stromal keratitis.

Rx : 3% Acyclovir ointment + Topical steroids.



Central
stromal
haze

Immune Stromal Keratitis

c. Endothelial lesion :

i. Disciform keratitis :

- Disc shaped areas of deep stromal/endothelial edema.
- Type IV Hypersensitivity reaction.

Hallmark of HSV Keratitis :

Corneal anaesthesia → metaherpetic keratitis/Neurotrophic ulcer : D/t 5th Cranial Nerve palsy.

HERPES ZOSTER OPHTHALMICUS/SHINGLES

- Affects dermatome supplied by ophthalmic branch of 5th cranial nerve.
- **Hutchinson's sign** : Rash involving tip of the nose → ↑ Chance of Herpes Zoster Ophthalmicus.

Clinical features :

1. Pseudodendritic ulcer : Branching pattern with no terminal buds.
 2. Nummular keratitis.
 3. mucous plaque keratitis : In chronic cases.
 4. Stromal keratitis.
 5. Cranial nerve palsy : 3rd cranial nerve-m/c.
 6. Sclerokeratitis : Least common finding.
 7. Post-herpetic neuralgia :
- Persistence of pain after 1 month of resolution.
 - Worse at night.

Treatment :

T. Acyclovir 800mg x 5 times/day x 2 weeks.



Shingles



Nummular Keratitis

Coin shaped sub-epithelial lesions

----- Active space -----

CORNEAL DYSTROPHIES, KERATOCONUS AND MISCELLANEOUS DISORDERS

Corneal Dystrophies

00:00:12

Features :

- Primary corneal disease.
- Present in 1st and 2nd decades.
- Non-inflammatory, progressive, B/L.
- Autosomal dominant inheritance.

Disease	Feature	
1. Epithelial		
Cogan's epithelial basement membrane dystrophy/ map dot	m/c corneal dystrophy.	
meesmann epithelial dystrophy	Non-progressive, d/t keratin gene mutation.	
Lisch epithelial dystrophy	AD/XLD.	
2. Bowman layer		
Reis Buckler's dystrophy/CDBI/GCD III	Fish net.	
Schnyder Central Crystalline dystrophy	Corneal lipid metabolism disorder.	
Thiel Behnke dystrophy / GCD III	Honeycomb pattern of opacities.	
3. Stromal		
Lattice corneal dystrophy I/ Biber Haab Dimmer lattice dystrophy II/ Finnish	Neuropathy seen 70-90 years.	Amyloid deposit : Stained by Congo red.
Lattice corneal dystrophy IIIA	AR, mulberry-like appearance.	
Gelatinous drop like dystrophy/Japanese	Hyaline deposit : Stained by masson Trichome stain.	
Granular corneal dystrophy I/ Groenaw granular corneal dystrophy II	Hyaline + Amyloid deposit	
macular dystrophy/Groenow II	AR, L/C corneal dystrophy	
Francois central cloudy dystrophy	Inborn error of keratin sulphate metabolism.	
4. Endothelial		
Fuch's endothelial dystrophy	Corneal guttata : Wart-like excrescences.	
Posterior polymorphous corneal dystrophy	-	
Congenital hereditary endothelial dystrophy I Congenital hereditary endothelial dystrophy II	Both have AR inheritance and have perinatal onset.	

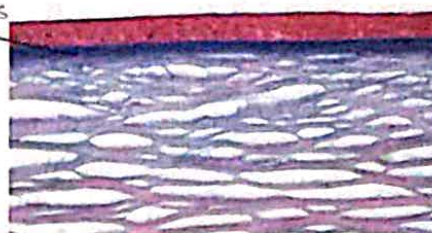
meesman epithelial dystrophy :



microcysts in corneal epithelium

macular dystrophy :

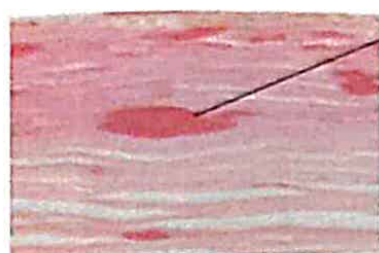
Acid mucopolysaccharidosis deposits



Alcian blue stain

----- Active space -----

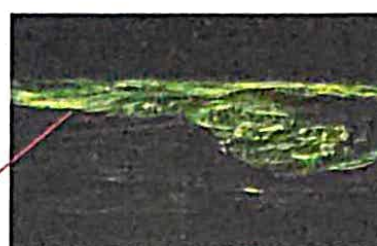
Lattice Corneal dystrophy -Type I :



PAS stain

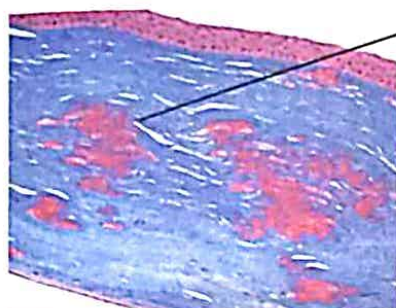
Amorphous deposit

Apple green birefringence



Congo red stain

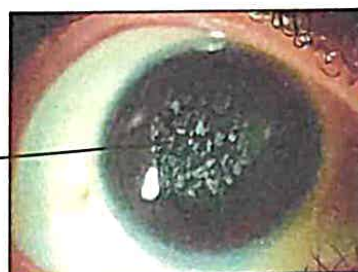
Granular dystrophy :



masson trichome stain

Hyaline deposits

Separated by well-demarcated clear space



Bread crumb like granules

Fuch's endothelial dystrophy :

- mutation of COXA2 gene.
- Descemet membrane + Endothelium damaged.
- Wart-like excrescences on cornea.



Corneal guttae (Diagnostic)

RECURRENT CORNEAL EROSIONS

Causes :

1. Cogan dystrophy.
2. Thiel Behnke dystrophy.
3. Reis Buckler dystrophy.
4. Lattice type I.

----- Active space -----

METABOLIC CAUSES OF DYSTROPHY

1. meesmann epithelial dystrophy.
2. macular dystrophy.
3. Schnyder central crystalline dystrophy.

TREATMENT OF CORNEAL DYSTROPHIES

If visual loss + : Keratoplasty/ Corneal transplantation.

Keratoconus

00:15:54

- Form of corneal ectasia.
- Non-inflammatory, progressive, usually B/L.

Etiology :

1. Congenital weakness.
2. Associated with :
 - Trauma.
 - Vernal keratoconjunctivitis.
 - Down syndrome.

Clinical features :

1. myopic astigmatism.
2. Hallmark : Stromal thinning (Central/ paracentral) → Inferior to the centre.
3. munson's sign : Bulging of lower eyelid on downgaze.
4. Rizzuti's sign : Light from temporal side → Arrowhead pattern of light over nasal limbus.
5. On retinoscopy : Scissoring reflex.
6. On ophthalmoscopy : Oil droplet reflex.
7. vogt's striae : Stromal stress lines.



munson's sign



Rizzuti's sign



Oil droplet reflex

Diagnosis :

IOC: Corneal topography.

Treatment :

- Rigid contact lens (may lead to opacity).
- Keratoplasty.
- C3R (corneal collagen cross linking using riboflavin and UVA radiation).

----- Active space -----

Miscellaneous

00:27:54

Band shaped keratopathy :

- D/T deposits of calcium in Bowman's layer.
- Treatment : Chelation with EDTA.

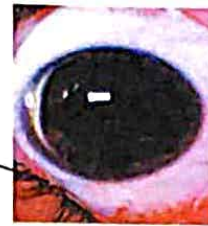


Band shaped keratopathy

Arcus senilis :

- Occurs in old age.
- D/t lipid deposition on Descemet's membrane near periphery.

Lucid interval of Vogt : Clear zone of cornea b/w arcus and limbus.



Arcus senilis

Interstitial keratitis :

Inflammation only of **corneal stroma**.

Causes :

- **Congenital syphilis**.
- Cogan syndrome (interstitial keratitis + deafness).
- Tuberculosis.
- Herpes Simplex virus (HSV).

Keratoplasty

00:32:57

AKA corneal transplantation.

TYPES

1. Penetrating keratoplasty (Full thickness).
2. Lamellar keratoplasty (Partial thickness)
 - DSEK (Descemet's stripping endothelial keratoplasty).
 - DALK (Deep anterior lamellar keratoplasty).

----- Active space -----

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00:15:54

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Rizzuti's sign



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----- Active space -----

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00:27:54

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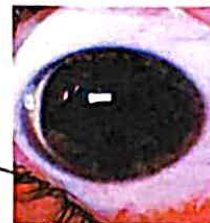


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Keratoplasty

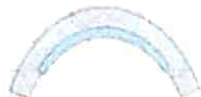
00:32:57

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----- Active space -----

	Penetrating keratoplasty	DSEK	DALK
Removal of	Epithelium to endothelium (i.e. all 5 layers)	Only endothelium & Descemet's membrane (i.e. posterior 2 layers).	Corneal tissue from epithelium upto stroma (i.e. anterior 3 layers).
Indication	Pseudophakic bullous keratopathy.	-	Acute hydrops in severe keratoconus
			

Procedure :

Host cornea cut using a trephine → Implant the graft cornea.

(Diameter : 7.5mm)

(Diameter : 0.25mm larger than host opening)



Trephine



Post-keratoplasty

Graft rejection :

- Endothelial rejection (m/c).
 - Forms Khodadoust line (Diagnostic).

Storage media for corneal transplantation :

Type	Duration	Other features
Short term	<48hrs	4°C, moist chamber/mcCarey Kaufman medium
Intermediate term	<2 weeks	-
Long term	upto 35 days	Cryopreservation (-70°C)

ANTERIOR UVEITIS

----- Active space -----

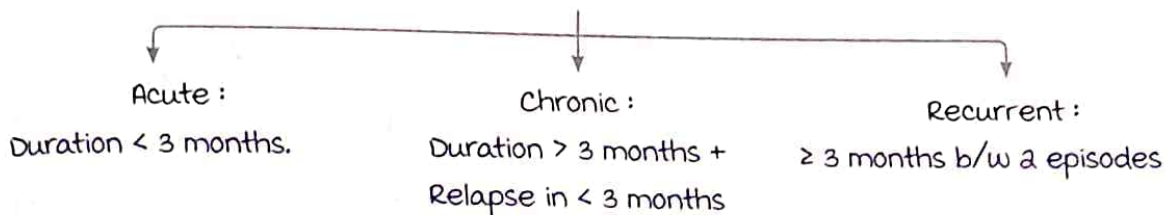
Classification of Uveitis

00:01:53

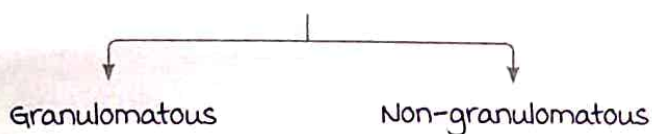
Anatomical Classification :

	Anterior uveitis (AU)	Intermediate uveitis	Posterior uveitis	Pan uveitis
AKA	Iridocyclitis.	Pars planitis/ Posterior cyclitis.	Chorioretinitis/ Retinochoroiditis	-
Involvement of	Iris : Iritis. Pars plicata : Cyclitis.	- Peripheral retina : Basal retinochoroiditis. - Vitreous : Vitritis/ Hyalitis.	Choroid & Retina	All layers of uvea.

Clinical Classification :



Etiological Classification :



Clinical features of Anterior Uveitis

00:08:29

SYMPTOMS

- Pain : Referred along branches of 5th nerve.
- Redness.
- Blurring of vision : D/t
 - Corneal haze
 - Aqueous turbidity
- Photophobia : Severe.
- Lacrimation.

----- Active space -----

SIGNS**1. Conjunctiva :**

Circumcorneal congestion.

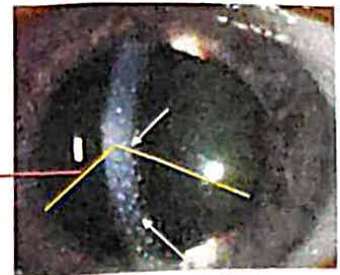
2. Cornea :

Corneal haze : D/t raised intraocular pressure (IOP)/ toxins.

Keratic Precipitates (KPs) :

- Cellular deposits on corneal endothelium.
- **Location** : Arlt's triangle (m/c).

Arlt's triangle



Keratic precipitates

Type of KP	Seen in	Other features
mutton fat	Granulomatous uveitis	Contain macrophages
Fine granules	Non-granulomatous uveitis	Contain lymphocytes
Pigmented	Chronic non-granulomatous uveitis	-
Stellate	Herpetic uveitis, Toxoplasmosis, Fuchs heterochromic iridocyclitis (FHI)	• Star-shaped • Diffuse distribution

Anterior Chamber :

1. Aqueous cells : markers of disease activity.

2. Aqueous flare :

- Turbidity of aqueous : D/t leakage of proteins.
- visible d/t Tyndall effect/Brownian movement.

3. Hypopyon :

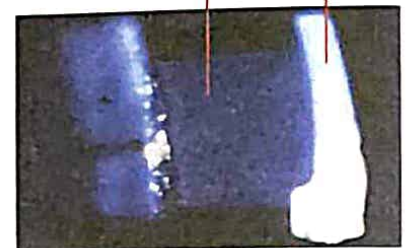
Hypopyon + Anterior uveitis : Behcet's disease.

4. Hyphema :

Hyphema + Anterior uveitis :

- Herpetic uveitis
- Toxoplasmosis
- Syphilis

Aqueous Cornea



Aqueous cells

Iris :

1. Pseudo-rubiosis : D/t dilated iris vessels.

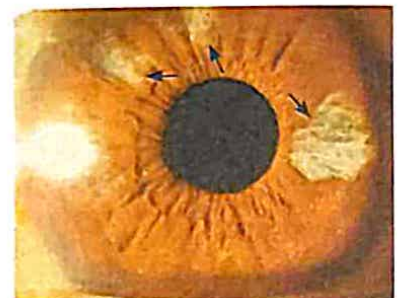
2. muddy iris :

- Seen in acute cases.
- Aka water logged iris.

3. Iris atrophy :

- Seen in chronic cases.

- Types :
 - Sectoral : Herpetic uveitis
 - Diffuse : FHI



Sectoral iris atrophy

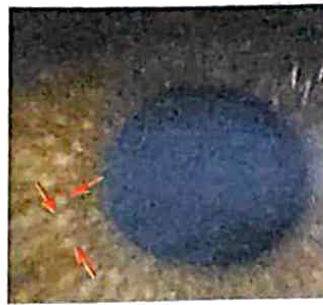
4. Iris nodules :

----- Active space -----

	Koepple nodules	Busacca nodules:
Location	Pupillary margin	Iris stroma
Found in	• Granulomatous uveitis • Non granulomatous uveitis	Granulomatous uveitis



Koepple nodules



Busacca nodules

Pupil :

1. Miosis : Acute cases.

2. Festooned pupil : Chronic cases.

Exudation → Formation of incomplete posterior synechiae
 (Adhesion b/w pupillary margin and lens).
 ↓ Dilates the pupil
 Irregular dilation.



Festooned pupil

3. Fixed pupil :

- Recurrent cases.
- Annular/Ring/Complete posterior synechiae (360° adhesion) → No aqueous flow from posterior chamber to anterior chamber → Secondary angle closure glaucoma.

4. Total Posterior Synechiae (TPS) :

Exudation in posterior chamber → Adhesion b/w posterior surface of iris and lens.

5. Cyclitic membrane : TPS + Exudates leak behind lens.

Covers ciliary processes, equator & posterior surface of lens → No aqueous secretion → ↓ IOP.

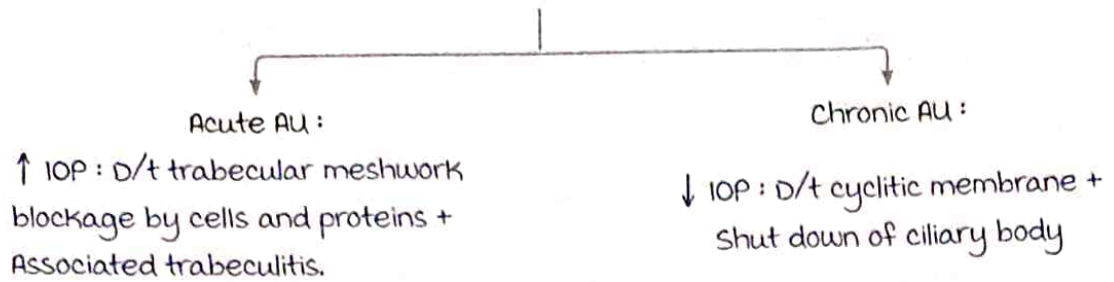
6. Occlusio pupillae : Exudates organised across anterior surface of pupil.



Occlusio pupillae

----- Active space -----

IOP :



Complications and Treatment of Anterior Uveitis

00:45:26

Complications :

1. Complicated Cataract : m/c

Bread crumb appearance

+ polychromatic lustre.

2. Secondary angle closure glaucoma :

m/c in recurrent cases.

3. Band shaped keratopathy :

Calcium deposits in **Bowman's layer** of cornea.

4. Cystoid macular edema :

m/c cause of visual loss d/t Posterior involvement in anterior uveitis (AU).

Note :

Causes of complicated cataract :

mnemonic : **UMAR**

- **U**veitis.
- **M**yopia.
- **A**ngle closure glaucoma.
- **R**etinitis pigmentosa.

Treatment :

Topical steroids :

- Prednisolone.
- Difluprednate.

Antimetabolites : Azathioprine.

- Given if inflammation + recurrent episodes requiring long term steroids.

Cyclopegics : Atropine

Action :

- Relax ciliary muscle → Pain relief.
- Break posterior synechiae.
- Prevent formation of posterior synechiae.

Note :

Side effects of steroids :

- Topical administration : Glaucoma.
- Systemic administration : Cataract.

Causes of Anterior Uveitis

00:52:39

----- Active space -----

GRANULOMATOUS

a. Tuberculosis.

b. Leprosy :

- m/c : Lepromatous leprosy.
- Iris pearls : Pathognomic feature.

c. Sarcoidosis :

- m/c ocular manifestation of sarcoidosis : *granulomatous anterior uveitis*.
- Lofgren syndrome: Erythema nodosum + B/L lymphadenopathy + Anterior uveitis.
- Heerfordt syndrome/uveoparotid fever.

d. Syphilis :

- m/c in late congenital syphilis.
- Presentation : Iris roseolas.

e. Herpes :

AU + *Raised IOP* + Hyphema + Stellate KPs + Sectoral iris atrophy.

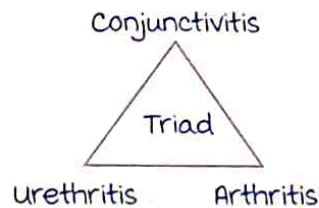
2. NON GRANULOMATOUS

Acute :

1. Idiopathic : m/c

2. HLA B27 associated disease :

- Ankylosing Spondylitis : Lower back pain + stiffness + recurrent W/L AU.
- *Reiter's disease* :



+ W/L AU + Keratoderma blenorrhagicum + Circinate balanitis + History of travel.

- Psoriatic arthritis : Sausage digits + B/L AU.

3. Behcet's disease :

- A/W HLA B5, B51.
- B/L recurrent AU + Cold mobile hypopyon + *Aphthous mouth ulcers*.

4. Inflammatory bowel disease :

ulcerative colitis (12%) > Crohn's disease (2.4%).

5. Drug induced uveitis :

----- Active space ----- Mnemonic : **CML Please Relapse Slowly.**

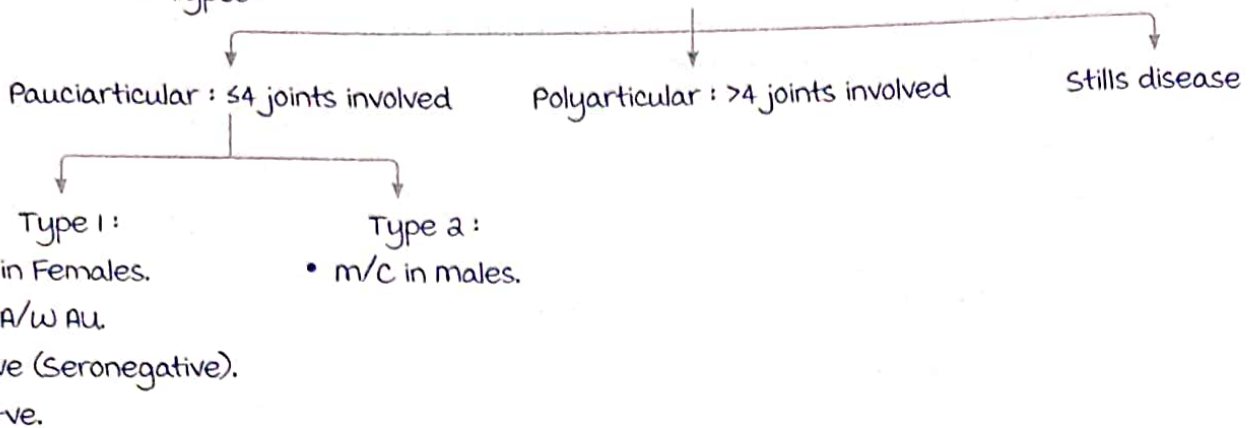
- Cidofovir.
- metipranolol.
- Latanoprost.

- Pilocarpine.
- Rifabutin.
- Sulfonamides.

Chronic :

Juvenile Idiopathic Arthritis (JIA) :

- m/c cause of AU in paediatric age group.
- Types :



2. Fuch's Heterochromic Iridocyclitis (FHC) :

- Clinical features :

Mnemonic : **NO FUCHS.**

- NO posterior synechiae.
- Floaters.
- unilateral.
- Cataract.

- Heterochromia iridis : Hypochromia in the affected eye d/t loss of iris crypts.
- Stellate keratic precipitates.

- Amsler's Sign : Bleeding on paracentesis.

INTERMEDIATE, POSTERIOR, PAN-UVEITIS AND MISCELLANEOUS DISORDERS

----- Active space -----

Intermediate Uveitis

00:00:07

Also known as pars planitis.

Causes :

1. Idiopathic (m/c).
2. Associated with:
 - Lyme's disease.
 - Sarcoidosis.
 - multiple sclerosis.
 - Tuberculosis.

Clinical Features :

Symptoms :

- B/L floaters.
- Blurring of vision.

Signs :

- Spillover anterior uveitis.
- **Snowballs** : Aggregation of cells in vitreous.

Treatment :

Steroids : Posterior subtenon injection

↓ If fails

Steroids : Systemic

↓ If fails

Immunomodulatory drugs : Azathioprine

↓ If fails

Pars plana vitrectomy + Photocoagulation of snowbank

- Snowbanking : Fibrovascular plaque.
- **Cystoid macular edema** : m/c cause of vision loss.

Posterior Uveitis/Chorioretinitis

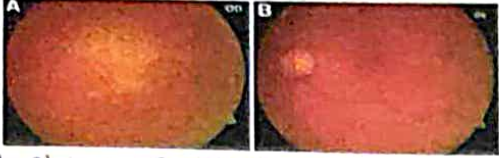
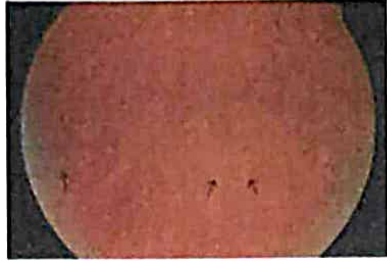
00:09:17

Inflammation of choroid & retina.

Clinical Features :

1. Floaters.
2. Painless loss of vision.
3. Scotoma.
4. metamorphopsia : Distortion of image.
5. Photopsia.

----- Active space ----- Causes :

Infectious	Non-infectious
1. most common : Toxoplasmosis. a. Active : Headlight in fog appearance (d/t associated active vitritis.)  b. Chronic : Punched out scar.  - Cats are definitive hosts. - Toxoplasmosis causes focal chorioretinitis. 2. Toxocariasis : - Unilateral LOV. - Leukocoria (white pupillary reflex). 3. CMV retinitis : - m/c cause of LOV in ocular HIV. a. Pizza pie appearance. b. Scrambled egg & ketchup appearance. c. Brushfire extension.   4. TB	1. Sarcoidosis : Candle wax dripping appearance. (d/t perivascular inflammation & sheathing).   - Lander's sign : Pre-retinal nodules. - Behcet's disease. - Birdshot chorioretinopathy : HLA A29 association. - Serpiginous choroiditis : HLA B7 association.

Focal chorioretinitis : Toxoplasma, Toxocariasis, CMV.

multifocal chorioretinitis : HSV, TB, syphilis.

Treatment :

----- Active space -----

1. Infectious cases : Treat the cause + systemic steroids.

2. Non-infectious cases : Systemic steroids

OR

- Adalimumab (visual studies I to II)
- Sirolimus (Sakura program).
- Voclosporin (Luminate trials).

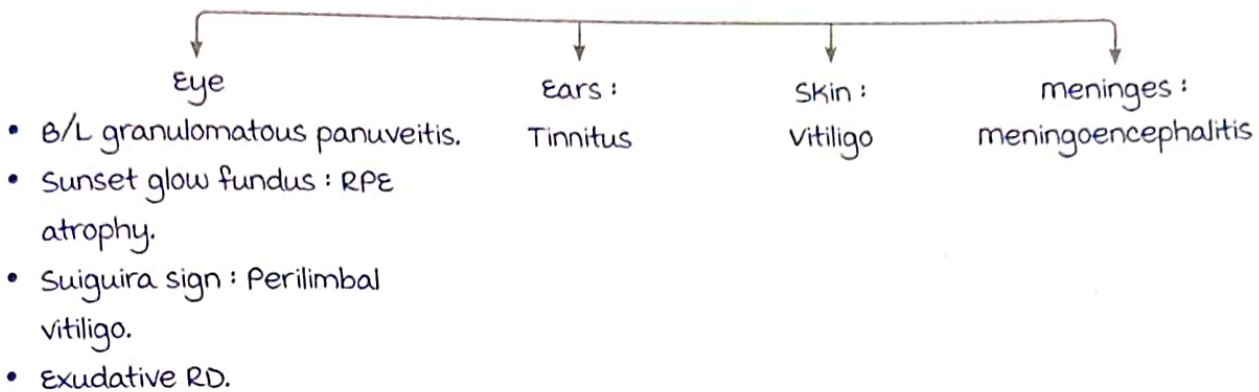
Pan-uveitis

00:22:27

Causes :

1. Vogt-Koyanagi-Harada (VKH) syndrome :

Affects :



2. Sympathetic ophthalmitis :

- Clinical feature : Granulomatous panuveitis.
- **Cause** : Accidental penetrating trauma affecting ciliary body.
- Pathogenesis : Trauma to one eye : Exciting eye

↓ >2 weeks

Sympathetic ophthalmitis in other eye (sympathizing eye).

Prevention : Enucleation of traumatic eye in 14 days.

Treatment : Steroids.

----- Active space -----

Miscellaneous

00:33:00

	Enucleation	Evisceration	Exenteration
Terminology	Removal of whole eyeball + optic nerve	Removal of all layers of eyeball except sclera.	Removal of eye and orbital contents.
Indications	- Retinoblastoma. - Trauma.	Endophthalmitis.	- Orbital mucormycosis. - Tumors metastasised to orbit.
c/i	Panophthalmitis	Retinoblastoma	-
Appearance			

Gyrate atrophy :

- Type of choroid atrophy.
- Autosomal recessive.
- mutation in ornithine aminotransferase gene : ↑ Levels of ornithine.
- Treatment : Arginine restriction in diet (ornithine is synthesised from arginine)

ANATOMY OF CONJUNCTIVA, TYPES OF CONJUNCTIVITIS AND PTERYGIUM

----- Active space -----

conjunctiva is transparent mucous membrane.
It does not cover cornea.

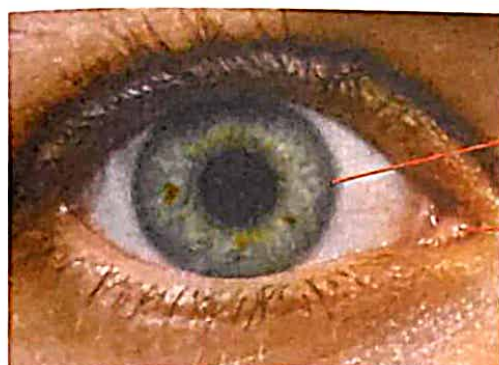
Anatomy

00:00:50

Parts of conjunctiva :



Bulbar (Covers sclera)
Forniceal
Palpebral (Lines inner surface of eyelids)



Plica semilunaris : Represents nictitating membrane
Caruncle : Contains hair follicles
Rudimentary structures

Histology :

1. Outermost layer : Epithelium (Non-keratinised stratified squamous).

Contains :

- Goblet cells : Secrete mucin (Forms innermost layer of tear film) on parasympathetic stimulation.
- melanocytes.
- Langerhans cells.

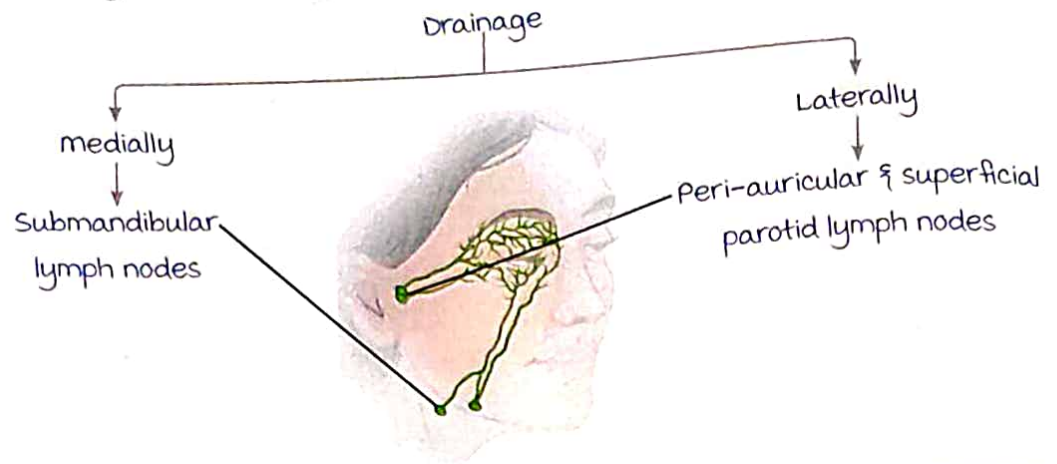
2. Adenoid/lymphatic layer : Develops after 3 months of age.

3. Fibrous layer : Contains nerve fibres & blood vessels.

----- Active space -----

Lymphatic drainage :

Only site of lymphatic drainage from the eye.



00:08:40

Conjunctivitis

AKA Eye flu.

Symptoms :

- Redness.
- Discharge sticking to eyelids.
- Foreign body sensation.
- Lacrimation.

- Itching (if allergic).
- Pain and loss of vision (LOV) : Corneal involvement.

Signs :

I. Conjunctival hyperemia (Redness) :

Differential diagnosis :

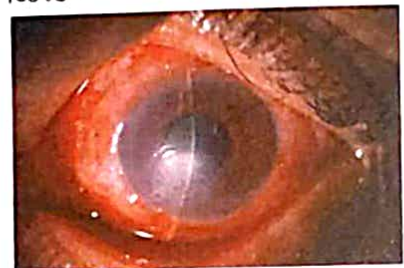
Circumcorneal redness seen in

- Glaucoma
- Uveitis
- Corneal ulcer



Conjunctival hyperemia

Note :



Circumcorneal hyperemia

- Seen in : Glaucoma, uveitis, corneal ulcer.

2. Subconjunctival hemorrhage :
No treatment required (Resolves spontaneously $\leq$ 2 weeks).



3. Chemosis : Swelling of bulbar conjunctiva.



----- Active space -----

4. Discharge : mucopurulent/watery.
• Causes colored halos & stickiness which resolves on washing eye.

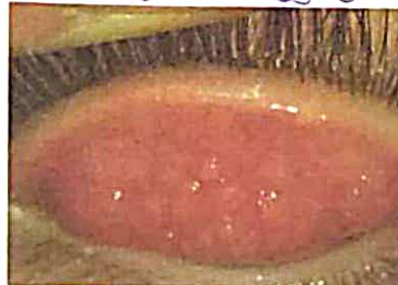


5. Inflammatory reactions of conjunctiva :

Papillae : Hypertrophied vessels.



Follicles : Lymphoid aggregates.



Etiology :

Discharge	Papillary reaction	Follicular reaction
mucopurulent	Bacterial	Chlamydia
Watery	Allergic	Viral

Bacterial conjunctivitis

00:17:22

Acute mucopurulent conjunctivitis :

m/c cause : *Staphylococcus aureus*.

Treatment : Topical antibiotics.

----- Active space -----

Acute purulent/hyperacute conjunctivitis/Blenorrhoea:m/c cause : *Neisseria gonorrhoeae*.

Route of transmission : Oculogenital.

Clinical features :

1. Copious purulent discharge.
2. Swelling of eyelids → Overhanging.
3. Intense pain.
4. Pre-auricular lymphadenopathy.



Blenorrhoea

Treatment :

I.m. Ceftriaxone 1g single dose

↓ followed by

Oral erythromycin x 2 weeks.

Acute membranous conjunctivitis :

m/c cause — { Overall/in adults : *Pneumococcus*.
 In unimmunised children : *Corynebacterium diphtheriae*.

Clinical features :

membrane formed → Fused with epithelium → Bleeds on removal.

Acute pseudomembranous conjunctivitis :

membrane does not fuse with epithelium

↓
No bleeding on removal.

Pseudomembrane peeling

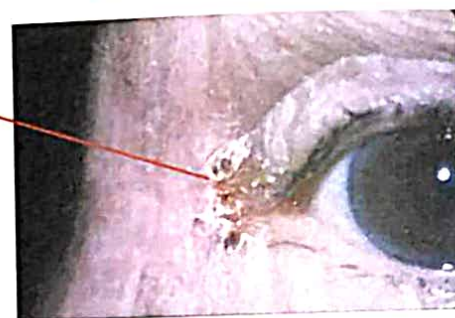
Angular/diplobacillary conjunctivitis :m/c cause : *Moraxella lacunata*/axenfeld.

Clinical features :

- Excoriation at canthi (Lateral > medial).
- Redness at intermarginal strip.
- Blepharitis (Inflammation of eyelid margin).

Treatment :

- Tetracycline 1% eye ointment for ≥ 2 wks.
- Zinc boric eye drops. (Zinc blocks proteolytic activity).

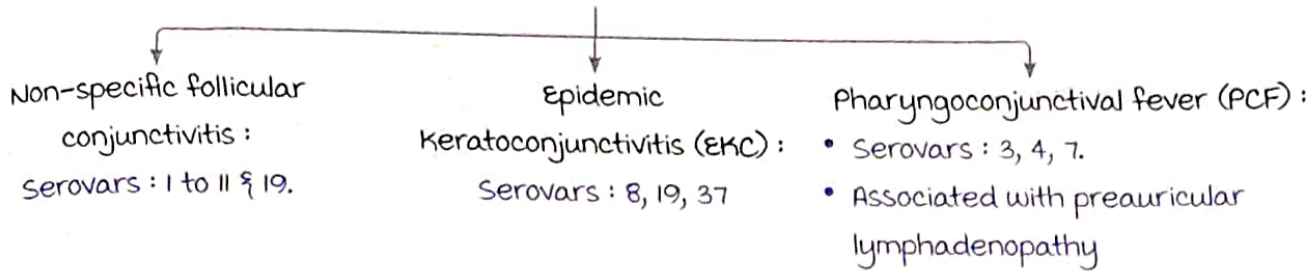


Viral Conjunctivitis

00:27:00

----- Active space -----

Adenoviral conjunctivitis :



Acute hemorrhagic conjunctivitis : Hemorrhage of palpebral & bulbar conjunctiva.

Causes :

1. Picornavirus : Enterovirus type 70,
Coxsackie A24 (m/c).
2. Adenovirus type 11.



molluscum contagiosum :

Clinical features :

- unilateral, painless nodule with umbilicated appearance.
- Nodule contains viral particles.

↓
Periodic shedding of virus

↓
Conjunctivitis

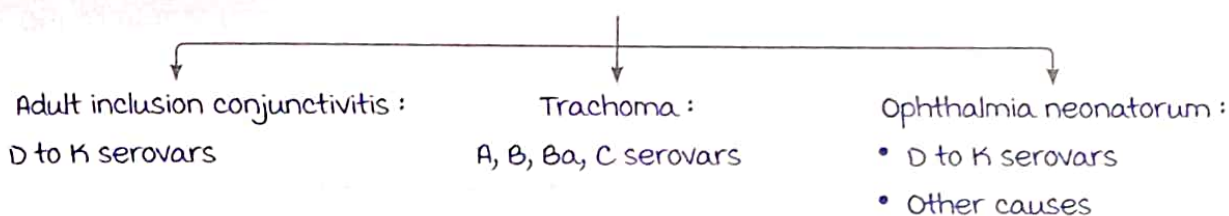


Umbilicated nodule

Treatment : Surgical excision [if cosmetic indication (+)].

Chlamydial Conjunctivitis

00:33:14



TRACHOMA

Aka Egyptian ophthalmia.

Route of transmission : mnemonic : 3F.

1. Fingers.
2. Flies.
3. Fomites.

----- Active space -----

Pathology :

Type IV hypersensitivity reaction : Active inflammation + cicatrization.

Age group : < 10 yrs.

Clinical signs :

1. Sago grain follicles : Necrosis & leber.

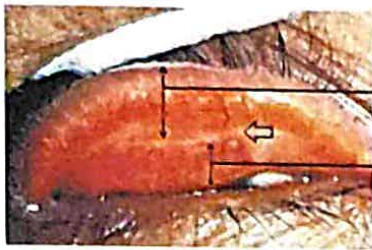
- m/c site : upper palpebral conjunctiva.
- Necrosis & leber cells seen.



Sago grain follicles

2. Cicatricial sign :

a. Arlt's line : Line of cicatrization.

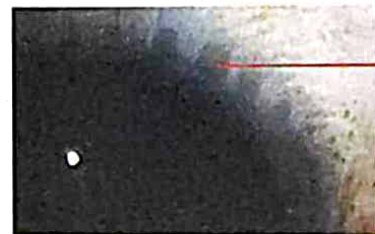


Lower 2/3rd

upper 1/3rd

b. Herbert's pits : Pathognomic.

- D/t healing of limbal follicles.



Herbert's pits

c. Pannus : Vascularisation of cornea superiorly.

**Complications :**

Trichiasis (Inturned eyelash) → Corneal ulcer → Scar formation → Loss of vision.

Treatment :

SAFE strategy (WHO, 1996) :

- Surgery for inturned eyelids.
- Antibiotics : Azithromycin Ig.
- Facial cleanliness.
- Environmental change.

Indications for treatment :

Prevalance in 1-9 yr old children	management
> 10 %	mass prophylaxis
5-10 %	Treatment to affected children & family
< 5 %	Facial cleanliness & environmental change advised

FISTO CLASSIFICATION (WHO)

----- Active space -----

	Features	management
TF : Trachomatous inflammation follicular	> 5 follicles +	Active stage Treatment : Antibiotics
TI : Trachomatous inflammation intense	Thickening of upper palpebral conjunctiva	
TS : Trachomatous scarring	Art's line	Inactive stage : No treatment
TT : Trachomatous trichiasis	Eyelash growing inwards	Requires surgery
CO : Corneal opacity	Visual impairment	-



Trachomatous inflammation follicular (TF)



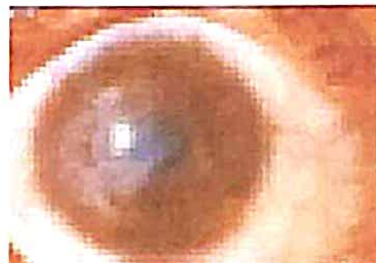
Trachomatous inflammation follicular and intense (TF)



Trachomatous scarring (TS)



Trachomatous trichiasis (TT)



Corneal opacity (CO)

OPHTHALMIA NEONATORUM

Conjunctivitis in neonates (< 28 days of onset).

Causes :

Onset (Hours after birth)	Cause
Within first 6h	Chemical conjunctivitis (Silver nitrate)
24 to 48h	Neisseria gonorrhoeae (most severe)
2 to 5 d	Other bacteria
5 to 7 d	HSV-II
> 1 wk	Chlamydia trachomatis (D to K) : m/c



Ophthalmia neonatorum

----- Active space -----

Prevention :

1. Crede's method : Topical 1% Silver nitrate (Not recommended d/t chemical conjunctivitis).
 2. 0.5% erythromycin.
 3. 1% tetracycline.
- } • Single application.
} • Within 1 hr of birth.

Allergic conjunctivitis

00:50:24

Presents with papillae reaction + watery discharge + itching.

VERNAL KERATOCONJUNCTIVITIS/SPRING CATARRH

Type I hypersensitivity reaction (Eg : pollen).

Incidence :

- Age : 5-15 yrs.
- Males > Females (Young boys).
- m/c in spring & summer.
- H/O atopy present.

Clinical signs :

1. Papillary hypertrophy :

- Cobble stone papillae.
- Giant papillae : > 1mm size.

a. Horner Trantas sign :

- Collection of eosinophils + epithelial debris.
- Site : On limbus.



Horner trantas sign

Pseudogerontoxon

3. Pseudogerontoxon : Paralimbal grey-white band (D/t lipid deposition) in children.

- Appearance similar to arcus senilis.

4. Shield ulcer (in cornea).

Note : Excessive itching

5. Dennie Morgan line : Extra lower-lid crease.

↑ incidence of keratoconus.

6. Maxwell Lyon sign : Ropy discharge.

Treatment :

1. Cold compress.
2. Topical anti-histamine.
3. Topical mast cell stabilisers.
4. **DOC** : Olopatadine or Alcaftadine.
5. **Topical steroids** in acute exacerbation (For symptomatic treatment).

----- Active space -----

ATOPIC KERATOCONJUNCTIVITISseen in 2nd to 5th decade.**Clinical features :**

- Eyelids : **Hertoghe sign** (Lateral eyebrow lost).
- Itching.
- Papillae.
- Watery discharge.

Treatment :

1. Topical anti-histamine.
2. Topical mast cell stabilisers.

GIANT PAPILLARY KERATOCONJUNCTIVITIS

- Papillae > 1mm.
- Type IV hypersensitivity reaction.

Etiology : mechanically induced by

- Contact lens
- Ocular prostheses
- Protruding sutures

Treatment : Anti-histamine + surgery to remove irritant (if required).

PHLYCTENULAR KERATOCONJUNCTIVITIS

Type IV hypersensitivity reaction in response to endogenous allergen.

Etiology :

- m/c in India : **TB**.
- m/c in western countries : **Staphylococcus aureus**.

Clinical features :

- **Phlycten** : Nodule near limbus.
 - Grows towards cornea → Ulcerates.
 - Gives rise to
 - Sacrofulvous ulcer
 - Fascicular ulcer
 - miliary ulcer

Treatment : Topical Steroids.



Phlycten

----- Active space -----

Pterygium

Triangular, fibrovascular growth of degenerative sub-conjunctival tissue over Cornea.

- Destroys Bowman's layer & superficial stroma.
- Associated with
 - UV ray exposure
 - Dust
 - Humidity

Cause : **Limbal stem cell deficiency** → Activation of matrix metalloproteinase.

Site : Nasal > Lateral.

Clinical features :

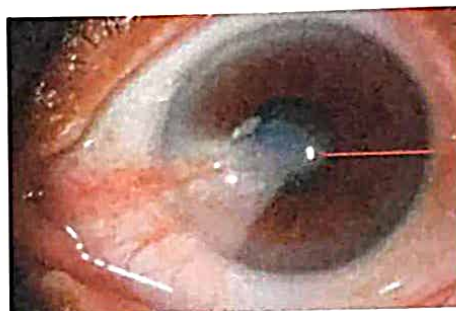
1. Triangular growth with apex towards cornea.
2. **Stocker's line** : Deposition of iron in front of apex.
3. Loss of vision.

Causes :

a. Corneal astigmatism.



b. Encroaching into visual axis.



Covers pupillary area

Treatment :

Surgical excision : ↑ rate of recurrence.

- To reduce recurrence :

1. Topical mitomycin C.
2. Autograft : Harvested from same eye.
 - Site : usually superior.
 - Fixed with sutures/fibrin glue/autologous serum.

Pinguecula & concretions

01:14:28

----- Active space -----

PINGUECULA

Elastic degeneration of collagen fibres in stroma of conjunctiva.

m/c site : Nasal part of conjunctiva.



Yellowish fat-like nodules

CONCRETIONS

Collection of epithelial debris & mucus.

Cause : Irritation d/t friction of eyelid against bulbar conjunctiva & cornea while blinking.

Treatment : Removal with 26g needle.

- Spontaneously resolves with good lubrication.

Note : Concretions do not contain calcium deposits.



minute yellowish-white elevations

----- Active space -----

EYELIDS : ANATOMY AND PATHOLOGIES

00:01:01

Anatomy

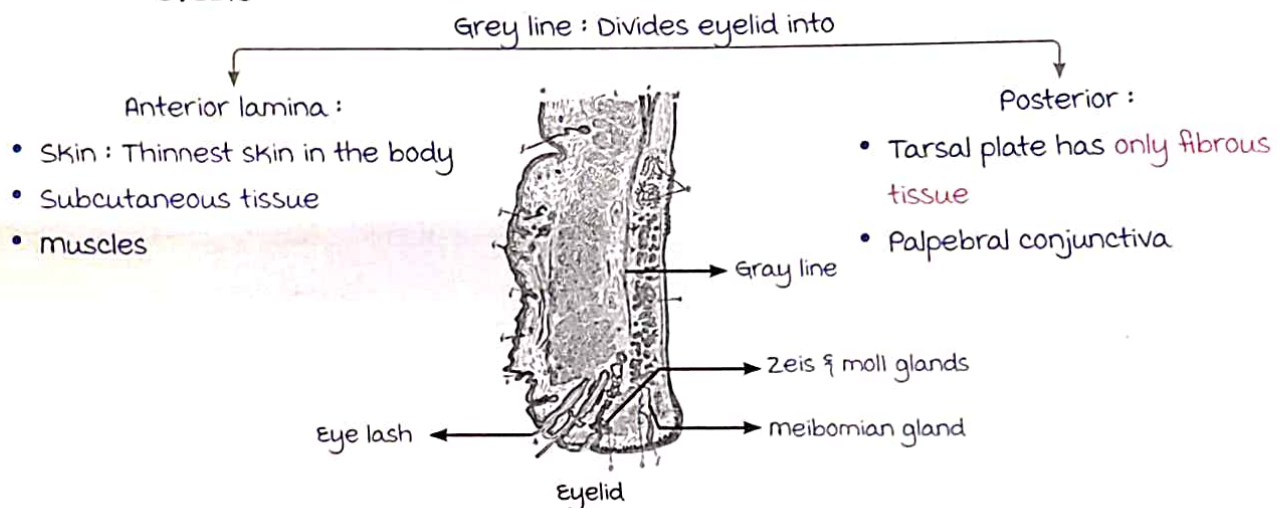
3 muscles :

muscle	Nerve	Function	Presentation in palsy
Levator palpebrae superioris	CN III	Elevation of upper lid	Ptosis
müller's muscle	Sympathetic fibres		
Orbicularis oculi	CN VII	Closure of eyelid	Lagophthalmos

3 glands :

Gland	Type of gland	Opens at
meibomian (tarsal)	modified sebaceous	Posterior to anterior surface of eyelid margin
Zeis		Base of lash follicles
moll	modified sweat	

EYELID



Pathologies

00:09:16

CHALAZION & HORDEOLUM

	Chalazion	External hordeolum / sty
Pathology	Lipogranulomatous inflammation of meibomian gland	Suppurative inflammation of Zeis gland
Infectivity	Non-infective	Infective : S. aureus

	Chalazion	External hordeolum / sty
Swelling	- Painless - Circumscribed swelling <i>away from lid margin</i>	- Painful - Diffuse swelling at the lid margin.
Rx	- Incision & curettage : - Horizontal incision on posterior part of eyelid (conjunctiva). - Intralesional triamcinolone	- Hot compresses - Oral antibiotics if recurrent.
Complications	Recurrence → Sebaceous cell carcinoma	—
Appearance		

----- Active space -----

Hordeolum internum :

Suppurative inflammation of meibomian gland.

TRICHIASIS

- Inward misdirection of eyelashes
- Causes corneal opacity



Trichiasis

Treatment

- Epilation every 1-2 months.
- Cryotherapy of lash base : Definitive.

ENTROPION & ECTROPION

	Entropion	Ectropion
Feature	Inward turning of eyelid margin	- Outward turning of eyelid margin - m/c cause : Senility
Rx :	 - Spastic → Botulinum toxin injection - Senile → modified wheeler operation, Weiss Operation - Cicatricial → modified Burrows operation, wedge resection	 - medial conjunctivoplasty - Buryon Smith operation

----- Active space -----

DISTICHIASIS

Extra posterior row of eye lashes.



Distichiasis

MADAROSIS

Loss of lateral 1/3rd of eye lashes & eyebrow d/t chronic rubbing of the eye.



madarosis

TYLOSIS

Thickening of eyelid margin.

ANKYLOBLEPHARON

- Fusion of eyelid margin of upper & lower lid.
- Post trauma.

CONGENITAL PTOSIS

D/t defective development of levator palpebrae superioris.

Clinical features :

- Absent upper lid crease.
- Lid lag on downgaze :
 - Failure of eyelid to go down on downward gaze.



Congenital Ptosis

Treatment :

Ptosis	LPS function	Treatment
moderate	Fair	LPS resection
Severe	Poor	Frontalis sling SX

Marcus Gunn Jaw winking syndrome :

D/t trigemino - oculomotor nerve synkinesis.



Closed jaw → ptosis Open jaw → lid moves up

ANATOMY OF ORBIT AND PROPTOSIS

----- Active space -----

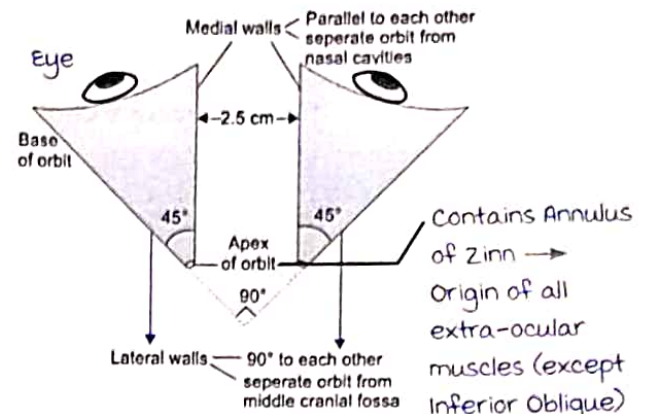
Bony cage surrounding eye.

Anatomy of Orbit

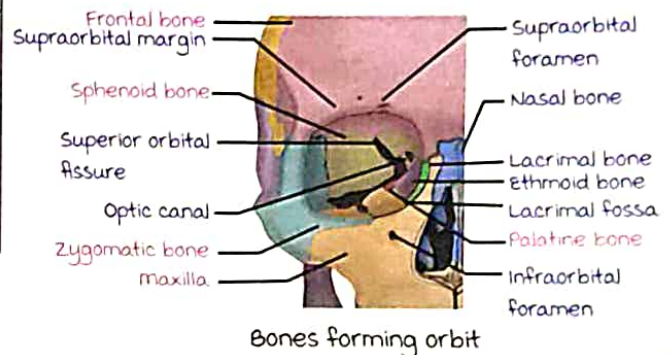
00:00:31

WALLS OF ORBIT

Wall	Constituent bones
Roof	- Frontal bone. - Lesser wing of sphenoid.
Lateral wall	- Zygomatic bone - Greater wing of sphenoid
Floor (m/c to fracture → Blow out fracture)	- maxilla. - Zygomatic bone. - Palatine bone
medial wall → Thinnest wall (Lamina papyracea)	Mnemonic: SMEL - Sphenoid : Body. - maxilla : Frontal process. - Ethmoid bone. - Lacrimal bone.



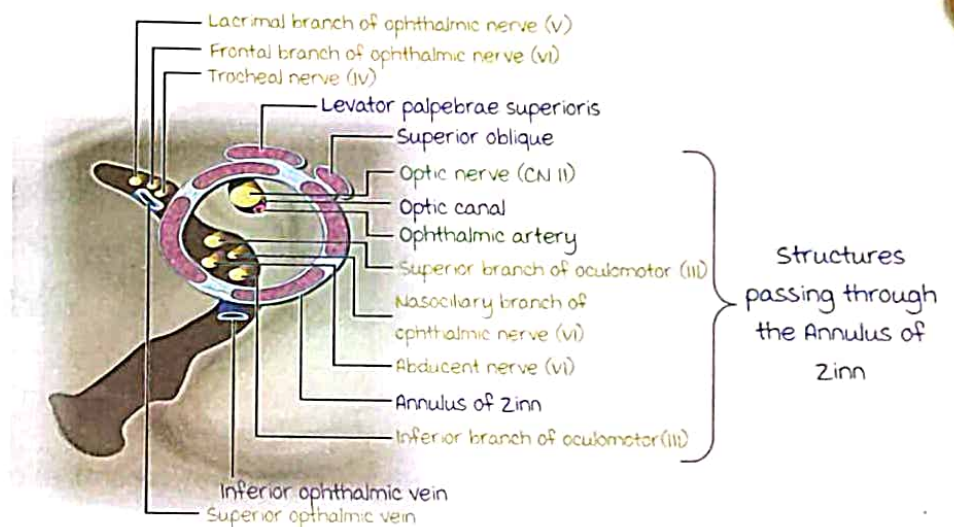
Orbit : Schematic



Note: Eyeball naturally protrudes slightly from lateral wall of orbit.

OPENINGS INTO ORBITAL APEX

- Optic canal.
- Superior orbital fissure.
(Between roof and lateral wall)
- Inferior orbital fissure.
(Between floor and lateral wall)



Structures passing through the openings

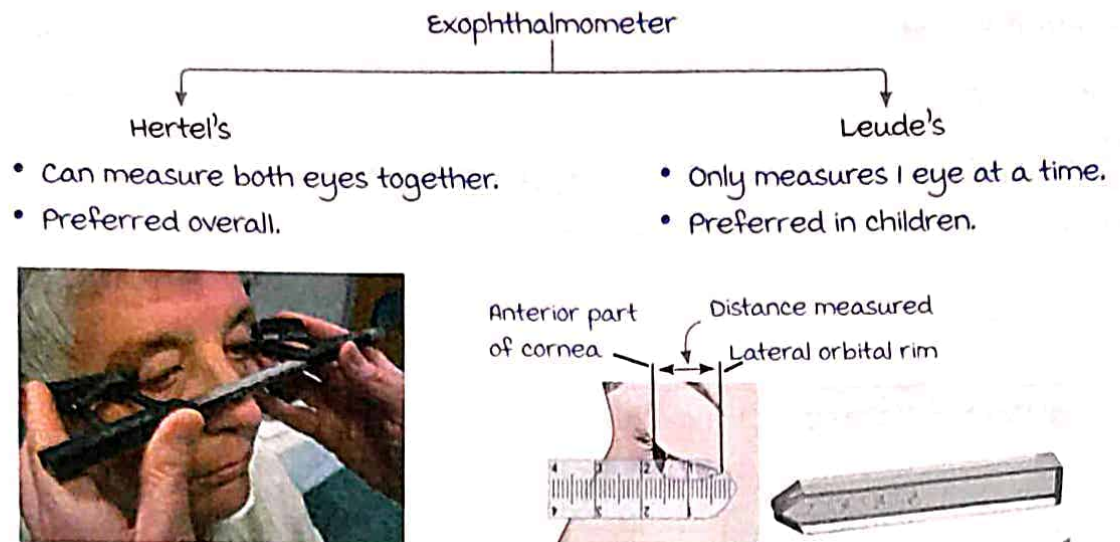
----- Active space -----

Proptosis/Exophthalmos

00:13:17

Protrusion of eyeball from lateral orbital rim :

- >21mm. OR
- 2mm difference between 2 eyes.

MEASUREMENT**CAUSES****Bilateral :****Painless :**

- Grave's disease.
- Cortico-cavernous fistula.
- Leukemia/ AML (in children)

Painful :

- Cavernous sinus thrombosis (CST).
- Grave's (restriction of ocular movement causing pain)

Unilateral :**Axial :**

- Along 1° axis.
- Outward, straight protrusion.

Abaxial :

- Away from 1° axis.
- Direction depends on cause.

Painful :

- Orbital cellulitis.
- CST

Painless :

- ON glioma.
- Cavernous hemangioma.

Inferomedial :

- Superotemporal lesions.
- Lacrimal gland tumors.
- Dermoid cysts.

upward :

- maxillary sinus tumor

Lateral :

- Lacrimal sac

m/c causes :

----- Active space -----

In adults : Grave's disease. (unilateral or bilateral proptosis.)

In children :

a. u/L : Orbital cellulitis.

b. b/L : Leukemia/AML.

Inflammatory Causes of Proptosis

00:23:23

CAVERNOUS SINUS THROMBOSIS & ORBITAL APEX SYNDROME

	Cavernous sinus thrombosis	Orbital apex syndrome
Structures involved (structures passing through)	CN III, IV, VI, V _p , V _a	- Contains structures passing through superior orbital fissure & optic canal. - CN II, III, IV, VI, V₁
Ophthalmoplegia : Restriction of ocular movement d/t CN III, IV, VI palsies	Sequential ophthalmoplegia : CN VI affected → CN III affected → CN IV affected	Concurrent ophthalmoplegia : All nerves affected together → All muscle actions lost at the same time.
Ophthalmic nerve (V ₁ palsy) : Corneal anaesthesia	Present	Present

	Cavernous sinus thrombosis	Orbital apex syndrome
Visual loss : D/t CN II palsy.	- No loss initially. - Can be affected in late stages d/t other causes.	Complete LOV.
Edema in mastoid area : D/t CN V ₂ palsy	Present	Absent
Chemosis & Proptosis	- b/L - marked/severe (presenting feature clinically)	- u/L - mild
Systemic signs (Eg : fever)	marked	mild
Presentation	Always b/L (starts u/L)	Always u/L.

----- Active space -----

ORBITAL CELLULITIS**Causes :**

1) Extension of paranasal sinusitis (m/c) :

- Ethmoid sinusitis (m/c).
- Staphylococcus and Streptococci : m/c etiological agents.
- m/c in children with upper respiratory tract infection.

2) Exogenous/Trauma

3) Hematogenous spread

Progression :

Pre-septal cellulitis $\xrightarrow{\text{If no Rx}}$ Orbital cellulitis $\xrightarrow{\text{If no Rx}}$ Subperiosteal abscess.

• Findings

- Eyelid edema.
- Pain.
- Inability to open eye.

• Rx : Topical/oral antibiotics.

• Findings

- Proptosis.
- Limitation of eye movement.

• Rx : IV antibiotics.

• Complications :

Compression of optic nerve $\longrightarrow$ Blindness.

Note : Orbital septum

- Formed by muscles of eyelid, fascia, medial and lateral canthal ligament.
- Separates orbit from pre-orbital area forming
 - Pre-septal.
 - Post-septal.

Other Causes & Types of Proptosis

00:37:44

VASCULAR CAUSES OF PROPTOSIS**Carotico Cavernous Fistula (CCF) :**

Arterio-venous fistula between carotid artery & cavernous sinus.

Cause : Trauma (75% cases), Spontaneous.

Feature :

• Proptosis :

- Bilateral
- Pulsatile
- Bruit & thrill present, stops on applying pressure to ipsilateral carotid artery.

• Ophthalmoplegia $\longrightarrow$ First affects CN VI.

IOC : Digital subtraction carotid angiography (DSCA).

Orbital varices :

Congenital enlargement of pre-existing venous channels.

----- Active space -----

Features : Proptosis

- Unilateral.
- Bag of worms consistency on palpation.
- Non-pulsatile.
- No bruit, no thrill.

Complication : Orbital hemorrhage/thrombosis.

TYPES OF PROPTOSIS**Intermittent proptosis :**

- ↑ on valsalva manoeuvre or bending forward : Orbital varices.
- ↑ with upper respiratory tract infection : Orbital lymphangiomas.
- ↑ on crying :
 - Children : Capillary hemangioma.
 - Infants : Encephalocele.

Pulsatile proptosis :

- Cortico cavernous fistula.
- Orbital roof fracture in :
 - NF-I
 - Trauma

Pseudo proptosis :

Appearance of eyeball protrusion.

Causes :

- Retraction of upper eyelid : Can be side effect of apraclonidine.
- Enophthalmos : Contralateral eye appears proptosed.
 - Seen in : Trauma, Horner's syndrome.
- Axial length > 26mm.
- Buphthalmos (enlarged eyeball).

Thyroid Associated Ophthalmology (TAO)

00:49:38

OVERVIEW

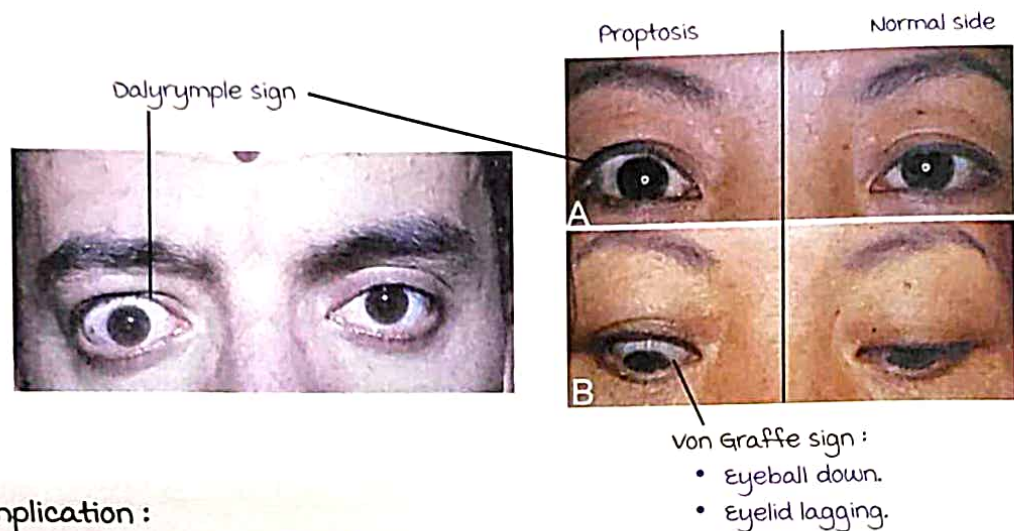
- m/c cause of unilateral and bilateral proptosis in adults.
- Can occur in any state of thyroid activity.

Pathogenesis :

- Inflammation → Enlargement of extra-ocular muscles.
- Inflammation and infiltration of orbital fat → Pushing of eyeball forward → Frank proptosis.

----- Active space ----- CLINICAL FEATURES

	Features
Goldzeiher's sign (earliest)	Congestion/hyperemia.
Dalrymple sign (m/c)	• upper eyelid retraction → pseudo proptosis. • D/t constriction of muller's muscle. • Seen in 92% of cases.
Proptosis (2 nd m/c)	• Axial unilateral or bilateral. • Seen in 60% cases.
Von Graefe sign	Lid lag : On downgaze, upper eyelid does not move down.
Kocher's sign	Staring/startled appearance.
Stellwags sign	Reduced eyelid blinking d/t upper eyelid retraction.
Restrictive squint	• Fibrosis of extra ocular muscles → Loss of movement • muscle affected first : Inferior rectus. (Elevation affected first.)

**Complication :**

Exposure keratitis (EK) d/t inability to close eye.

Management :

Ensure euthyroid state.

Supportive treatment :

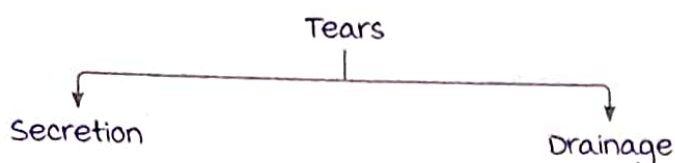
- For EK : Lubricants.
- For upper lid retraction : Botox injection into muller's muscle.
- For inflammation : Systemic steroids.

LACRIMAL APPARATUS : ANATOMY, WATERING EYE AND DRY EYE

----- Active space -----

Anatomy

00:00:48



SECRETION

Glands :

main lacrimal gland (1) :

- Divided by *levator palpebrae superioris* (2)
 - Orbital part (3) : Superior
 - Palpebral part (4) : Inferior
- Function : *Reflex tear* secretion.

Accessory lacrimal glands of Krause & Wolfring :

- Located in fornix (B/w sclera & palpebral conjunctiva).
- Function : Basal tear secretions.

Parasympathetic innervation :

Secretomotor function.

Centre : Superior salivatory nucleus in pons.

Pre-ganglionic N. : Greater superficial petrosal N. (Branch of facial N.)

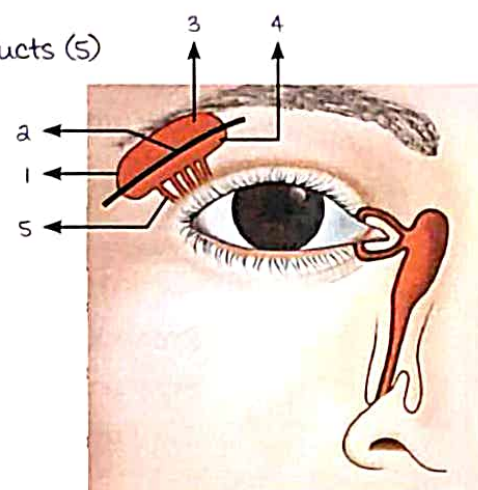
Ganglion : Pterygopalatine.

Post ganglionic N. : Zygomatic N. (Branch of maxillary branch of trigeminal N.).

↓ Stimulates

Lacrimal gland

↓ Secretion



Lacrimal apparatus

Sympathetic innervation :

Inhibitory function.

----- Active space -----

DRAINAGE

Puncta (1) : upper & lower.

Canaliculi : upper (2) & Lower (3)

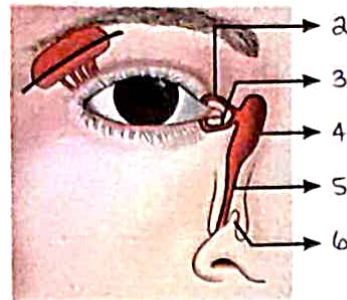
Lacrimal sac (4)

- Surrounded by orbicularis oculi

↓ On blinking

Nasolacrimal duct - NLD (5) opens in inferior meatus of nose (6).

- Opening of NLD guarded by **valve of Hasner**.
- m/c **block** in congenital dacryocystitis.



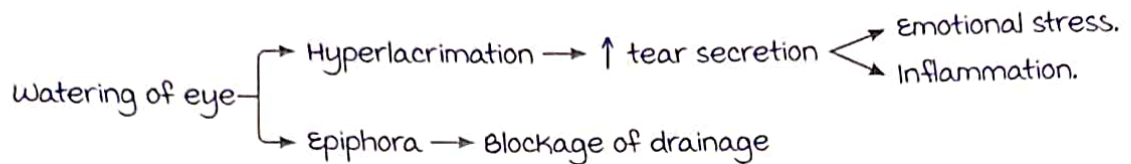
Lacrimal apparatus



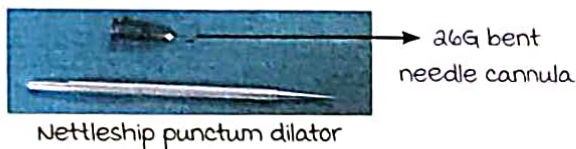
Lower puncta (1)

Watering of Eye & Tumours

00:09:00

**TESTING OF WATERING EYE**

Instruments :



Nettleship punctum dilator

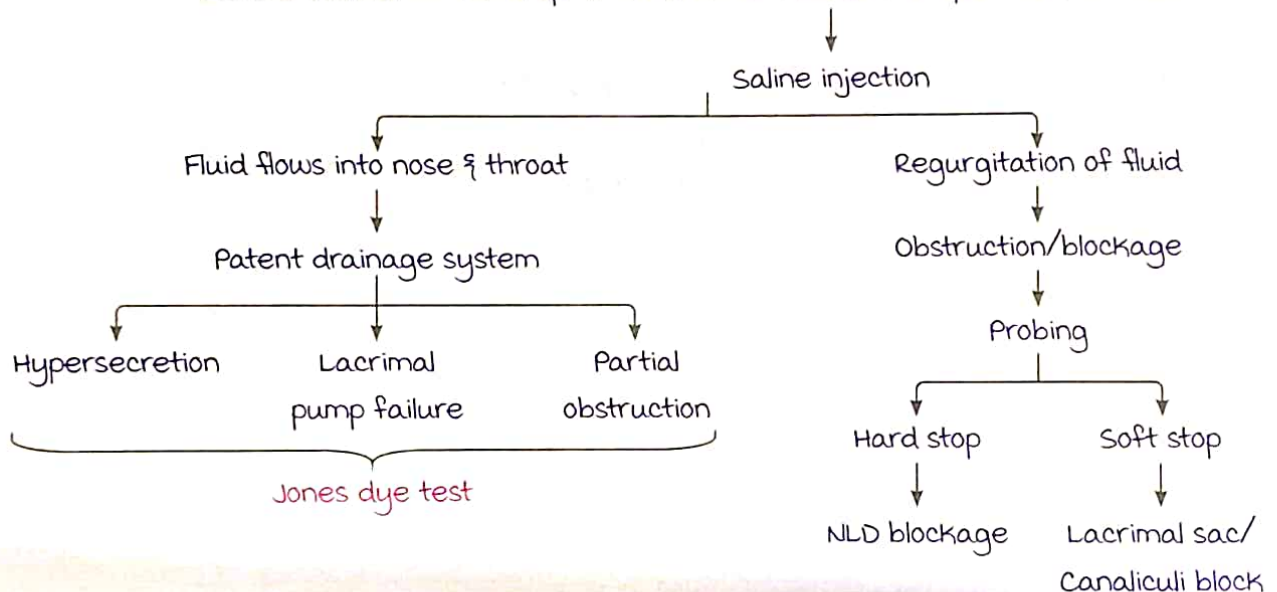


Bowman's lacrimal probe

Procedure :

Syringing :

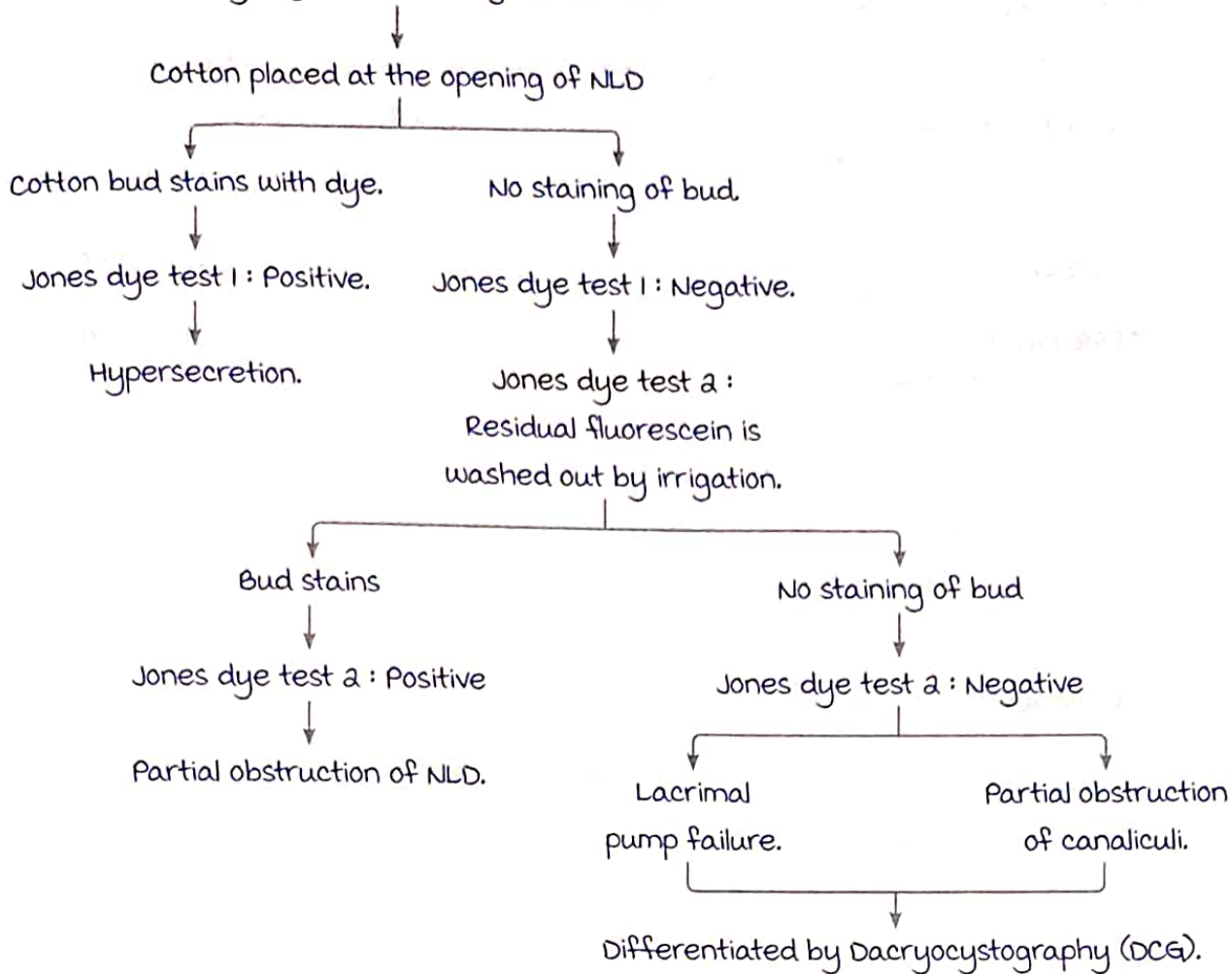
Puncta dilated → Blunt tip of 26G needle inserted into puncta & canaliculi.



----- Active space -----

Jones dye test I :

2% fluorescein dye injected into conjunctival sac

**DACRYOCYSTITIS**

Inflammation of lacrimal sac.

Types :**Congenital :**

Presentation : c/o watering since birth.

Cause : Non-canalization of NLD.

- m/c site of occlusion :
Valve of Hasner.

Rx :

- Lacrimal sac massage + Topical antibiotics (Tobramycin, erythromycin).
- 9-12 months : Syringing.
- >1 yr : Probing.
- >4 yrs : Dacryocystorhinostomy (DCR).

Acquired :

Cause : Streptococcus hemolyticus.

Rx :

- Oral antibiotics.
- DCR : If fistula (+).

----- Active space -----

TUMOURS OF LACRIMAL GLAND

Pleomorphic adenoma :

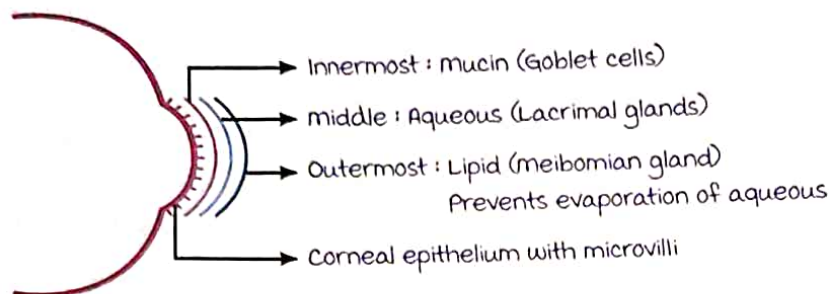
Benign tumour

Adenoid cystic carcinoma

malignant tumour

Displacement of eye **downwards and medially**
d/t location of lacrimal gland.

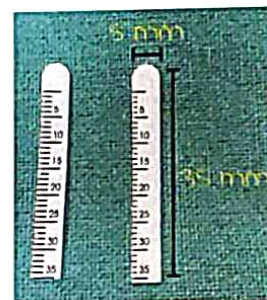
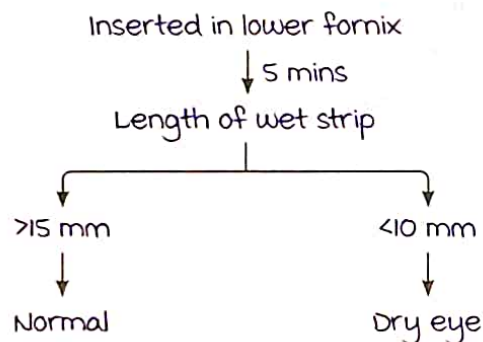
00:26:45

Dry Eye**TEAR FILM****DISORDERS**

	Aqueous deficiency	Lipid deficiency	mucin deficiency
AKA	keratoconjunctivitis sicca	Evaporative dry eye	-
Eg.	Eg. Sjögren syndrome	- meibomian gland disease - Contact lens wearer	vit A deficiency

TESTS FOR DRY EYES**I. Schirmer test :**

- Quantitative test to measure tear production.
- uses Whatman's filter paper No. 41.

Procedure :

whatman's filter paper



Schirmer test

2. TBUT (Tear Film Break-up Time) :

2% Fluorescein instilled in conjunctival sac → Illuminated with blue light.

Interval b/w a blink & appearance of 1st spot of dryness of cornea



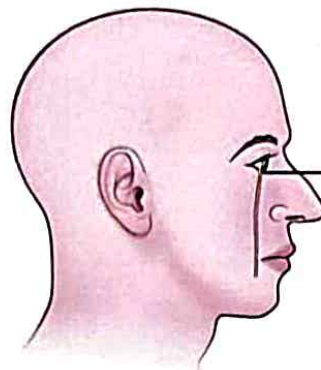
3. Phenol red thread test :

Thread impregnated with phenol red dye :

Yellow in colour

↓ + tears

Red in colour



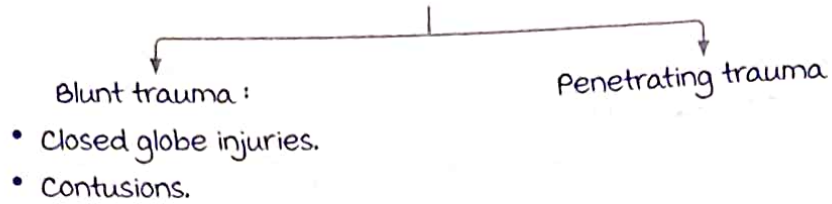
Phenol red thread test

----- Active space -----

OCULAR TRAUMA

00:00:44

Mechanical Ocular Trauma



BLUNT TRAUMA

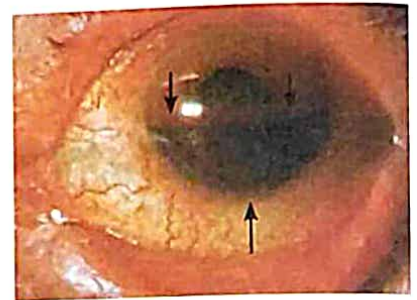
Cause :

- Injury by fist.
- Tennis ball.
- Champagne cork.

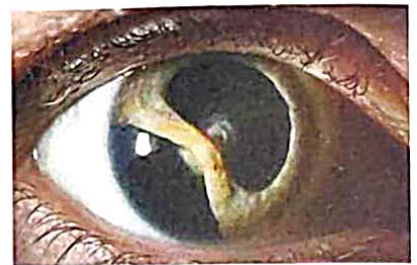
Signs :

1. Black eye : Periorbital hematoma.
2. Subconjunctival hemorrhage : Common in trivial trauma.
3. Hyphema : Collection of blood in anterior chamber
 - m/c sign of blunt ocular trauma.
 - Source of bleeding : Circulus Arteriosus major.
 - Rx :
 - Topical steroids : To prevent re-bleeding & associated inflammation.
 - Atropine (Optional).
 - Paracentesis/washing of anterior chamber : If no resorption until 4 days.
4. In Iris :
 - Iridodonesis : Tremulous/trembling iris.
 - Iridodialysis : Detachment of root of iris from ciliary body.

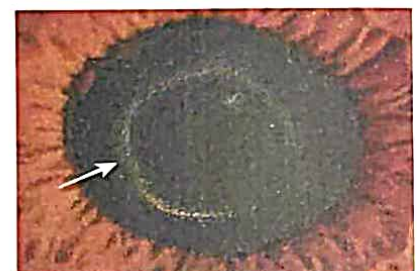
} D-shaped pupil.
5. In lens :
 - Rosette shaped cataract : Posterior subcapsular.
 - Ectopia lentis/Subluxation of lens :
 - m/c cause : Blunt trauma.
 - Vossius ring : Imprint of miotic pupil on anterior surface of lens.
 - Phacodonesis/tremulous lens.



Hyphema



Iridodialysis



Vossius ring

6. Fundus :

- Cherry red spot at macula.
 - **Berlin's edema** : Fluid collection d/t blunt trauma.
- Comotio retinae : Cloudy grey appearance.
- Traumatic optic neuropathy : Self-limiting.
- Choroidal rupture : **Bucket handle configuration**.

----- Active space -----



Scleral show d/t choroidal rupture

7. Globe rupture :

- Occurs d/t fall on a knob-like object.
- Site : Supero-nasal limbus near Schlemm's canal.

PENETRATING TRAUMA

- Sympathetic ophthalmitis (Panuveitis).
- Intra-ocular foreign body.

Intra-ocular foreign body :

- **Iron, steel** : m/c foreign body.
 - IOC : **CT scan**.
 - C/I : MRI.
- Deposition of iron → **Siderosis bulbi**.
 - Deposition of rust ring on anterior capsule of lens.
 - Hyperpigmentation of iris.
 - 2° open angle glaucoma.
 - Dx : Prussian blue reaction → Stains iron deposits.

Note :

Ophthalmia nodosa : Contact with Caterpillar hair → uveitis.

Orbital Trauma

00:24:04

BLOW OUT FRACTURE

- Fracture of floor of orbit.
- **m/c fracture in orbital trauma**.

Clinical feature :

- Infraorbital nerve anesthesia.
- Double diplopia : Restrictive squint.
 - Fractured bone pieces → muscle entrapment of **inferior oblique & inferior rectus** → Diplopia in upgaze & downgaze.

----- Active space ----- Investigation :

X-ray : Tear drop sign.



Tear drop sign

FRACTURE OF ROOF OF ORBIT

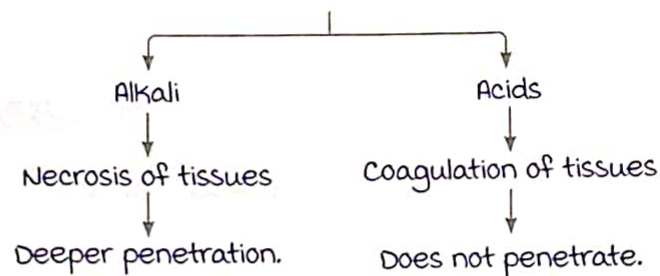
Presentation : Raccoon eyes/Panda eyes.



Raccoon eyes

Chemical injuries

00:29:49



Classification :

Dua's classification.

Roper-Hall classification.

} Grading of ocular surface burns.

management :

1. Copious irrigation :

Balanced salt solution/Ringer Lactate by double eversion of eyelids.

2. Topical antibiotics.

3. Atropine : To relieve ciliary spasm.

4. Lubricating eye drops.

5. Topical sodium citrate : Prevents proteolysis.

COMMUNITY OPHTHALMOLOGY

----- Active space -----

NPCB & VI

00:00:52

Amended from NPCB (Started in 1976) to NPCBVI in 2017.

DEFINITIONS OF BLINDNESS

Clinically : No perception of light.

According to NPCBVI (As amended in 2017) and

WHO :

- Presenting visual acuity $<3/60$ in the better eye with best possible correction.

or

- Limitation of visual field to less than 10° from center of fixation.



Visual Impairment :

Term	Definition according to NPCB & VI
Blindness	Presenting visual acuity $<3/60$ in better eye with available correction.
Severe visual impairment (SVI)	Presenting visual acuity $<6/60 - 3/60$ in better eye with available correction.
Moderate visual impairment (MVI)	Presenting visual acuity $<6/18 - 6/60$ in better eye with available correction.
Early visual impairment (EVI)	Presenting visual acuity $<6/12 - 6/18$ in better eye with available correction.
Functional low vision	• Visual acuity $<6/18$ or • Visual field $<10^\circ$ from fovea. • Able to/potentially able to use vision to plan/execute a task.
Blindness (BCVA/pinhole $<3/60$)	Best corrected visual acuity $<3/60$ in better eye.

RECENT FACTS

Prevalence of blindness :

0.36% (2019) according to National Blindness & Visual Impairment Survey (2015-19).

Goal of program :

Reduce prevalence of avoidable blindness (Preventable + curable) to $<0.25\%$ by 2025.

----- Active space ----- Causes and prevalence :

	≥50yrs of age	0-49 yrs
Blindness	- Cataract (m/c) : 66.2% - Corneal opacity due to any cause (2nd m/c) : 8.2%	- Corneal opacity (m/c) : 37.5% - Amblyopia : (2nd m/c) : 25%
Visual impairment	- Cataract (m/c) : 71.2% - Refractive errors (2nd m/c) : 13.4%	- Refractive errors (m/c) : 29.6% - Cataract (2nd m/c) : 25%

Vision 2020

00:17:05

OVERVIEW

- Launched in 1999 at Geneva, Switzerland.
- Adopted by India in 2001 at Goa.

Aims :

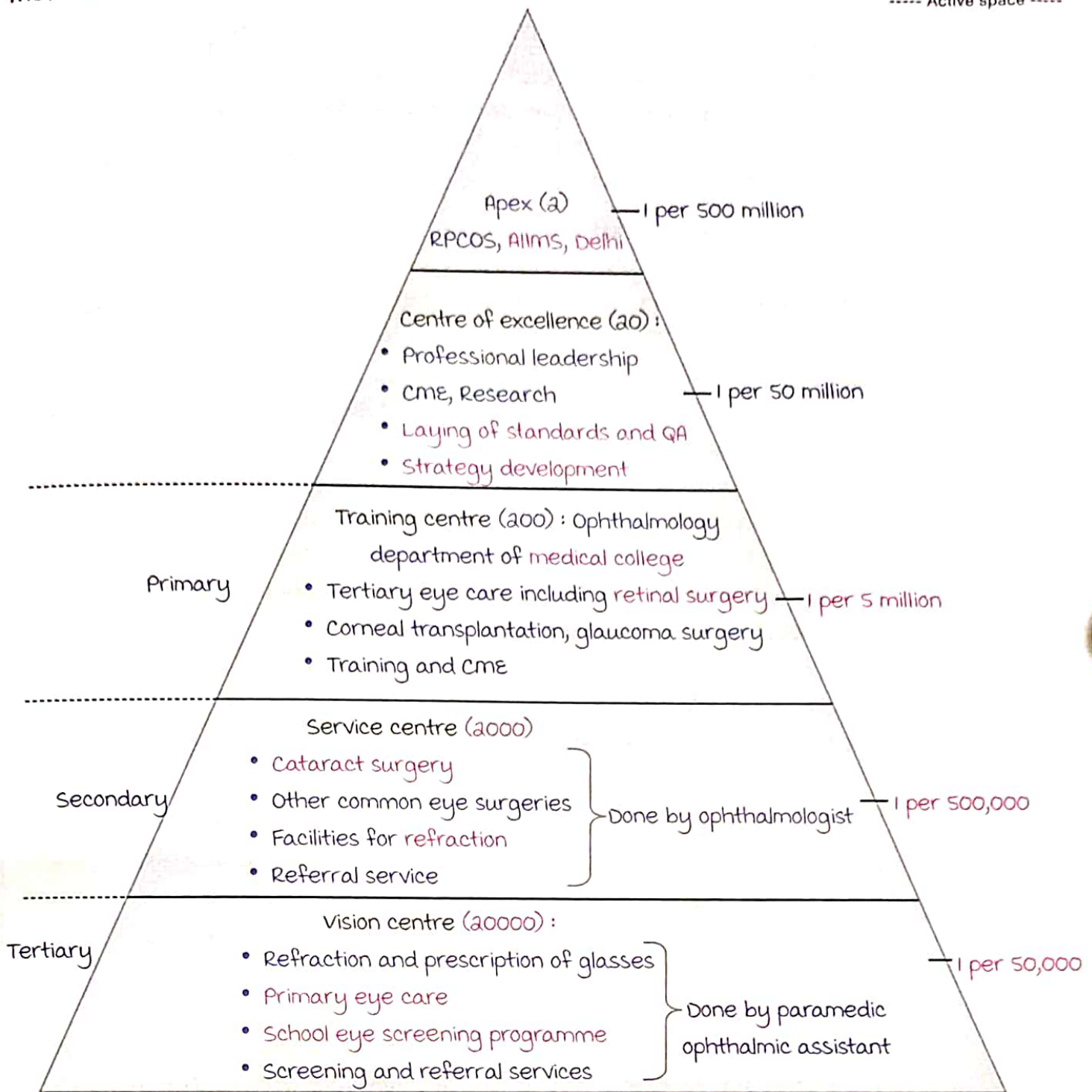
- Reduce blind people to 25 million by 2020.
- Eliminate avoidable blindness.

Diseases covered :

India	Globally
- Cataract - Refractive errors - Childhood blindness - Trachoma (Endemic) - Glaucoma - Diabetic retinopathy - Corneal blindness	- Cataract - Refractive errors - Childhood blindness - Trachoma (Endemic) - Onchocerciasis

METRICS PROVIDED BY VISION 2020

----- Active space -----



School eye screening programme:

Target: Class V - VIII (10-14 yrs age).

Performed by: Teacher → 1 teacher per 150 students.

method: Vision screening cards + measurement tape

↓
Record visual acuity (VA)↓
If VA $\leq 6/9$ in either eye: Refer to nearest PHC (Primary health centre)

Rashtriya Bal Swasthya Karyakram (RBSK)

00:26:53

Overview

- Launched under National health mission (NHM).
- Screening and early intervention for children (Birth - 18 yrs of age).

SCREENING

Health conditions screened (4 D's) :

4 D's :

- Defects at birth.
- Deficiency.
- Diseases.
- Developmental delays.

Health conditions screened under this program :	
Defects at birth : 1. Neural tube defect. 2. Down's syndrome. 3. Cleft lip & palate/cleft palate alone. 4. Talipes (Club foot). 5. Developmental dysplasia of the hip. 6. Congenital cataract. 7. Congenital deafness. 8. Congenital heart diseases. 9. Retinopathy of Prematurity.	Deficiencies : 10. Anaemia especially severe anaemia. 11. Vitamin A deficiency (Bitot spot). 12. Vitamin D Deficiency (Rickets). 13. Severe acute malnutrition. 14. Goiter.
Childhood diseases : 15. Skin conditions (Scabies, fungal infection & eczema). 16. Otitis media. 17. Rheumatic heart disease. 18. Reactive airway disease. 19. Dental caries. 20. Convulsive disorders.	Developmental delays : 21. Vision impairment. 22. Hearing impairment. 23. Neuro-motor impairment. 24. Motor delay. 25. Cognitive delay. 26. Language delay. 27. Behaviour disorder (Autism). 28. Learning disorder. 29. Attention deficit hyperactivity disorder. 30. Congenital hypothyroidism, Sickle cell anaemia, Beta thalassemia (Optional).
31. Tuberculosis. 32. Leprosy.	

} Not specific to children.

mechanism of screening :

----- Active space -----

Categories	Age group	Estimated coverage	Screening performed by
Babies born at public health facilities (PHC & CHC) and home.	Birth to 6 weeks.	2 crores.	mobile health team.
Preschool children in rural areas and urban slum.	6 weeks to 6 yrs.	8 crores.	Anganwadi & ANM workers.
School children enrolled in class 1 st to 12 th in government and government aided schools.	6 yrs to 18 yrs.	17 crores.	A school supported by district hospital : Performed via questionnaires.

INTERVENTION

1. 0-6 yrs : District Early Intervention Center (DEIC) → Referral to appropriate tertiary centre.
2. 6-18 yrs : Existing public health facility.

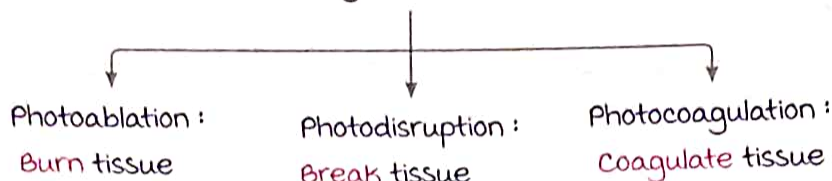
----- Active space -----

LASERS IN OPHTHALMOLOGY

Light Amplification by Stimulated Emission of Radiation (LASER)

Principle :

Laser field → Produce effects in eye



Types of Lasers

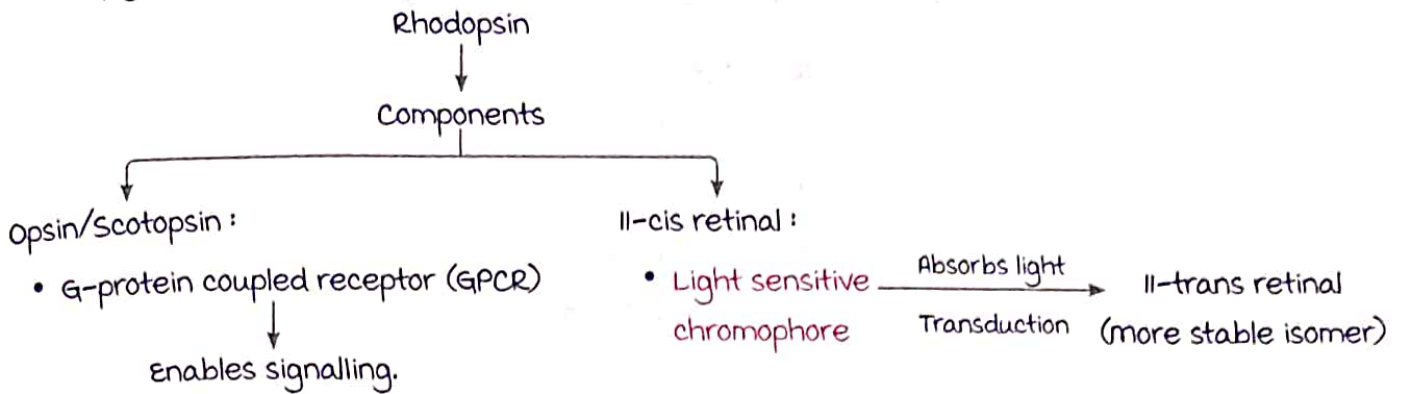
00:02:21

Based on tissue effects caused	Lasers causing photoablation	Lasers causing photocoagulation	Lasers causing photodisruption
mechanism	Producing chemical effects : breaks chemical bonds ↓ vaporizing tissue.	Thermal effect ↓ denaturation of proteins.	3 Effects working simultaneously : • Stripping electrons from nuclei → Ionization of tissue. • Production of plasma-like materials → Breakage of chemical bonds. • Generation of acoustic shock waves → mechanical effect on tissue.
wavelength used	ultraviolet (150nm - 300nm)	visible spectrum (400nm - 580nm)	Infrared : invisible ↓ Helium-neon beam (red) used to view and aim beam.
Examples	Excimer laser: Argon & Fluoride (193 nm)	- Argon green laser: 514nm - Frequency doubled Nd: YAG laser : 532nm - Diode laser : 810 nm (preferred in retinopathy of prematurity). - Ruby Red laser : 694nm	- Nd : YAG (Neodymium Yttrium : Aluminium Garnet) : 1064nm. - ND : Glass : 1053nm.
Uses	Corneal refractive procedures : • LASIK • PRK	- Pan retinal photocoagulation for: - Diabetic retinopathy - CRVO - Laser trabeculoplasty for POAG. - Focal retinal laser photocoagulation for: - Diabetic macular Edema - CSR - ARMD	- Nd: Glass : • FLACS for Cataract sx. • SMILE for refractive error correction. - Nd: YAG: • Posterior Capsulotomy for PCO. • Peripheral iridotomy for ACG.

PHYSIOLOGY OF VISION

----- Active space -----

Visual pigment :



Visual Cycle

00:02:22

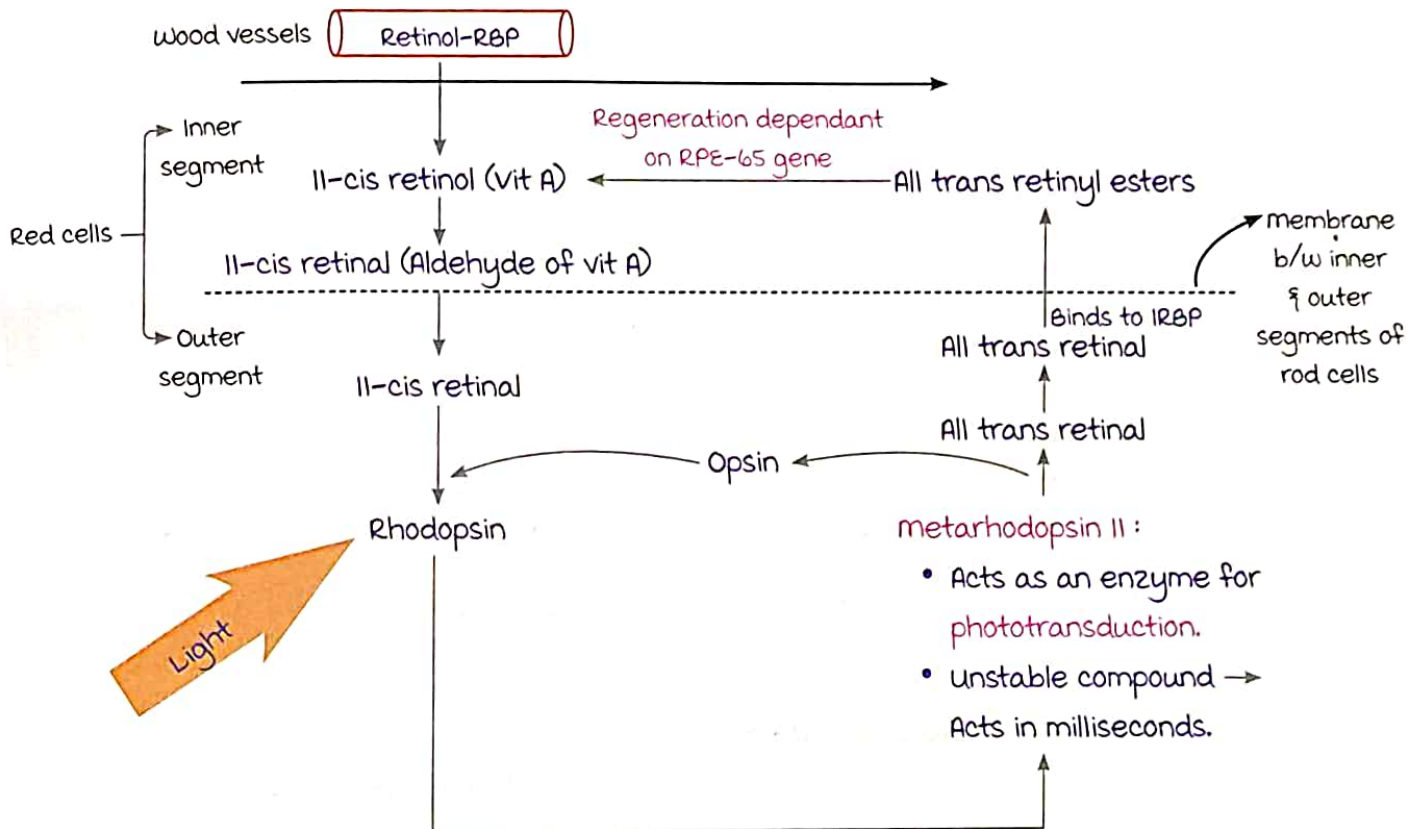
Site : Photoreceptors (Rod cells).

RBP : Retinol binding protein

Binds retinol in blood.

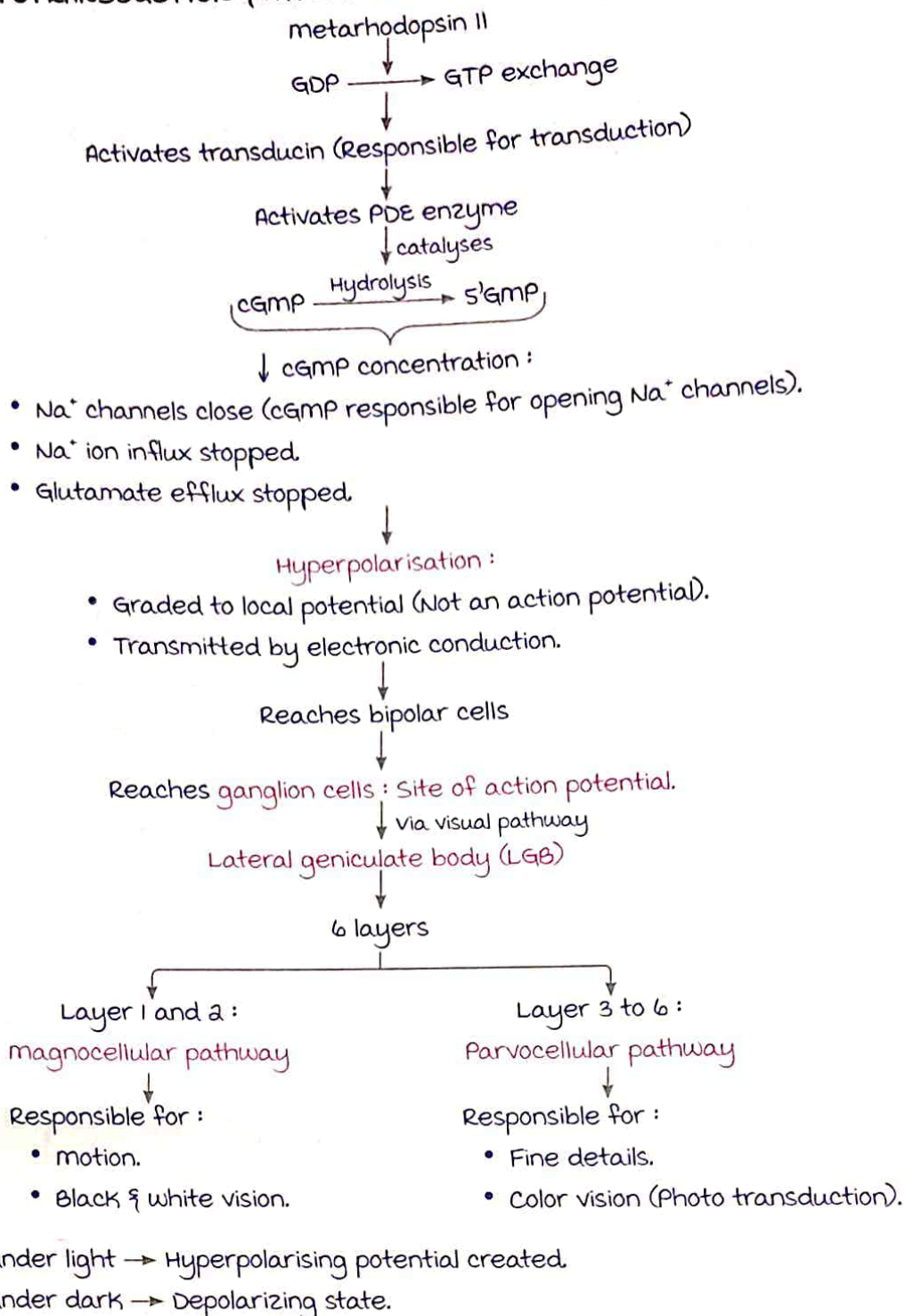
PATHWAY

IRBP : Intra Retinal Binding Protein.



----- Active space -----

PHOTOTRANSDUCTION & HYPERPOLARISATION



DEFECT IN VISUAL PATHWAY

